Numerical Techniques + Applied Numerical Analysis.

Introduction:

$$\int_1^2 \sin x^2 dx \quad , \quad \int_0^1 e^{x^3} dx \ldots$$

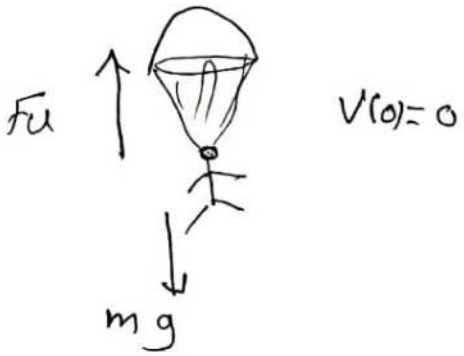
$$2x - 6 = 0 \rightarrow x = 3$$

$$x^3 + x - 3 = 0$$

$$x = e^{-x}$$

Modeling

Parachutist Problem.



m : mass (kg)

V: velocity

$Fu \propto V \rightarrow Fu = cV$

c: drag coefficient kg/s

$F_{resultant} = ma$ $\quad a = \frac{dv}{dt}$

$$mg - cV = m\frac{dv}{dt}$$ Differential Equation.

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$$m\frac{dv}{dt} = mg - cv$$

$$m\frac{dv}{dt} + cv = mg$$

$\div m \rightarrow$ $\boxed{\frac{dv}{dt} + \frac{c}{m}v = g}$ First order LDE

$* \rightarrow e^{\int \frac{c}{m} dt} = e^{\frac{c}{m}t}$

$$e^{\frac{c}{m}t}\frac{dv}{dt} + \frac{c}{m}e^{\frac{c}{m}t}v = g e^{\frac{c}{m}t}$$

$y' + P(x)y = g(x)$

$\rightarrow * \; e^{\int P(x)dx}$

$$\frac{d}{dt}\left(e^{\frac{c}{m}t}v\right) = g e^{\frac{c}{m}t}$$

Integrate $\rightarrow$ $e^{\frac{c}{m}t}v = g \cdot \frac{m}{c} e^{\frac{c}{m}t} + K$

$\div e^{\frac{c}{m}t} \rightarrow$ $v = \frac{gm}{c} + K e^{-\frac{c}{m}t}$

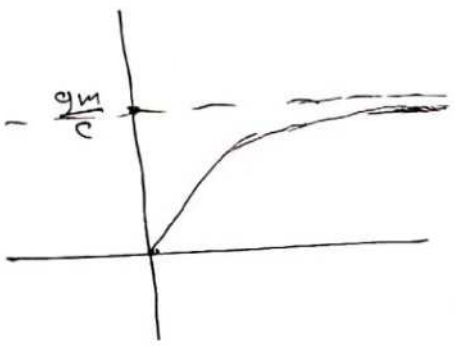
But $v(0) = 0 \Rightarrow \frac{gm}{c} + K = 0$

$$\Rightarrow K = -\frac{gm}{c}$$

$$\therefore v = \frac{gm}{c} - \frac{gm}{c} e^{-\frac{c}{m}t}$$

$$\boxed{v = \frac{gm}{c}\left(1 - e^{-\frac{c}{m}t}\right)}$$

as $t \rightarrow \infty \Rightarrow v \rightarrow \frac{gm}{c}$



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Taylor and Maclaurin series:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \cdots$$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

Taylor series of $f(x)$ about $x=a$.

$a = 0 \longrightarrow$ $f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)\,x^k}{k!}$$

Maclaurin series

---

e.g: Find Taylor Expansion of $f(x) = e^x$ about $x = 1$.

$$f(x) = f(1) + f'(1)(x-1) + \frac{f''(1)(x-1)^2}{2!} + \frac{f'''(1)(x-1)^3}{3!} + \cdots$$

$f(x) = e^x \rightarrow f(1) = e$

$f'(x) = e^x \rightarrow f'(1) = e$

$f''(x) = e^x \rightarrow f''(1) = e$

$$\therefore f(x) = e + e(x-1) + \frac{e(x-1)^2}{2!} + \frac{e(x-1)^3}{3!} + \cdots$$

$$e^x = \sum_{k=0}^{\infty} \frac{e(x-1)^k}{k!}$$

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e.g: Find Maclaurin series of the given function.

1) $f(x) = e^x$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$$

$f(x) = e^x \longrightarrow f(0) = e^0 = 1$

$f'(x) = e^x \longrightarrow f'(0) = 1$

$f''(x) = e^x \longrightarrow f''(0) = 1$

$\vdots$

$$\therefore f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

2) $f(x) = \sin x$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$$

$f(x) = \sin x \longrightarrow f(0) = 0$

$f'(x) = \cos x \longrightarrow f'(0) = 1$

$f''(x) = -\sin x \longrightarrow f''(0) = 0$

$f'''(x) = -\cos x \longrightarrow f'''(0) = -1$

$f^{(4)}(x) = \sin x \longrightarrow f^{(4)}(0) = 0$

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$$f(x) = 0 + 1x + \frac{(0)x^2}{2!} + \frac{(-1)x^3}{3!} + \frac{0x^4}{4!} + \frac{1x^5}{5!} + \cdots$$

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

Rules:

1) $$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \qquad x \in \mathbb{R}$$

2) $$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

3) $$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

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4) $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$

$$= \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

5) $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$

$$= \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$$

$a + ar + ar^2 + \cdots = \frac{a}{1-r}$ $\quad |r|<1$

6) $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$

$|x|<1$

$\Leftrightarrow -1 < x < 1$

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \qquad |x|<1$$

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(7)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \qquad x \in \mathbb{R}$$

$$x = 1 \rightarrow e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots$$

$$x = \frac{1}{2} \rightarrow e^{\frac{1}{2}} = \sqrt{e} = 1 + \left(\frac{1}{2}\right) + \frac{\left(\frac{1}{2}\right)^2}{2!} + \frac{\left(\frac{1}{2}\right)^3}{3!} + \cdots$$

$$\int_0^1 e^{x^2} dx$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots$$

$$e^{x^2} \sim 1 + x^2$$

$$\int_0^1 e^{x^2} dx = \int_0^1 (1 + x^2) dx = \cdots$$

END

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Recall

1) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

$= \sum_{k=0}^{\infty} \frac{x^k}{k!}$

2) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$

3) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$

~~4) $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$~~

4) $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$, $|x| < 1$

$= \sum_{k=0}^{\infty} x^k$

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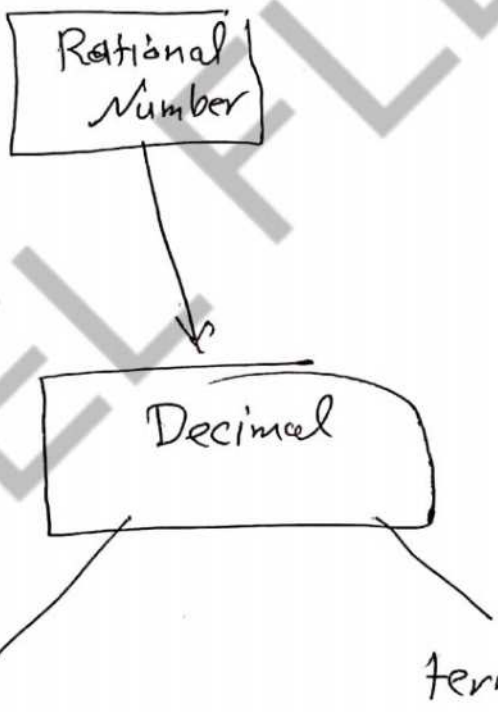
Set of Numbers

$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, 3, \ldots\}$ set of integers

$\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ set of rationals

$\hookrightarrow \frac{2}{5}, -\frac{1}{4}, \frac{22}{7}, 8, 0,$

$2.5, 0.\overline{3} = 0.3333\ldots$



$0.4444\ldots = 0.\overline{4}$

$0.9999\ldots = 0.\overline{9}$

$2.3275275275\ldots = 2.3\overline{275}$

1.42, 7.6, 3, 2.31750

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$\bar{Q}$: irrational numbers

- $\sqrt{2}, \sqrt{3}, \sqrt{5}, \cdots$
- $\pi \approx \frac{22}{7}$
- $e \approx 2.7$
- $5.141441444144441\cdots$

$\mathbb{R} = Q \cup \bar{Q}$ = real numbers

$Q \cap \bar{Q} = \phi$

e.g: Show that $0.\bar{9} = 1$?

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots \quad |x| < 1$$

$0.\bar{9} = 0.9999999\cdots$

$= 0.9 + 0.09 + 0.009 + \cdots$

$\frac{0.09}{0.9} = 0.1$, $\frac{0.009}{0.09} = 0.1, \cdots$

G.S $r = 0.1$

$$a + ar + ar^2 + \cdots = \frac{a}{1-r}$$

$a = 0.9$

$$0.\bar{9} = \frac{0.9}{1 - 0.1} = \frac{0.9}{0.9} = 1$$

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e.g: Write $0.\overline{4}$ in the form $\frac{a}{b}$, $a, b \in \mathbb{Z}$

$0.\overline{4} = 0.4 + 0.04 + 0.004 + \cdots$

g.s $a = 0.4$, $r = 0.1$

$$0.\overline{4} = \frac{0.4}{1 - 0.1} = \frac{0.4}{0.9} = \frac{4}{9}$$

e.g: write $0.\overline{23}$ in the form $\frac{a}{b}, a, b \in \mathbb{Z}$?

$0.\overline{23} = 0.23 + 0.0023 + 0.000023 + \cdots$

g.s $r = 0.01$, $a = 0.23$

$$\therefore 0.\overline{23} = \frac{0.23}{1 - 0.01}$$

$$= \frac{0.23}{0.99}$$

$$= \frac{23}{99}$$

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Accuracy and Precision:

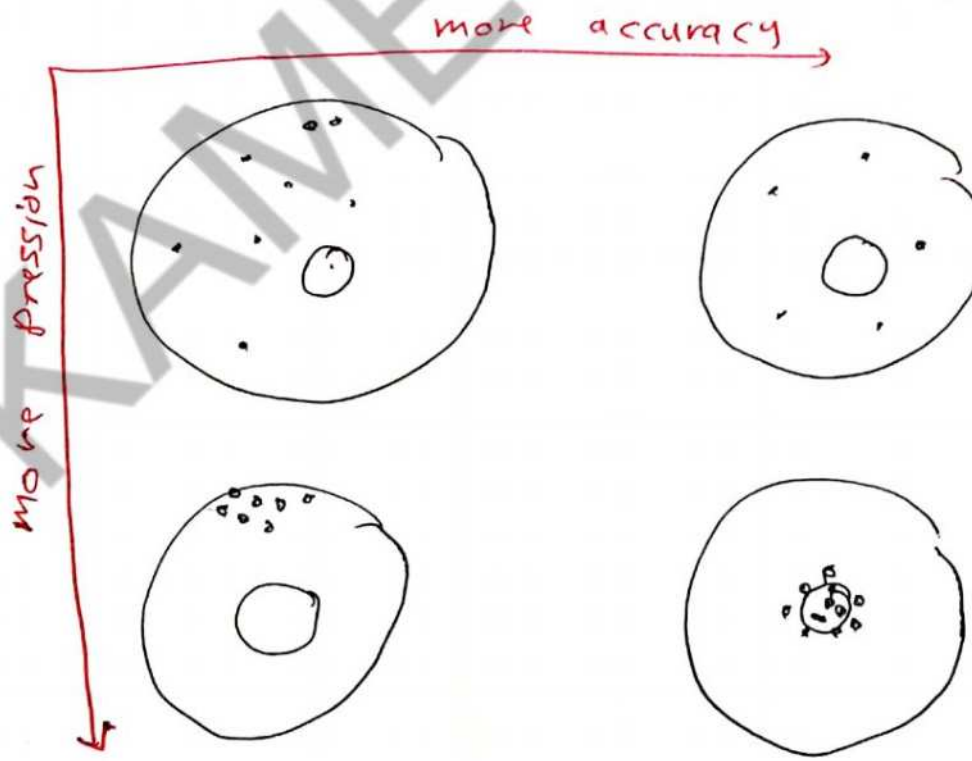
Accuracy: The closeness of computed or measured values to the true value

Precision: The closeness of computed or measured value to each other.

True length of the pen = 10 cm

10.1, 9.9, 10.01, 9.95 high accuracy and high precision

20.1, 20.2, 19.9, 20.01 Low ~~high~~ accuracy and high precision



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Error Definitions:

If X is the true value and $\hat{X}$ is the approximation, then

1) True Error: $E_t = X - \hat{X}$

2) Absolute True Error: $|E_t| = |X - \hat{X}|$

3) True Relative Error: $\epsilon_t = \frac{X - \hat{X}}{X}$

4) Absoule True Relative Error: $|\epsilon_t| = \left|\frac{X - \hat{X}}{X}\right|$

5) $|\epsilon_t| = \left|\frac{X - \hat{X}}{X}\right| * 100\%$

Absolute true percent relative error.

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e.g: If the true weight of car is 1000 Kgs and the approximation was 999 kgs.

Also, the true weight of a boy 20 kgs and the approximation was 19 kgs.

Find the true error and the true percent relative error for the car and the boy.

Sol:

Car

True value = 1000 kg = X

Approximated value = 999 kgs = $\tilde{X}$

$$E_t = X - \hat{X} = 1000 - 999 = 1 kg$$

$$\epsilon_t = \frac{X - \hat{X}}{X} * 100\% = \frac{1}{1000} * 100\% = 0.1\%$$

Boy:

True value: $Y = 20$ kgs

Approximated value $\hat{Y} = 19$ kg.

$$E_t = Y - \hat{Y} = 20 - 19 = 1 kg$$

$$\epsilon_t = \frac{Y - \hat{Y}}{Y} * 100\% = \frac{1}{20} * 100\% = 5\%$$

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Def: If X^{New} and X^{old} are two approximations for a quantity then

1) The Approximate Error : $E_a = X^{New} - X^{old}$

2) The Relative Approximate Error

$$\epsilon_a = \frac{X^{New} - X^{old}}{X^{New}}$$

$$= \frac{X^{New} - X^{old}}{X^{New}} * 100\%$$

e.g: $X^{current} = 5.5$, $X^{previous} = 3.5$

find ϵ_a

$$\epsilon_a = \frac{5.5 - 3.5}{5.5} = \frac{2}{5.5} \simeq 0.36 = 0.36 * 100\%$$

$$= 36\%$$

END

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Errors (Part 2)

ϵ : Tolerance

ϵ_s : Tolerance related to significant figures.

Scarborough criterion:

To get a ~~beta~~ result to be correct to at least n significant figures, do iterations till you get $|\epsilon_a| < \epsilon_s$ (i.e stop).

where $\epsilon_s = (0.5 \times 10^{-n}) * 100\%$

$$\epsilon_s = (0.5 \times 10^{2-n})\%$$

Recall $\epsilon_a\% = \left| \frac{\text{current app.} - \text{previous app.}}{\text{current approximation}} \right| * 100\%$

$$\epsilon_t\% = \left| \frac{\text{true value} - \text{approximation}}{\text{true value}} \right| * 100\%$$

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e.g: Approximate the value of $e^{0.3}$ by using (2) Maclaurin Series of e^x, start by using 2 terms and on of the series.

Use Scarborough Criterion so that your answer to be correct to at least 3 significant figures.

Sol:

The answer will be correct to at least 3 significant figure, $\epsilon_a\% \leq (0.5 \times 10^{2-3})\% = \epsilon_s$

i.e stop when $\epsilon_a\% < (0.05)\%$

Now $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

if $x = 0.3 \Rightarrow$ $e^{0.3} = 1 + 0.3 + \frac{(0.3)^2}{2!} + \frac{(0.3)^3}{3!} + \frac{(0.3)^4}{4!} + \cdots$

1st iteration (2 terms):

$e^{0.3} \simeq 1 + 0.3$

$e^{0.3} \simeq 1.3$

we cant compute ϵ_a here.

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2nd iteration (3 terms)

$$e^{0.3} \simeq 1 + 0.3 + \frac{(0.3)^2}{2!}$$

$$e^{0.3} \simeq 1.3 + \frac{(0.3)^2}{2!}$$

$$\therefore \boxed{e^{0.3} \simeq 1.345}$$

$$\epsilon_a \% = \left|\frac{1.345 - 1.3}{1.345}\right| * 100\%$$

$$\simeq 3.3457\% \not< 0.05\% = \epsilon_s$$

3rd iteration (4 terms):

$$e^{0.3} \simeq 1 + 0.3 + \frac{(0.3)^2}{2!} + \frac{(0.3)^3}{3!}$$

$$e^{0.3} \simeq 1.345 + \frac{(0.3)^3}{3!}$$

$$\boxed{e^{0.3} \simeq 1.3495}$$

$$\epsilon_a = \left|\frac{1.3495 - 1.345}{1.3495}\right| * 100\%$$

$$\simeq 0.333 \not< 0.05\% = \epsilon_s$$

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4th iteration (5 terms):

$$e^{0.3} \cong 1 + 0.3 + \frac{(0.3)^2}{2!} + \frac{(0.3)^3}{3!} + \frac{(0.3)^4}{4!}$$

$$\therefore e^{0.3} \cong 1.34495 + \frac{(0.3)^4}{4!}$$

$$\boxed{e^{0.3} \cong 1.3498375}$$

$$\epsilon_a = \left| \frac{1.3498375 - 1.3495}{1.3498375} \right| * 100\%$$

$$\epsilon_a \cong 0.02\% < 0.05\% \rightarrow \text{Stop}$$

$$\therefore \boxed{e^{0.3} \cong 1.3498375}$$

To find the (true value)? use ~~calculator~~ Calculator

Could be used as true value → 1.349858808

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e.g: Using the first three non-zero terms of Maclaurin series for $\sin x$, approximate the value of the integral

$$\int_1^{1.2} \frac{\sin x}{x}\,dx$$ (round off your answer to 5 decimal places).

And then compute the relative true error ($\epsilon_t\%$)

Solution

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\sin x \cong x - \frac{1}{6}x^3 + \frac{1}{120}x^5$$

$$\Rightarrow \frac{\sin x}{x} \cong 1 - \frac{1}{6}x^2 + \frac{1}{120}x^4$$

$$\Rightarrow \int_1^{1.2} \frac{\sin x}{x}\,dx \cong \int_1^{1.2} \left(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4\right) dx$$

$$\Rightarrow \int_1^{1.2} \frac{\sin x}{x}\,dx \cong 0.16204 \quad \text{(to 5 decimal places)}$$

True value: $\int_1^{1.2} \frac{\sin x}{x}\,dx = 0.161964128\,6$

$$\therefore \epsilon_t \cong \left|\frac{0.16196 - 0.16204}{0.16196}\right| * 100\% = 0.04\%$$

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Chopping and Rounding off:

Def: A number X is said to be in normalized decimal form (decimal floating point form) if

$$X = \underbrace{(0.d_1d_2d_3\cdots)}_{\text{fraction part (mantissa)}} \times \underbrace{10^{n}}_{\text{exponential part}}$$

where $d_1, d_2, \cdots \in \{0, 1, 2, \cdots, 9\}$

and $d_1 \neq 0$ (non-zero), $n \in \mathbb{Z}$ ← integer

i.e $X = 0.\square\square\square\cdots \times 10^{n}$

e.g: Normalize the following numbers.

1) 145.7632×10^{0}

$= 0.1457632 \times 10^{3}$

2) 12.3×10^{-10}

$= 0.123 \times 10^{-10+2}$ $= 0.123 \times 10^{-8}$

3) 0.00027×10^{-15}

$= 0.27 \times 10^{-15-3}$

$= 0.27 \times 10^{-18}$

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e.g: After normalizing, chop the following numbers to 4 significant digits.

1) $0.0007247783 = 0.7247783 \times 10^{-3}$

$= 0.7247 \times 10^{-3}$

$= 0.0007247$

2) 7.8135×10^{5}

$= 0.78135 \times 10^{5+1}$

$= 0.78135 \times 10^{6}$

$= 0.7813 \times 10^{6}$

3) 0.0085482

$= 0.8548|2 \times 10^{-2}$

$= 0.854 \times 10^{-2}$

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e.g: Round off the following numbers to 4 significant digits.

1) 0.624783×10^{6}

$\simeq 0.6248 \times 10^{6}$

2) 0.179231×10^{-7}

$\simeq 0.1792 \times 10^{-7}$

3) 0.854851×10^{20}

$= 0.8549 \times 10^{20}$

4) 0.85485×10^{-5}

$= 0.8548 \times 10^{-5}$.

- END -

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Truncation Errors and Taylor series

Truncation error occurs if we approximate mathematical procedure.

examples:

1) $e^{0.3} = \underbrace{1 + 0.3 + \frac{(0.3)^2}{2!}}_{\text{if approximation of } e^{0.3}} + \underbrace{\frac{(0.3)^3}{3!} + \cdots}_{\text{Truncation error}}$

2) If $f(x) = x^2$

$f'(1) = 2$ ← $f'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$

if $h = 0.1 \Rightarrow f'(1) \simeq \frac{f(1.1) - f(1)}{0.1}$

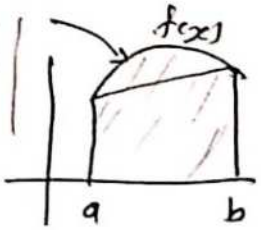
$\simeq \frac{1.21 - 1}{0.1}$

$\simeq \frac{0.21}{0.1}$.

$\simeq 2.1$

Truncation error

3) $\int_a^b f(x)dx \simeq$ area of trapezoid



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Taylor series (Taylor Expansion)

If f is a smooth function ($f^{(n)}$ exists for any n)

$a \in \mathbb{R}$ then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \cdots$$

$$+ \frac{f^{(n)}(a)(x-a)^n}{n!} + R_n$$

where $R_n = \frac{f^{(n+1)}(\xi)(x-a)^{n+1}}{(n+1)!}$ (Lagrange remainder form)

where ξ between a and x.

Now ~~Let~~ if $a = x_i$ and $x = x_{i+1}$

and $x_{i+1} - x_i = h$

we get

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!}$$

$$+ \frac{f^{(4)}(x_i)h^4}{4!} + \cdots + \frac{f^{(n)}(x_i)h^n}{n!} + R_n$$

or

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots + \frac{f^{(n)}(x_i)h^n}{n!} + O(h^{n+1})$$

$R_n = O(h^{n+1})$ ≡ truncation error

↗ order of error

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$$* f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots$$

$f(x_i)$ $\rightarrow P_0(x)$ ← Taylor polynomial of order 0.

$f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots \rightarrow R_0 = O(h)$

$$* f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots$$

$f(x_i) + f'(x_i)h \rightarrow P_1(x)$ ← Taylor polynomial of order 1

$\frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots \rightarrow R_1 = O(h^2)$

$$* f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \dots$$

$f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} \rightarrow P_2(x)$ ← Taylor polynomial of order 2

$\frac{f'''(x_i)h^3}{3!} + \dots \rightarrow R_2 = O(h^3)$

⋮

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e.g.: Use zero-through second order Taylor series expansions to approximate $f(2.1)$ and where $f(x) = 2.5x^2 + 3x + 7$ take $x_i = 2$. Compute ϵ_a % in each step (if possible).

Sol:

$x_i = 2$, $x_{i+1} = 2.1$ want to predict $f(x_{i+1})$

$h = x_{i+1} - x_i = 2.1 - 2$

$\Rightarrow h = 0.1$

$\therefore f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots$

$\Rightarrow f(2.1) = f(2) + f'(2)(0.1) + \frac{f''(2)(0.1)^2}{2!} + \cdots$

1st iteration: (order 0 $\equiv$ 1 term)

$f(2.1) \cong f(2)$

$f(2.1) \cong 23$

$f(x) = 2.5x^2 + 3x + 7$

$f'(x) = 5x + 3$

2nd iteration (order 1 $\equiv$ 2 terms)

$f(2.1) \cong f(2) + f'(2)(0.1)$

$\cong 23 + 13(0.1)$

$\cong 23 + 1.3$

$f(2.1) \cong 24.3$

$\epsilon_a\% = \left| \frac{24.3 - 23}{24.3} \right| * 100\% \cong 5.35\%$

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3rd iteration (order 2 ≡ 3 terms)

$$f(2.1) \simeq \underbrace{f(2) + f'(2)(0.1)} + \frac{f''(2)(0.1)^2}{2!}$$

$f''(x) = 5$

$$\simeq 24.3 + \frac{(5)(0.1)^2}{2}$$

$$\simeq 24.325$$

$$\epsilon_a\% = \left|\frac{24.325 - 24.3}{24.325}\right| * 100\% \simeq 0.1\%$$

$$f(x) = 2.5x^2 + 3x + 7$$

$f(2.1) = 24.325$ i.e $\epsilon_t = 0\%$

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e.g: Using Taylor polynomial of order (2), approximate $\sin(\frac{\pi}{6})$ taking $x_i = \frac{\pi}{4}$.

Then compute ϵ_t %.

$x_{i+1} = \frac{\pi}{6}$, $x_i = \frac{\pi}{4} \Rightarrow h = x_{i+1} - x_i$

$h = \frac{\pi}{6} - \frac{\pi}{4}$

$h = -\frac{\pi}{12}$, $f(x) = \sin x$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots$$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$$\therefore \sin\frac{\pi}{6} \simeq \sin\frac{\pi}{4} + (\cos\frac{\pi}{4})(\frac{-\pi}{12}) + \frac{(-\sin\frac{\pi}{4})(\frac{-\pi}{12})^2}{2!}$$

$$\sin\frac{\pi}{6} \simeq 0.4978544914$$

$$\epsilon_t = \frac{0.5 - 0.4978544914}{0.5} * 100\ \%$$

$$\epsilon_t = 0.45\ \%$$

END

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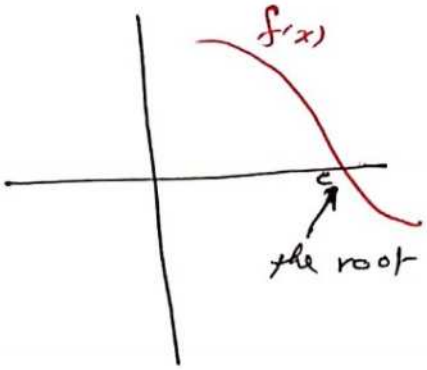
Bracketing Methods

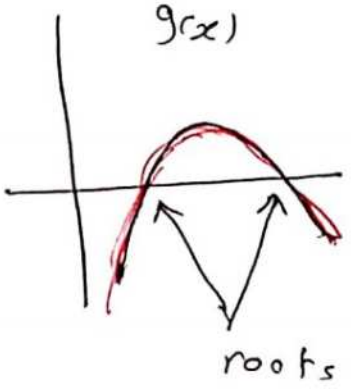
Introduction:

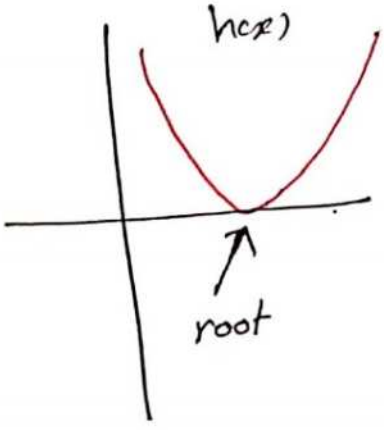
* The equation $f(x)=0$, has a root (C) if

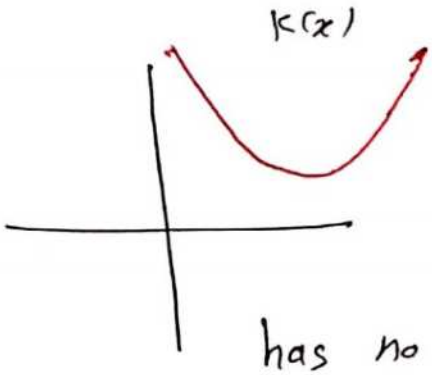
$f(c)=0$.

Graphically the graph of $y=f(x)$ meets the x-axis at c. (c: real root)



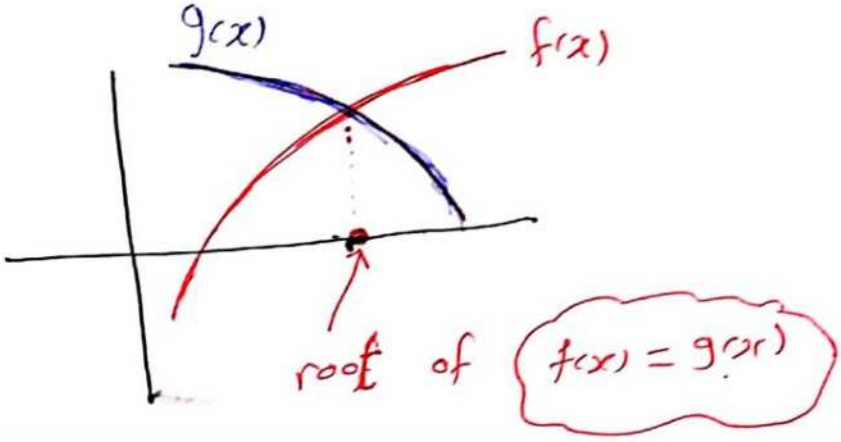




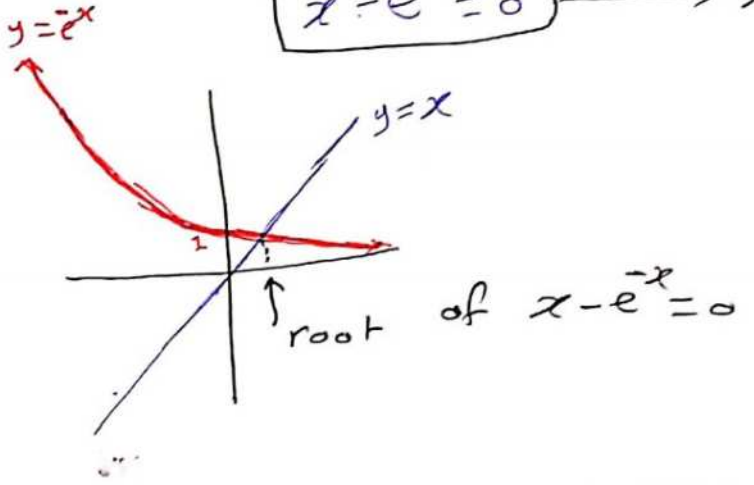


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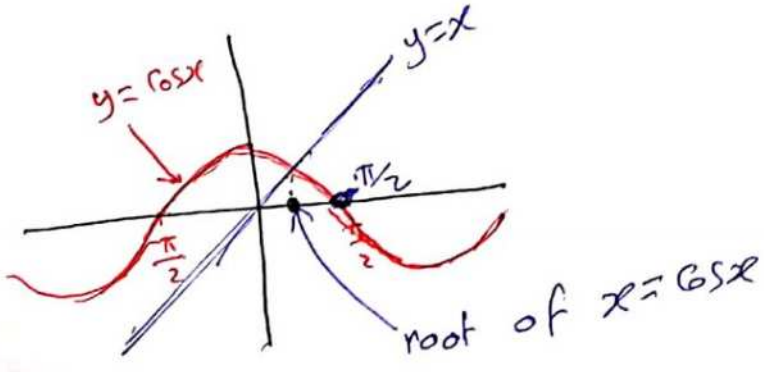
* $f(x) = g(x)$ this equation has root(s) if any at x-coordinates of the point(s) of intersection.



$x - e^{-x} = 0 \longrightarrow x = e^{-x}$



$x - \cos x = 0 \longrightarrow x = \cos x$



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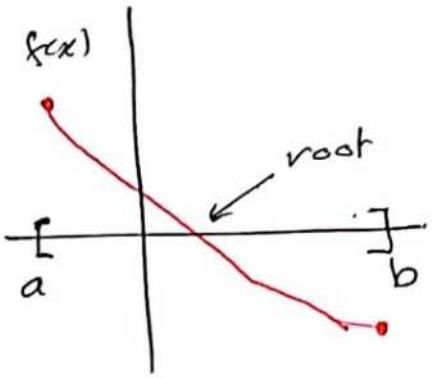
Bolzano Theorem:

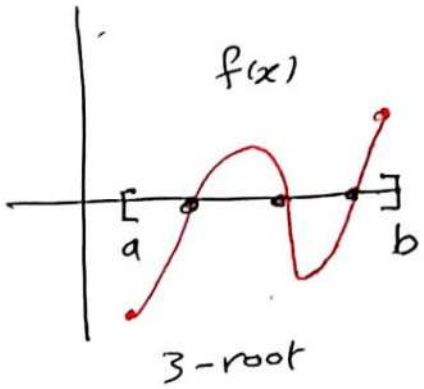
If $f(x)$ is continuous on $[a,b]$, (1)

$$f(a)\,f(b) \le 0 \quad (2)$$

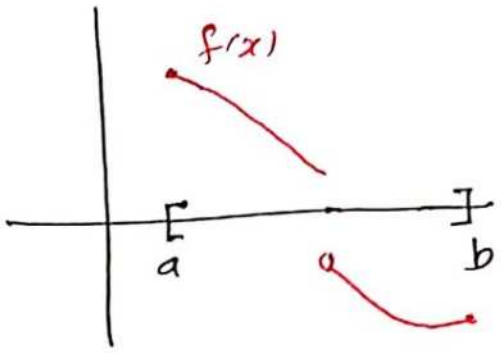
then $\exists\, c \in (a,b)$ such that $f(c)=0$

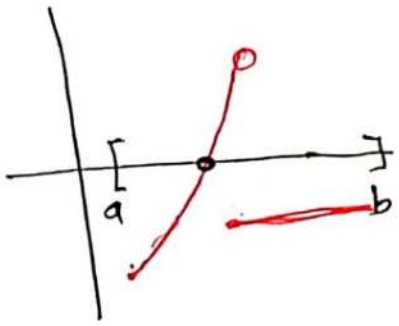
i.e c is a root of $f(x)=0$





3-root





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Bolzano Theorem:

If $f(x)$ is continuous on $[a,b]$,

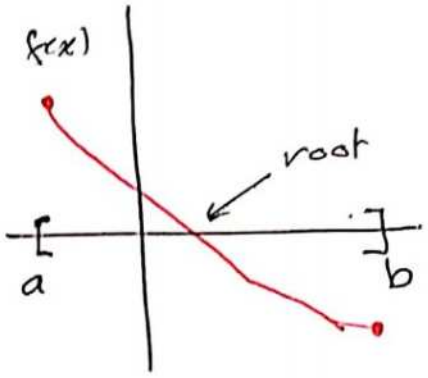
(1)

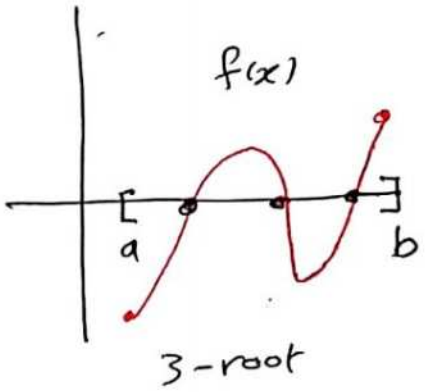
$$f(a)\,f(b) < 0$$

(2)

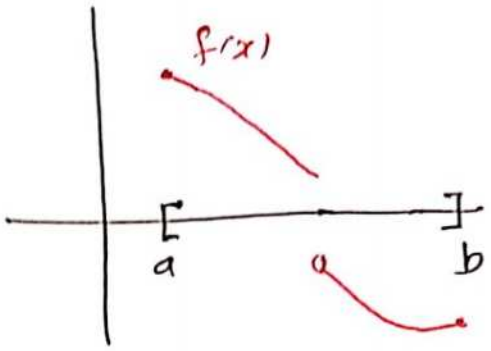
then $\exists\, c \in (a,b)$ such that $f(c)=0$

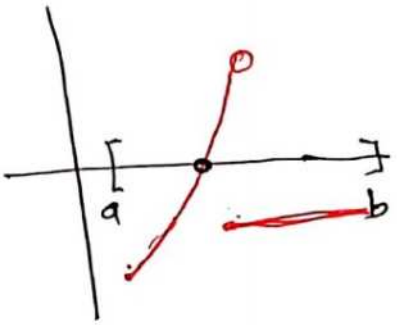
i.e c is a root of $f(x)=0$





3-root





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e.g.: Show that $x^3+x^2-3=0$ has a root in the interval $(1,2)$

Sol:

Let $f(x) = x^3+x^2-3$, $x \in [1,2]$

f is continuous on $[1,2]$

$f(1) = -1$

$f(2) = 9$

$f(1)\,f(2) = -9 < 0$

$\therefore$ By Bolzano theorem

$\exists\ c \in (1,2)$ such that $f(c)=0$

$\Rightarrow c^3+c^2-3=0$, $c \in (1,2)$

i.e there exists a root $c \in (1,2)$ for the equation $x^3+x^2-3=0$

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1) Bisection Method

If $f(x)$ satisfies Bolzano conditions

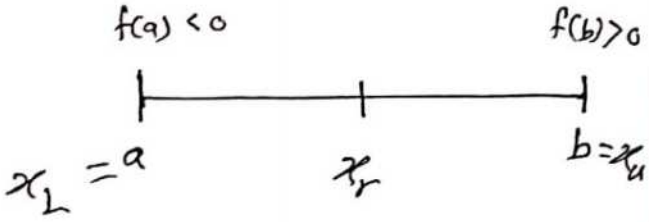
i.e f is continuous on $[a,b]$ and $f(a)f(b)<0$

then f has a root between a and b

Let $x_L = a$ and $x_u = b$

1st iteration:

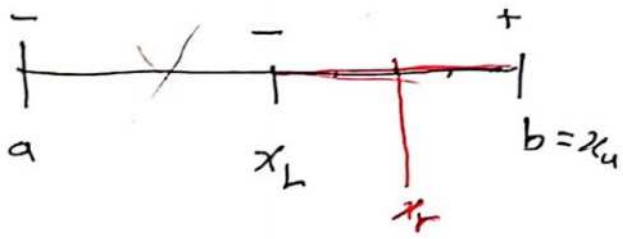
$$x_r = \frac{x_L + x_u}{2}$$



$$|x_r - \text{true root}| < \frac{b-a}{2}$$

assume $f(x_r) < 0$

2nd iteration



$$x_r = \frac{x_L + x_u}{2}$$

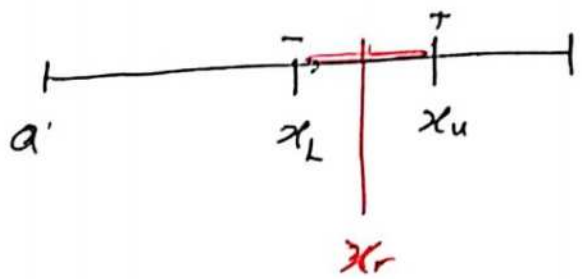
$$|x_r - \text{true root}| < \frac{b-a}{2^2}$$

assume $f(x_r) > 0$

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3rd iteration :

$$x_r = \frac{x_L + x_u}{2}$$



$$|\text{true root} - x_r| < \frac{b-a}{2^3}$$

⋮

n^{th} iteration

$$|\text{true root} - x_r| < \frac{b-a}{2^n}$$

if ϵ ~~the~~ is the Tolerance for true error

and $\frac{b-a}{2^n} \leq \epsilon \Rightarrow |\text{true root} - x_r| \leq \epsilon$

$$\Rightarrow \ln\left(\frac{b-a}{2^n}\right) \leq \ln \epsilon$$

$$\Rightarrow \ln(b-a) - n\ln 2 \leq \ln \epsilon$$

$$n \ln 2 \geq \ln(b-a) - \ln \epsilon \Rightarrow \boxed{n \geq \frac{\ln(b-a) - \ln \epsilon}{\ln 2}}$$

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2] False-Position Method:

If f is continuous on $[a,b]$

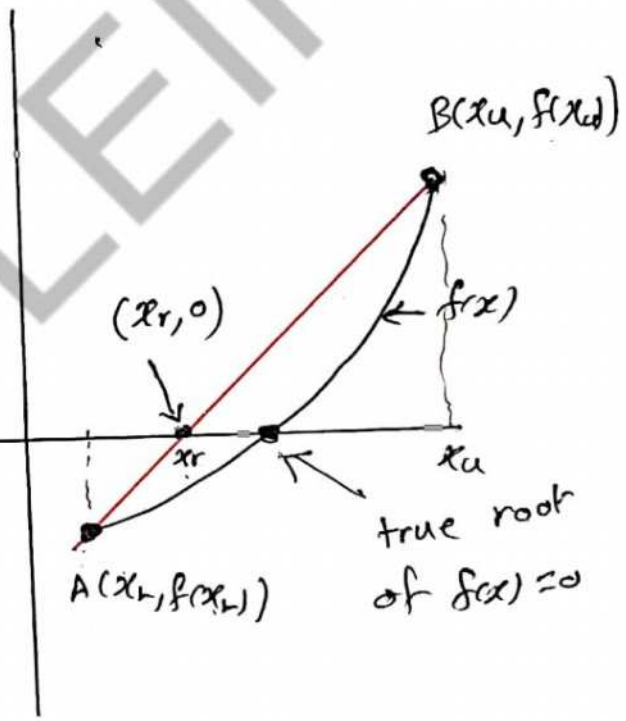
and $f(a)f(b)<0 \Rightarrow \exists c \in (a,b) : f(c)=0$

According to the figure

$f(x_L) < 0$
$f(x_u) > 0$] $f(x_L)f(x_u) < 0$

f is continuous.

to find x_r (the approximation of the true root of $f(x)=0$)



slope AB : $\frac{f(x_L) - f(x_u)}{x_L - x_u}$ ----①

Also slope AB : $\frac{f(x_u) - 0}{x_u - x_r} = \frac{f(x_u)}{x_u - x_r}$ --②

from ① and ②

$$\frac{f(x_L) - f(x_u)}{x_L - x_u} = \frac{f(x_u)}{x_u - x_r}$$

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$$\Rightarrow \quad \frac{x_L - x_u}{f(x_L) - f(x_u)} = \frac{x_u - x_r}{f(x_u)}$$

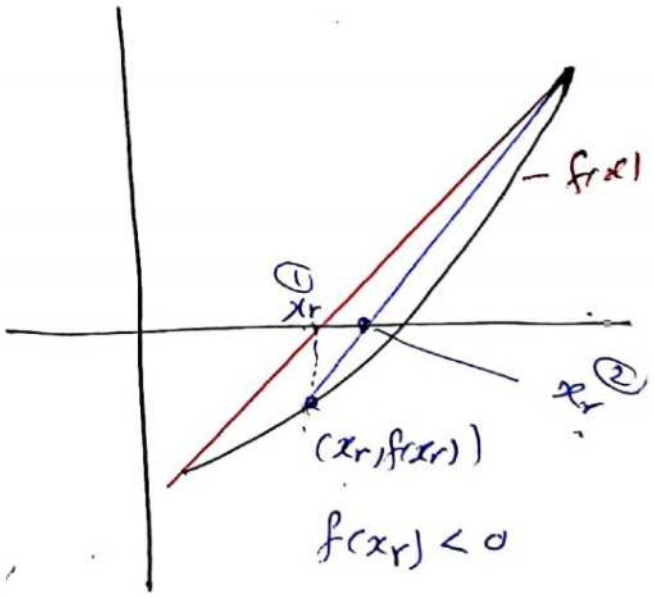
$$\xrightarrow{* f(x_u)} \quad \frac{f(x_u)(x_L - x_u)}{f(x_L) - f(x_u)} = x_u - x_r$$

$$\Rightarrow \quad \boxed{x_r = x_u - \frac{f(x_u)(x_L - x_u)}{f(x_L) - f(x_u)}}$$

2nd iteration

$x_L = x_r \rightarrow f(x_L) < 0$

$x_u = x_u \rightarrow f(x_u) > 0$



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e.g: Show that the equation $x^3 + x^2 - 3 = 0$ has a root in the interval $(1, 1.5)$, then use false position method to approximate the root of this equation by rounding off the answer of second iteration to (4) decimal places.

Sol:

Let $f(x) = x^3 + x^2 - 3$, $x \in [1, 1.5]$

f is continuous on $[1, 1.5]$

$f(1) = -1$

$f(1.5) = 2.625$

$\Big]\ f(1)f(1.5) < 0$

$\Rightarrow \exists\ c \in (1, 1.5)$ such that $f(c) = 0$

$\Rightarrow c^3 + c^2 - 3 = 0$ (i.e has root in $(1, 1.5)$)

1st iteration:

$x_L = 1 \longrightarrow f(x_L) = -1$

$x_u = 1.5 \longrightarrow f(x_u) = 2.625$

$\Big]\ f(x_L)f(x_u) < 0$

$$x_r = x_u - \frac{f(x_u)(x_L - x_u)}{f(x_L) - f(x_u)}$$

$$x_r = 1.5 - \frac{2.625(1 - 1.5)}{-1 - 2.625}$$

$\Rightarrow x_r = 1.137931$

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$$f(x) = x^3 + x^2 - 3$$

$x_r = 1.137931$

$f(x_r) = -0.231621$

2nd iteration:

$x_L = 1.137931 \Longrightarrow f(x_L) = -0.231621$

$x_u = 1.5 \longrightarrow f(x_u) = 2.625$

$f(x_L)\, f(x_u) < 0$

$$x_r = 1.5 - \frac{2.625(1.137931 - 1.5)}{-0.231621 - 2.625}$$

$x_r \cong 1.1673$

True root 1.174559411

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5

e.g: Use false position method, approximate the root of $x + \sin x = 1$ $\quad x \in [0, \frac{\pi}{2}]$ by doing one iteration.

Sol:

$x + \sin x - 1 = 0$

$f(x) = x + \sin x - 1$

f is Cont. on $[0, \frac{\pi}{2}]$

$f(0) = -1$

$f(\frac{\pi}{2}) = 1.5707963$

$\Rightarrow f(0) f(\frac{\pi}{2}) < 0$

$\Rightarrow \exists\, c \in (0, \frac{\pi}{2}) : f(c) = 0$

1st iteration:

$x_L = 0 \longrightarrow f(x_L) = -1$

$x_u = \frac{\pi}{2} \longrightarrow f(x_u) = 1.5707963$

$\Rightarrow f(x_L)\, f(x_u) < 0$

$$x_r = \frac{\pi}{2} - \frac{1.5707963\,(0 - \frac{\pi}{2})}{0 - \frac{\pi}{2}}$$

$x_r = 0.61101559$

True value

0.5109734

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Open Methods

1) Simple fixed point method:

$f(x) = 0$

Rearrange ↓

$x = g(x)$

initial guess given. (x_0)

For example

$x^2 - 2x - 3 = 0$

$x = \sqrt{2x+3}$ ← $x = g_1(x)$

$x = \frac{x^2-3}{2}$ ← $x = g_2(x)$

$x^2 = 2x + 3$

$\div x \rightarrow x = \frac{2}{x} + \frac{3}{x}$ ← $x = g_3(x)$

$x_{i+1} = g(x_i)$ (x_0 given) (initial guess)

$x_0, x_1, x_2, \ldots$ → converges / diverges

this sequence will converge if

$|g'(\text{near the root})| < 1$.

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e.g: To, find a root of $x^2-2x-3=0$

use $x_{i+1} = \sqrt{2x_i+3}$ $\quad x_0 = 0$

$x_1 = \sqrt{2x_0+3}$

$x_2 = \sqrt{2x_1+3}$

$x_3 = \sqrt{2x_2+3}$

$\epsilon_a = \left|\frac{x_i^{New} - x_i^{old}}{x_i^{New}}\right| * 100\%$

$\epsilon_t = \left|\frac{3 - x_i}{3}\right| * 100\%$

i	x_i	ϵ_a %	ϵ_t %
0	0	-	100%
1	1.732051	100%	42.3%
2	2.542460		
3	2.843399		
4	2.947338		
5	2.982394		

true root 3

3

Notice that $g(x) = \sqrt{2x+3}$

and $g'(x) = \frac{2}{2\sqrt{2x+3}}$

$g'(x) = \frac{1}{\sqrt{2x+3}} < 1$

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e.g: To find a root of the equation $x^3 - x + 1 = 0$. by using simple fixed point method and

using $x_{i+1} = x_i^3 + 1$, $x_0 = 1$

find $x_0, x_1, x_2, \ldots$ (is this sequence convergent).

sol:

$$x_{i+1} = x_i^3 + 1$$

$x_0 = 1$, $x_1 = 2$, $x_2 = 9$, $x_3 = \ldots$

$x_4 =$, x_5 $= x_6 =$

diverges

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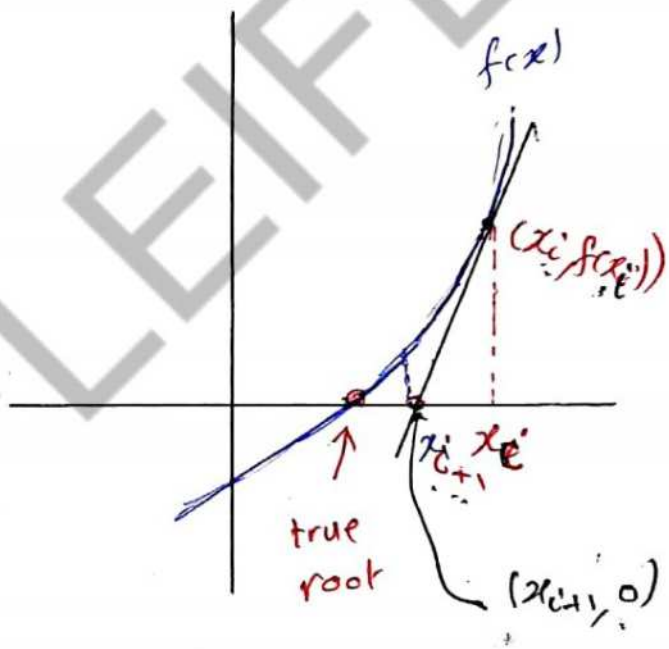
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2 Newton - Raphson Method:

If f is differentiable function

to find the root of $f(x)=0$ (if it exists)

taking x_0 as initial guess



$$\frac{f'(x_i)}{1} = \frac{f(x_i) - 0}{x_i - x_{i+1}}$$

~~$f(x_i) = f'$~~

$$\frac{1}{f'(x_i)} = \frac{x_i - x_{i+1}}{f(x_i)}$$

$f(x_i)$

$$\Longrightarrow \quad \frac{f(x_i)}{f'(x_i)} = x_i - x_{i+1}$$

$$\boxed{x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\vdots$$

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E.g Use Newton - Raphson method, to solve the equation $x = e^{-x}$ take the initial guess $x_0 = 0$, do 4 iterations and compute $\epsilon_a\%$, $\epsilon_t\%$ where it is possible.

Solution

$$f(x) = x - e^{-x} = 0$$

$$f'(x) = 1 + e^{-x}$$

$$x_{i+1} = x_i - \frac{x_i - e^{-x_i}}{1 + e^{-x_i}} \qquad x_0 = 0$$

$$x_0 = 0$$

$$x_1 = 0 - \frac{0 - e^{0}}{1 + e^{0}} = 0.5$$

x_2

x_3

x_4

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i	x_i	ϵ_t %	ϵ_a %
0	0	100 %	—
1	0.5	$\frac{\text{true} - 0.5}{\text{true}} \times 100$	$\frac{0.5 - 0}{0.5} \times 100$
2	0.566311		
3	0.567143		
4	0.567143		

True value :

0.567143290 4

END -

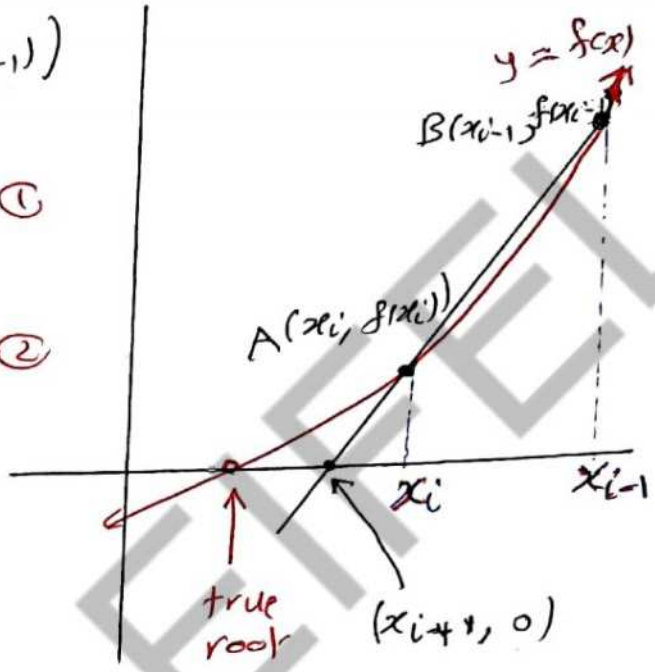
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3) secant - Method

x_{i-1} , x_{i-2} two initial guesses

$A(x_i, f(x_i))$, $B(x_{i-1}, f(x_{i-1}))$



$$\text{slope } AB = \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i} \quad \text{--- (1)}$$

$$\text{Also slope } AB = \frac{f(x_i) - 0}{x_i - x_{i+1}} \quad \text{--- (2)}$$

from (1) and (2)

$$\frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i} = \frac{f(x_i)}{x_i - x_{i+1}}$$

$$\Rightarrow \frac{x_{i-1} - x_i}{f(x_{i-1}) - f(x_i)} = \frac{x_i - x_{i+1}}{f(x_i)}$$

$$\xrightarrow{* f(x_i)} \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} = x_i - x_{i+1}$$

$$\Rightarrow \boxed{x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}} \quad i = 0, 1, 2, 3, \ldots$$

x_{-1}, x_0 are given

x_{-1}, x_0, x_1, x_2, x_3, ---

→ Conv.

→ div.

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e.g: Use secant method to estimate the root of $e^{-x} - x = 0$. Start with initial guesses $x_{-1} = 0$ and $x_0 = 1$, do 3 iterations

Sol:

Let $f(x) = e^{-x} - x$

$x_{i-1} = 0$

$x_i = 1$

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

$$x_{i+1} = x_i - \frac{(e^{-x_i} - x_i)(x_{i-1} - x_i)}{(e^{-x_{i-1}} - x_{i-1}) - (e^{-x_i} - x_i)}$$

$x_{i-1} = x$
$x_i = y$
لاستخدام الآلة

$$x_{i+1} = y - \frac{(e^{-y} - y)(x - y)}{(e^{-x} - x) - (e^{-y} - y)}$$

1st iteration

$x = x_{i-1} = 0$

$y = x_i = 1$

$x_{i+1} = 0.6126998$

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2nd iteration

$x \cong x_{i-1} = 1$

$y = x_i \cong 0.6126998$

$x_{i+1} \cong 0.5638384$

3rd iteration

$x = x_{i-1} = 0.6126998$

$y = x_i = 0.5638384$

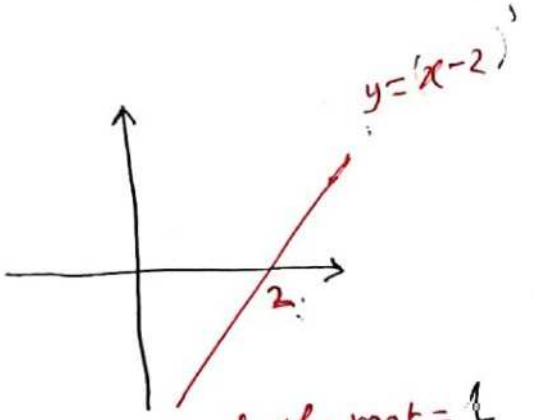
$x_{i+1} \cong 0.5671704$ The answer

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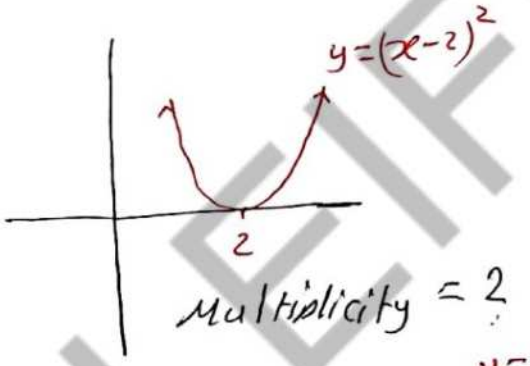
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4] Modified - Newton Raphson Method:

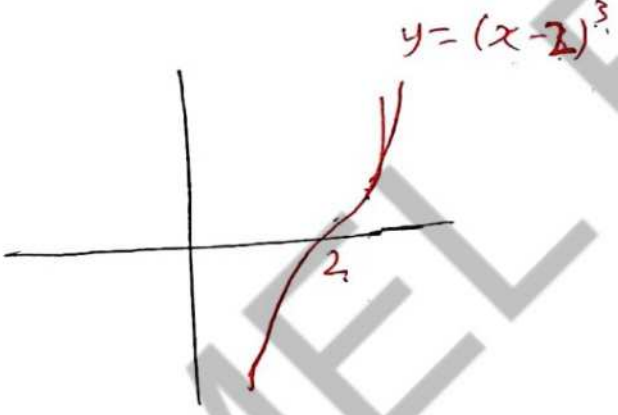
Introduction:



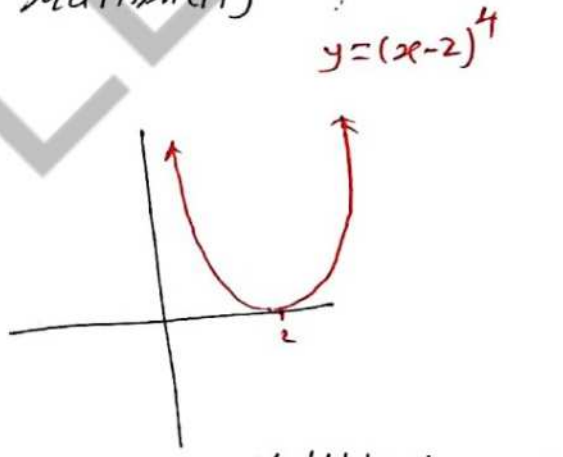
Multiplicity of the root = 1
(simple root)



Multiplicity = 2



Multiplicity = 3



Multiplicity = 4

Def: If $f(x)$ is a polynomial and $f(a) = 0$ then the ~~root~~ multiplicity of the root a is (n) if

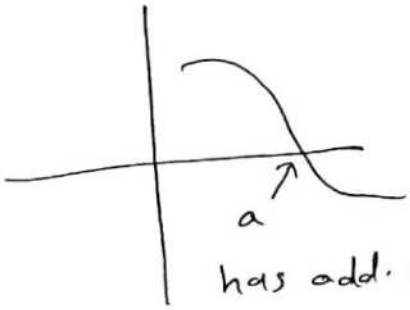
$$f(x) = (x-a)^n g(x) \quad \text{such that } g(a) \neq 0$$

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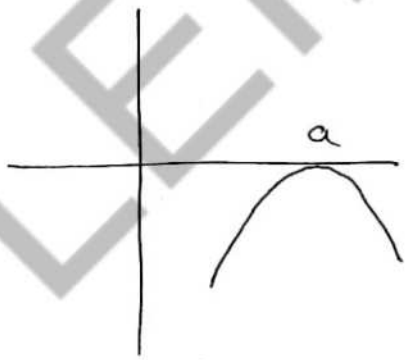
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If a is a root with odd multiplicity then the curve $f(x)$ intersects the x-axis (crosses) at $x = a$.

But if a with even multiplicity then f touches the x-axis at $x = a$.



has odd. multiplicity



has even multiplicity.

e.g.: If $f(x) = (x-a)^2 g(x)$ where $g(a) \neq 0$ (the multiplicity of a is 2) and g is twice differentiable, find $f(a)$, $f'(a)$, $f''(a)$.

Sol: $f(a) = 0$

$$f'(x) = (x-a)^2 g'(x) + 2(x-a) g(x)$$

$$f'(a) = 0$$

$$f''(x) = (x-a)^2 g''(x) + 2(x-a) g'(x) + 2(x-a) g'(x) + 2 g(x)$$

$$f''(a) = 2g(a) \neq 0$$

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In geral If f is a polynomial,

a is root of $f(x) = 0$

then the multiplicity of (a) is (n) iff

$$f(a) = f'(a) = f''(a) = \cdots = f^{(n-1)}(a) = 0 \text{ but}$$

$$f^{(n)}(a) \neq 0 .$$

e.g: Find the roots of $f(x) = (x^2-1)^2(x^2-x)^3$. and determine the multiplicity of each.

Sol:

$$f(x) = \big((x-1)(x+1)\big)^2 \big(x(x-1)\big)^3$$

$$= (x-1)^2(x+1)^2 x^3 (x-1)^3$$

$$= x^3(x+1)^2(x-1)^5$$

The roots are:

0 and its multiplicity = 3

-1 " " " = 2

1 " " = 5

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e.g: Given that 1 is a root of $x^4 - 2x^3 + 2x - 1 = 0$

find its multiplicity.

Sol:

Let $f(x) = x^4 - 2x^3 + 2x - 1$

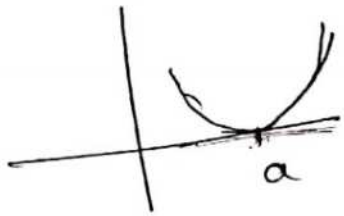
then $f(1) = f'(1) = f''(1) = 0$

but $f'''(1) = 12 \neq 0$ (check that)

So the multiplicity of 1 is (3).

Recal (Newton - Raphson Method)

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$



Modified Newton - Raphson Method

$$x_{i+1} = x_i - \frac{f(x_i)\, f'(x_i)}{(f'(x_i))^2 - f(x_i)\, f''(x_i)}$$

x_0 = initial guess.

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e.g: Using Modified Newton-Raphson method approximate the root of the equation: $x^3 - 4x^2 + 5x - 2 = 0$

take $x_0 = 0.5$ (initial guess).

Do 3 iterations (i.e find x_3) and round off your answer to 6 d.p's.

Sol

Let $f(x) = x^3 - 4x^2 + 5x - 2$

$f'(x) = 3x^2 - 8x + 5$

$f''(x) = 6x - 8$

$$x_{i+1} = x_i - \frac{f'(x_i)\, f(x_i)}{(f'(x_i))^2 - f(x_i) f''(x_i)}$$

$$x_{i+1} = x_i - \frac{(3x_i^2 - 8x_i + 5)(x_i^3 - 4x_i^2 + 5x_i - 2)}{(3x_i^2 - 8x_i + 5)^2 - (x_i^3 - 4x_i^2 + 5x_i - 2)(6x_i - 8)}$$

$x_0 = 0.5$

$x_1 = 1.052631579$

$x_2 = 1.001540832$

$x_3 = 1.000001191$

$x_3 \cong 1.000001$ to 6 d.p's.

- END -

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Muller Method:

Aim: $f(x) = 0$

x_0, x_1, x_2 are guesses want to find x_3

Introduction.

To solve: $ax^2 + bx + c = 0$

we can use $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

or we could use ??

$$x = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

Check: $x^2 - 2x - 3 = 0$

$a = 1, b = -2, c = -3$

$$x = \frac{-2(-3)}{-2 \pm \sqrt{4 - 4(1)(-3)}}$$

$$= \frac{6}{-2 \pm 4}$$

$$x_1 = \frac{6}{-2+4} \quad , x_2 = \frac{6}{-2-4}$$

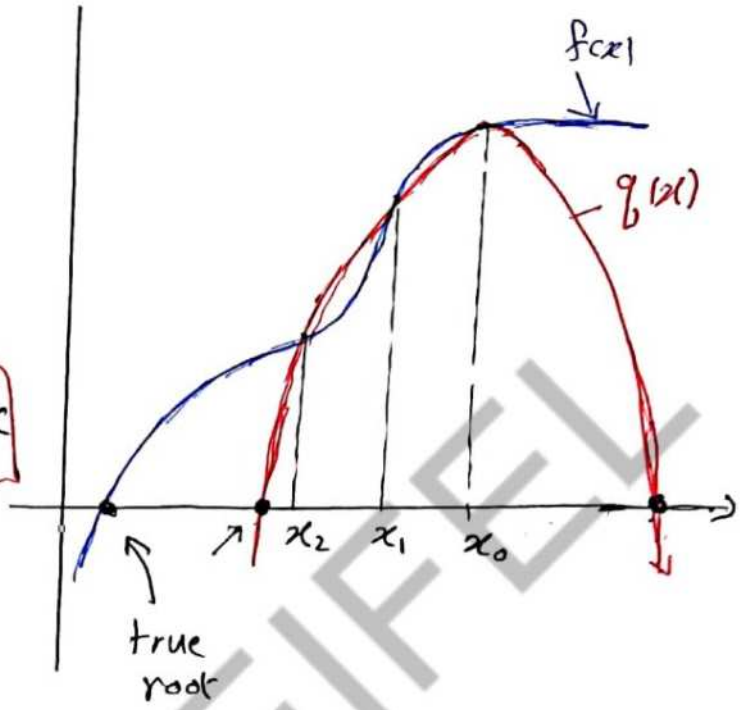
$$x_1 = 3 \quad , x_2 = -1$$

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$f(x) = 0$

x_0, x_1, x_2 are initial guesses



Let $\boxed{q(x) = a(x-x_2)^2 + b(x-x_2) + c}$

such that

$q(x_0) = f(x_0)$

$q(x_1) = f(x_1)$

$q(x_2) = f(x_2)$

Now $q(x_2) = f(x_2) \longrightarrow \boxed{c = f(x_2)}$

$q(x_0) = f(x_0) \longrightarrow \boxed{a(x_0-x_2)^2 + b(x_0-x_2) + c = f(x_0)}$

$q(x_1) = f(x_1) \longrightarrow \boxed{a(x_1-x_2)^2 + b(x_1-x_2) + c = f(x_1)}$

So we can find a, b and c

Now solve $a(x-x_2)^2 + b(x-x_2) + c = 0$

$$\Rightarrow x - x_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$x = x_2 - \frac{2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$\Rightarrow \boxed{x_3 = x_2 - \frac{2c}{b + sgn(b)\sqrt{b^2 - 4ac}}}$$

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Also we can find a, b and c using the following formulas

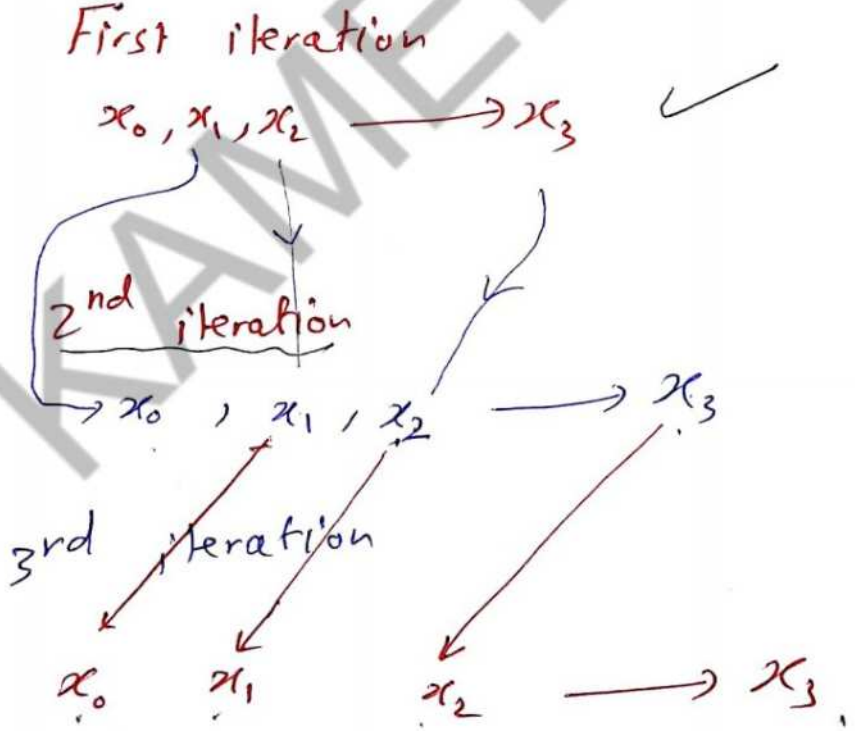
$$h_0 = x_1 - x_0 \qquad h_1 = x_2 - x_1$$

$$\delta_0 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \qquad \delta_1 = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$a = \frac{\delta_1 - \delta_0}{h_1 + h_0}, \quad b = a h_1 + \delta_1, \quad c = f(x_2)$$

then use

$$x_3 = x_2 - \frac{2c}{b + \operatorname{sgn}(b)\sqrt{b^2 - 4ac}}$$



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e.g (From the Text)

Use Muller's method

to solve $x^3 - 13x - 12 = 0$ take $x_0 = 4.5$,

$x_1 = 5.5$, $x_2 = 5$

Do 2 iterations.

Solution

Let $\boxed{f(x) = x^3 - 13x - 12}$.

$x_0 = 4.5 \longrightarrow f(x_0) = 20.625$

$x_1 = 5.5 \longrightarrow f(x_1) = 82.875$

$x_2 = 5 \longrightarrow f(x_2) = 48$

Let $\boxed{q(x) = a(x-5)^2 + b(x-5) + c}$

$f(x_2) = q(x_2) \Longrightarrow c = f(x_2) \longrightarrow \boxed{c = 48}$

$f(x_0) = q(x_0) \Longrightarrow 0.25a + 0.5b + c = 20.625$

$\boxed{0.25a - 0.5b = 27.375}$ — ①

$f(x_1) = q(x_1) \Rightarrow 0.25a + 0.5b + c = 82.875$

$\Rightarrow \boxed{0.25a + 0.5b = 34.875}$ — ②

use calculator or any method else,

find a, b

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$\Rightarrow$ $a = 15$, $b = 62.25$, $c = 48$

$$x_3 = x_2 - \frac{2c}{b + \text{sgn}(b)\sqrt{b^2 - 4ac}}$$

$$x_3 = 5 - \frac{2(48)}{62.25 + \sqrt{62.25^2 - 4(15)(48)}}$$

$$x_3 = 3.976487042$$

$$x_3 \tilde{=} 3.976487$$

2nd iteration

$x_0 = 5.5 \longrightarrow f(x_0) = 82.875$

$x_1 = 5 \longrightarrow f(x_1) = 48$

$x_2 = 3.976487 \Longrightarrow f(x_2) = -0.816334$

$\vdots$

$x_3 \tilde{=} 4.00105$

- END -

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Solving systems of Non-linear Equations:

(by fixed point-iteration approach):

$u(x,y) = 0$
$v(x,y) = 0$
] 2-equations with 2 unknows | initial guesses x_0, y_0

$u(x,y) = 0 \longrightarrow x = g_1(x,y)$

$v(x,y) = 0 \longrightarrow y = g_2(x,y)$

1st iteration: x_0, y_0

$x_1 = g_1(x_0, y_0)$

$y_1 = g_2(x_1, y_0)$

2nd iteration

$x_2 = g_1(x_1, y_1)$

$y_2 = g_2(x_2, y_1)$

3rd iteration

$x_3 = g_1(x_2, y_2)$

$y_3 = g_2(x_3, y_2)$

⋮

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e.g: Solve the non-linear system

$$x^2 + xy = 10$$

$$y + 3xy^2 = 57$$

by using

$$x = \sqrt{10 - xy}$$

$$y = \sqrt{\frac{57 - y}{3x}}$$

take $x_0 = 1.5$ and $y_0 = 3.5$

Do 3 iterations give your answer to 5 d.p's.

$$x = \sqrt{10 - xy} \leftarrow g_1(x,y)$$

$$y = \sqrt{\frac{57 - y}{3x}} \leftarrow g_2(x,y)$$

1st iteration:

$x_0 = 1.5$, $y_0 = 3.5$

$$x_1 = \sqrt{10 - (1.5)(3.5)} = 2.179449$$

$$y_1 = \sqrt{\frac{57 - 3.5}{3(2.179449)}} = 2.860506$$

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2nd iteration

$x_1 = 2.179449$, $y_1 = 2.860506$

$$x_2 = \sqrt{10 - (2.179449)(2.860506)}$$

$$x_2 = 1.940534$$

$$y_2 = \sqrt{\frac{57 - 2.860506}{3(1.940534)}}$$

$$y_2 = 3.049551$$

3rd iteration

$x_2 = 1.940534$, $y_2 = 3.049551$

$$x_3 = \sqrt{10 - (1.940534)(3.049551)}$$

$$x_3 = 2.020456$$

$$y_3 = \sqrt{\frac{57 - 3.049551}{3(2.020456)}}$$

$$y_3 = 2.983405$$

$\therefore x \cong 2.02046$, $y = 2.98340$ to 5 d.p's

→ Given that the true solution $x = 2$, $y = 3$

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e.g: Approximate the solution of simultaneous system of non-linear equations

$$y = x^2 + 1$$
$$y = 2\cos x$$

by using fixed-point method use the equations

$$x = \sqrt{y-1} \quad , y = 2\cos x$$

taking $x_0 = 1$ and $y_0 = 1.5$. Do 2 iterations

Sol:

$$x = \sqrt{y-1}$$
$$y = 2\cos x$$

Note
set up your calculator to radian mod

1st iteration:

$x_0 = 1 \quad , y_0 = 1.5$

$$x_1 = \sqrt{1.5-1} = 0.707167812$$

$$y_1 = 2\cos(0.707167812) = 1.520489194$$

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2nd iteration:

$x_1 = 0.707167812$

$y_1 = 1.520489194$

$x_2 = \sqrt{1.520489194 - 1} = 0.721449704$

$y_2 = 2\cos(0.721449704)$

$y_2 = 1.501698494$

$\therefore x \cong 0.721449704$

$y \cong 1.501698494$

END

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Matrices and determinants:

Notation $A = [A] \longleftrightarrow$ matrix A

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 5 & 0 & 7 \end{bmatrix} = \begin{pmatrix} 2 & -1 & 3 \\ 5 & 0 & 7 \end{pmatrix}$$

A is 2×3 - matrix

- 2×3 → size or dimensions
- 2 → no of rows
- 3 → no of columns

$a_{11} = 2$ $\quad a_{12} = -1$ $\quad a_{13} = 3$

$a_{21} = 5$ $\quad a_{22} = 0$ $\quad a_{23} = 7$

entries (elements) of the matrix A

$$B = \begin{bmatrix} 5 & 8 \\ 4 & -2 \\ 0 & 10 \end{bmatrix} \longrightarrow 3 \times 2 \text{ - matrix}$$

$b_{12} = 8$, $b_{32} = 10$

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Def: If A and B are two matrices then $A = B$ if

1) A and B have the same size

2) $a_{ij} = b_{ij}$ for any i and j.

e.g: Given that

$$\begin{bmatrix} x^2 & 5 \\ 7 & 2Z \end{bmatrix} = \begin{bmatrix} 25 & y \\ 7 & -2 \end{bmatrix}$$

find the values of x, y and Z ?

$x^2 = 25 \longrightarrow x = \pm 5$

$5 = y \longrightarrow y = 5$

$2Z = -2 \longrightarrow Z = -1$

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If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nm} \end{bmatrix}$ is

$n \times m$ - matrix then A could be written as

$$A = [a_{ij}] \qquad 1 \leq i \leq n \;,\; 1 \leq j \leq m$$

and A is called a square matrix if $n = m$, otherwise it is called a rectangular matrix

Notice that

If A is a square matrix

$A = \begin{bmatrix} a_{11} & & & & \\ & a_{22} & & & \\ & & \ddots & & \\ & & & & a_{nn} \end{bmatrix}$

$i < j$ (above the diagonal), $i = j$ (the diagonal), $i > j$ (below the diagonal)

diagonal

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$i < j$ $\quad i = j$ $\quad i > j$

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$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & a_{3n} \\ \vdots & & & \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

→ upper triangular matrix

all the elements are Zeros

upper triangular matrix ≡ square matrix such that all the elements below the diagonal are Zeros

For any $i > j$
$a_{ij} = 0$

---

all elements are Zeros

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ a_{n1} & a_{n2} & \cdots & & a_{nn} \end{bmatrix}$$

→ Lower triangular matrix

Lower triangular matrix ≡ square matrix such that all elements above the diagonal are Zeros.

For any $i < j$
$a_{ij} = 0$

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$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & -1 & 3 \\ 0 & 0 & 7 \end{bmatrix}, \quad \begin{bmatrix} 4 & 8 \\ 0 & -10 \end{bmatrix}, \quad \begin{bmatrix} 2 & 3 & 1 & 8 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

are upper triangular matrices.

$$\begin{bmatrix} 2 & 0 & 0 \\ 5 & -3 & 0 \\ 0 & -1 & 8 \end{bmatrix}, \quad \begin{bmatrix} 5 & 0 \\ -1 & 6 \end{bmatrix}, \quad \begin{bmatrix} 4 & 0 & 0 & 0 \\ -1 & 10 & 0 & 0 \\ -4 & 5 & 8 & 0 \\ 0 & -7 & 3 & 0 \end{bmatrix}$$

are lower triangular matrix

Any matrix (with any size) such that all its elements are Zeros is called a Zero matrix

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

are Zero matrices.

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Identity (unit) matrix (square matrix)

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \ldots$$ are identity (unit) matrices

$I_n = I \cong$ square matrix such that all the elements on the diagonal are 1's, and all the elements outside the diagonal are 0's.

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Def:

If $A = [a_{ij}]$ is $n \times m$-matrix then the transpose of A is $A^T = [a_{ji}]$ ($m \times n$-matrix)

Def: A matrix is called symmetric if $A^T = A$.

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e.g: Find the transpose of each matrix then determine whether the matrix is symmetric.

1) $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & 8 \end{bmatrix}$ 2×3

$\rightarrow A^T = \begin{bmatrix} 2 & 4 \\ -1 & 0 \\ 3 & 8 \end{bmatrix}$ 3×2

2) $B = \begin{bmatrix} 4 & 5 \\ 1 & -2 \end{bmatrix}$ 2×2

$B^T = \begin{bmatrix} 4 & 1 \\ 5 & -2 \end{bmatrix}$ 2×2 $\quad B^T \neq B$

$\rightarrow$ B is not symmetric

3) $C = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ 3×1

$C^T = \begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$ 1×3

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4) $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 7 & -5 \\ 1 & -5 & 0 \end{bmatrix}$ 3×3

$$A^T = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 7 & -5 \\ 1 & -5 & 0 \end{bmatrix}$$

$\Rightarrow A^T = A \rightarrow A$ is symmetric

5) $B = \begin{bmatrix} 4 & -1 \\ -1 & 5 \end{bmatrix}$

$$B^T = \begin{bmatrix} 4 & -1 \\ -1 & 5 \end{bmatrix}$$

$\rightarrow B^T = B \rightarrow B$ is symmetric

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Operation on Matrices:

Def: If $A=[a_{ij}]$, $B=[b_{ij}]$ are two $m\times n$-matrices then

1) $A \mp B = [a_{ij} \mp b_{ij}]$ $m\times n$-matrix

2) $kA = [ka_{ij}]$ $m\times n$-matrix k : scalar $k \in \mathbb{R}$

e.g: Find if possible

1) $\begin{bmatrix} 2 & -1 & 3 \\ 0 & 5 & 8 \end{bmatrix} + \begin{bmatrix} -7 & 5 & 1 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 4 & 4 \\ 3 & 7 & 9 \end{bmatrix}$

2×3 — 2×3

2) $\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ (undefined)

2×1 3×1

3) $[-1 \quad 3] - [1 \quad 5] = [-2 \quad -2]$

1×2 1×2

4) $5\begin{bmatrix} -1 & 2 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} -5 & 10 \\ 5 & -15 \end{bmatrix}$

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5) $$2\begin{bmatrix} -1 & 4 \\ 5 & 3 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 8 \\ 10 & 6 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 8 \\ 10 & 13 \end{bmatrix}$$

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Matrix - Multiplication:

If A is $n \times m$ - matrix,
B is $m \times p$ - matrix then

$$\underset{n\times m}{A}\ \underset{m\times p}{B} = \underset{n\times p}{C}$$ is $n \times p$ - matrix.

where $$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$

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e.g: Find if possible

1) $\begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 0 & -3 \\ 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix}$

$2\times 3 \qquad 3\times 2$ ✓

2×2-matrix

$$= \begin{bmatrix} (2)(5)+(1)(0)+(-1)(4) & (2)(1)+(1)(-3)+(-1)(-3) \\ (3)(5)+(0)(0)+(2)(4) & (3)(1)+(0)(-3)+(2)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 2 \\ 23 & 5 \end{bmatrix}$$

2) $\begin{bmatrix} 5 & 1 \\ 0 & -3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 13 & 5 & -3 \\ -9 & 0 & -6 \\ 11 & 4 & -2 \end{bmatrix}$

$3\times 2 \qquad 2\times 3$ ✓

$AB \neq BA$

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e.g: Given that $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -3 \\ -2 & 0 \end{bmatrix}$

Find

1) $AB = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & -3 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} -6 & 3 \\ -2 & -9 \end{bmatrix}$

$2\times 2 \quad 2\times 2$

2) $BA = \begin{bmatrix} 2 & -3 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$

$= \begin{bmatrix} -11 & -8 \\ 2 & -4 \end{bmatrix}$

3) $A^2 = AA = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$

$= \begin{bmatrix} 7 & 6 \\ 9 & 22 \end{bmatrix}$

4) $A^T + B^T = \begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & -2 \\ -3 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ -1 & 4 \end{bmatrix}$ ✓

5) $(A+B)^T$

$A+B = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & -3 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 4 \end{bmatrix}$

$\Rightarrow (A+B)^T = \begin{bmatrix} 1 & 1 \\ -1 & 4 \end{bmatrix}$ ✓

$(A+B)^T = A^T + B^T$

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6) $(A-B)^T$

7) $A^T - B^T$

8) $A^T . B^T$

9) $B^T A^T$

10) $(AB)^T$

EX

$(AB)^T = B^T A^T$

$(ABC)^T = C^T B^T C^T$

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Answers

6 $\begin{bmatrix} -3 & 5 \\ 5 & 4 \end{bmatrix}$ 7) $\begin{bmatrix} -3 & 5 \\ 5 & 4 \end{bmatrix}$ 8) $\begin{bmatrix} -11 & 2 \\ -8 & -4 \end{bmatrix}$ 9) $\begin{bmatrix} -6 & -2 \\ 3 & -9 \end{bmatrix}$

10) $\begin{bmatrix} -6 & -2 \\ 3 & -9 \end{bmatrix}$

In general $(AB)^T = B^T A^T$

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Determinants: (For square matrices)

Def: (For 2×2-matrix)

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then the determinant of A is denoted by

$$\det(A) = |A| = ad - bc$$

Notes For any square matrix (of any size)

* $\det(A) = 0 \equiv$ A is singular
  $\equiv$ A has no inverse (A^{-1})
  $\equiv$ A is not invertible



* $\det(A) \neq 0 \equiv$ A is not singular.
  $\equiv$ A has inverse (A^{-1})
  $\equiv$ A is invertible



---

* $IA = AI = A$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

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e.g: Find the determinante of the given matrix

1) $A = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix} \rightarrow$.

2) $B = \begin{bmatrix} 2 & 4 \\ 6 & 12 \end{bmatrix}$

3) $C = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

Sol: 1) $\det(A) = (2)(5) - (-1)(3) = 10 + 3 = 13$

2) $\det(B) = (2)(12) - (4)(6)$
$= 24 - 24 = 0$

3) $\det(C) = \cos^2\theta - (-\sin^2\theta)$
$= \cos^2\theta + \sin^2\theta$
$= 1$.

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A is 3x3

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$(-1)^{i+j}$

e.g: If $A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 2 \\ 4 & -2 & 1 \end{bmatrix}$ find determinante of A. (det(A) or |A|).

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 1 & 2 \\ 4 & -2 & 1 \end{vmatrix}$$

$$= +(2)\begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} - (3)\begin{vmatrix} 0 & 2 \\ 4 & 1 \end{vmatrix} + (-1)\begin{vmatrix} 0 & 1 \\ 4 & -2 \end{vmatrix}$$

$$= 2(1+4) - 3(0-8) - 1(0-4)$$

$$= 10 + 24 + 4$$

$$= 38$$

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$$\begin{vmatrix} 2 & 3 & -1 \\ 0 & 1 & 2 \\ 4 & -2 & 1 \end{vmatrix}$$

$$= +2(1+4) - 0(3-2) + 4(6+1)$$

$$= 10 - 0 + 28$$

$$= 38$$

- END -

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Linear System of Equations

Def: The following set of linear equations

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

m - equation
n - variables

is called Linear System of Equations

The above system could be written (using matrices) in two ways

① $$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$m \times n$ $\quad n \times 1$ $\quad m \times 1$

Simply $\boxed{AX = B}$

A: matrix of coefficients
X: matrix of variables (unknowns)
B: matrix of constants

② Augmented matrix $[A : B]$

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$$

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e.g: Write the following linear system of equations, using matrices in the two different ways.

$$2x_1 - x_2 + 3x_3 = 5$$
$$x_1 + x_2 + x_4 = -1$$
$$7x_1 + x_3 + x_4 = 8$$

Sol:

1) $AX = B$

$$\begin{bmatrix} 2 & -1 & 3 & 0 \\ 1 & 1 & 0 & 1 \\ 7 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 8 \end{bmatrix}$$

A $\quad$ X $\quad$ B

3×4 $\quad$ 4×1 $\quad$ 3×1

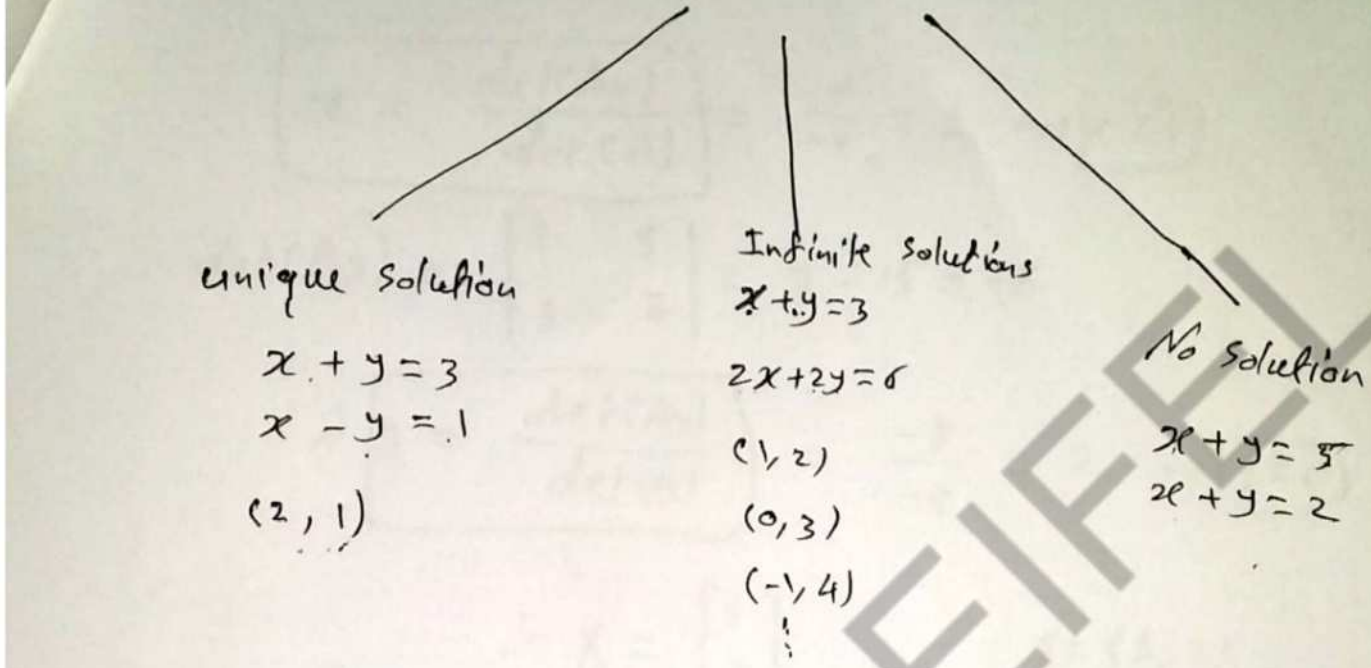
2) $[A \,;\, B] \Rightarrow$

$$\left[\begin{array}{cccc|c} 2 & -1 & 3 & 0 & 5 \\ 1 & 1 & 0 & 1 & -1 \\ 7 & 0 & 1 & 1 & 8 \end{array}\right]$$

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Linear system



Cramer's Rules

$AX = B$

A: square matrix.

$\det(A) \neq 0$

e.g: Use Cramer's rule to solve the linear system?

$$x + 2y = 5$$
$$3x + 2y = 7 .$$

Sol:

$$AX = B \longrightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$A \quad X \quad B.$

$$\det(A) = (1)(2) - (2)(3)$$
$$= 2 - 6 = -4 \neq 0$$

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$$\det(A_x) = \begin{vmatrix} 5 & 2 \\ 7 & 2 \end{vmatrix} = 10 - 14 = -4$$

$$x = \frac{\det(A_x)}{\det(A)} = \frac{-4}{-4} = 1 \rightarrow \boxed{x = 1}$$

$$\det(A_y) = \begin{vmatrix} 1 & 5 \\ 3 & 7 \end{vmatrix} = 7 - 15 = -8$$

$$\therefore y = \frac{\det(A_y)}{\det(A)} = \frac{-8}{-4} = 2 \rightarrow \boxed{y = 2}$$

$$\therefore X = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad X = (1, 2)$$

e.g.: Using Cramer's rule, solve

$$x + y + z = 6$$
$$2x - y + z = 3$$
$$y + z = 5$$

Sol:

$$AX = B \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

A X B

$$\det(A) = +0(1 - (-1)) - (1)(1 - 2) + (1)(-1 - 2)$$
$$= 0 + 1 - 3 = -2 \neq 0$$

$$\det(A) = -2 \neq 0$$

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$$\det(A_x) = \begin{vmatrix} 6 & -1 & 1 \\ 3 & -1 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

5

$$= +(6)(-1-1) - (1)(3-5) + 1(3-(-5))$$

$$= -12 + 2 + 8$$

$$= -2$$

$$\therefore x = \frac{\det(A_x)}{\det(A)} = \frac{-2}{-2} = 1$$

$$x = 1$$

$$\det(A_y) = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 0 & 5 & 1 \end{vmatrix} =$$

$$= +1(3-5) - (2)(6-5) + 0$$

$$= -2 - 2 = -4$$

$$\therefore y = \frac{\det(A_y)}{\det(A)} = \frac{-4}{-2} = 2 \rightarrow y = 2$$

$$\det(A_z) = \begin{vmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 0 & 1 & 5 \end{vmatrix}$$

$$= 0 - 1(3-12) + 5(-1-2)$$

$$= 9 - 15 = -6$$

$$\therefore Z = \frac{\det(A_z)}{\det(A)} = \frac{-6}{-2} = 3$$

$$\therefore X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

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Inverse method:

A is n×n-matrix.

$$\det(A) \neq 0 \longrightarrow \exists A^{-1} \quad (A^{-1}A = I)$$

$$AX = B \longrightarrow A^{-1}AX = A^{-1}B$$

$$\longrightarrow IX = A^{-1}B$$

$$\therefore \boxed{X = A^{-1}B}$$

e.g.: Use the inverse method to solve, the linear system

$$x + 2y = 5$$
$$3x + 2y = 7$$

sol:

$$AX = B \longrightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$
$$A \quad X \quad B$$

$$\boxed{X = A^{-1}B}$$

$$\det(A) = 2 - 6 = -4 \neq 0$$

$$A^{-1} = \frac{1}{-4}\begin{bmatrix} 2 & -2 \\ -3 & 1 \end{bmatrix} \rightarrow A^{-1} = \begin{bmatrix} \frac{-2}{4} & \frac{-2}{-4} \\ \frac{+3}{4} & \frac{-1}{4} \end{bmatrix}$$

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$$\therefore A^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ +\frac{3}{4} & -\frac{1}{4} \end{bmatrix}$$

$$\therefore X = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ +\frac{3}{4} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$2 \times 2 \quad 2 \times 1$

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \rightarrow \begin{matrix} x = 1 \\ y = 2 \end{matrix}$$

Gauss - Jordan method

Def: A matrix A is called in the reduced row echelon form (RREF), if it satisfies the following conditions

1) Zero rows (if any) at the bottom.

2) the first non-zero element in any non-zero row is 1 (called leading 1)

3) For any two non-zero rows, the leading 1 of the lower row to the right of the leading 1 of the upper one.

4) the elements in the column of any leading 1 are zeros (except the leading 1)

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If A satisfies ①, ② and ③ only it is called in row echelon form (REF)

e.g: which of the following in RREF, REF or neither.

1) $\begin{bmatrix} \textcircled{1} & 0 & 1 & 0 & 3 \\ 0 & \textcircled{1} & 0 & 0 & 3 \\ 0 & 0 & 0 & \textcircled{1} & 5 \end{bmatrix}$ RREF.

2) $\begin{bmatrix} \textcircled{1} & 2 & 1 & 4 \\ 0 & \textcircled{1} & -1 & 2 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$ REF but not in RREF

3) $\begin{bmatrix} \textcircled{1} & 0 & 0 & 1 \\ \textcircled{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ Not REF
Not in RREF

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Elementary row operations

1. Multiplying any row by non-zero constant
2. Multiplying a row by a constant and adding the results to another row.
3. interchanging any two rows.

e.g: By Gauss-Jordan method, solve the linear system

$$x + y + z = 6$$
$$2x - y + z = 3$$
$$y + z = 5$$

Sol:

$$[A|B] \xrightarrow[\text{row operations}]{\text{elementary}} \text{RREF}$$

$[A|B]$

$$\begin{bmatrix} \textcircled{1} & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 0 & 1 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} \textcircled{1} & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & 1 & 5 \end{bmatrix} -2R_1 + R_2$$

$$\sim \begin{bmatrix} \textcircled{1} & 1 & 1 & 6 \\ 0 & \textcircled{1} & 1 & 5 \\ 0 & -3 & -1 & -9 \end{bmatrix} \begin{matrix} \\ R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & \textcircled{2} & 6 \end{bmatrix} \begin{matrix} -R_2 + R_1 \\ \\ 3R_2 + R_3 \end{matrix}$$

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$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & \textcircled{1} & 3 \end{bmatrix} \frac{1}{2}R_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix} -R_3 + R_2$$

$$\left.\begin{array}{l} x=1 \\ y=2 \\ z=3 \end{array}\right\} \longmapsto X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Part 2

e.g: Solve, using Gauss-Jordan.

$$x + y + z = 6.$$
$$2x + y + z = 7$$
$$3x + 2y + 2z = 5$$

Sol:

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 7 \\ 3 & 2 & 2 & 5 \end{bmatrix}$$

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e.g: Solve, by Gauss-Jordan method

$$x + y + z = 6$$
$$2x + y + z = 7$$
$$3x + 2y + 2z = 5$$

sol:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 7 \\ 3 & 2 & 2 & 5 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -5 \\ 0 & -1 & -1 & -13 \end{array}\right] \begin{array}{l} \\ -2R_1 + R_2 \\ -3R_1 + R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & -1 & -1 & -13 \end{array}\right] -R_2 \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & -8 \end{array}\right] \begin{array}{l} -R_2 + R_1 \\ \\ 1R_2 + R_3 \end{array}$$

$$0x + 0y + 0z = -8$$
$$0 = -8 \quad \text{contradiction}$$

$\therefore$ The system has no solution

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e:9 : Using Gauss-Jordan, solve the system

$$2x_1 + x_2 + x_3 - 2x_4 = 4$$
$$x_1 + 2x_2 + 2x_3 - 4x_4 = 5$$
$$3x_1 + 3x_2 + 3x_3 - 6x_4 = 9$$

sol:

$$\left[\begin{array}{cccc|c} 2 & 1 & 1 & -2 & 4 \\ 1 & 2 & 2 & -4 & 5 \\ 3 & 3 & 3 & -6 & 9 \end{array}\right]$$

$$\sim \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 2 & -4 & 5 \\ 2 & 1 & 1 & -2 & 4 \\ 3 & 3 & 3 & -6 & 9 \end{array}\right] \begin{matrix} R_1 \\ R_2 \\ \\ \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & -4 & 5 \\ 0 & \textcircled{-3} & -3 & 6 & -6 \\ 0 & -3 & -3 & 6 & -6 \end{array}\right] \begin{matrix} \\ -2R_1 + R_2 \\ -3R_1 + R_3 \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & -4 & 5 \\ 0 & \textcircled{1} & 1 & -2 & 2 \\ 0 & -3 & -3 & 6 & -6 \end{array}\right] \begin{matrix} \\ -\frac{1}{3}R_2 \\ \\ \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{matrix} -2R_2 + R_1 \\ \\ 3R_2 + R_3 \end{matrix}$$

→ Infinitely many solutions

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$x_1 = 1$

$x_2 + x_3 - 2x_4 = 2$

Let $x_4 = r$, $r \in R$

$x_3 = s$ $s \in R$

$x_2 + s - 2r = 2$

$\rightarrow x_2 = 2 + 2r - s$

$$\therefore X = \begin{bmatrix} 1 \\ 2+2r-s \\ s \\ r \end{bmatrix} \qquad r, s \in R$$

$$\left.\begin{matrix} r = 0 \\ s = 1 \end{matrix}\right\} \longrightarrow X = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\left.\begin{matrix} r = 1 \\ s = 1 \end{matrix}\right] \longrightarrow X = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} \begin{matrix} \rightarrow x_1 \\ \rightarrow x_2 \\ \rightarrow x_3 \\ \rightarrow x_4 \end{matrix}$$

$\vdots$

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Gauss- Elimination Method

$$AX = B$$

$$[A : B] \longrightarrow REF$$

e.g: Using Gauss elimination method, solve

$$x + y + z = 4$$
$$3x + y + 2z = 8$$
$$x - y + 5z = 10$$

sol:

$$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & 1 & 4 \\ 3 & 1 & 2 & 8 \\ 1 & -1 & 5 & 10 \end{array}\right] \quad \text{Pivot}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & \textcircled{-2} & -1 & -4 \\ 0 & -2 & 4 & 6 \end{array}\right] \begin{matrix} \\ -3R_1 + R_2 \\ -R_1 + R_3 \end{matrix}$$

$$\frac{-(-2)}{-2} = -1$$

(Pivot)

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & -1 & -4 \\ 0 & 0 & 5 & 10 \end{array}\right] \begin{matrix} \\ \\ -1R_2 + R_3 \end{matrix}$$

$$5z = 10 \rightarrow \boxed{z = 2}$$

$$-2y - z = -4 \rightarrow -2y - 2 = -4 \rightarrow -2y = -2 \rightarrow \boxed{y = 1}$$

$$x + y + z = 4$$
$$x + 1 + 2 = 4 \rightarrow x = 1$$

$$\therefore X = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

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e.g: Find all the value(s) of (if any) for which, the system $AX = B$

where $[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & a^2-4 & a-2 \end{array}\right]$

has, a) infinite solutions b) Non solution

c) unique solution.

Sol:

$a^2 - 4 = 0 \longrightarrow a = \pm 2$

$a - 2 = 0 \longrightarrow a = 2$

a) $a = 2$

b) $a = -2$

c) $a \in \mathbb{R} \setminus \{-2, 2\}$

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e.g: Solve, by Gauss-Jordan method

$$x + y + z = 6$$
$$2x + y + z = 7$$
$$3x + 2y + 2z = 5$$

sol:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 1 & 7 \\ 3 & 2 & 2 & 5 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & -1 & -5 \\ 0 & -1 & -1 & -13 \end{array}\right] \begin{array}{l} \\ -2R_1 + R_2 \\ -3R_1 + R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & -1 & -1 & -13 \end{array}\right] -R_2 \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & -8 \end{array}\right] \begin{array}{l} -R_2 + R_1 \\ \\ 1R_2 + R_3 \end{array}$$

$$0x + 0y + 0z = -8$$
$$0 = -8 \quad \text{contradiction}$$

$\therefore$ The system has no solution

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e:9 : Using Gauss-Jordan, solve the system

$$2x_1 + x_2 + x_3 - 2x_4 = 4$$
$$x_1 + 2x_2 + 2x_3 - 4x_4 = 5$$
$$3x_1 + 3x_2 + 3x_3 - 6x_4 = 9$$

sol:

$$\left[\begin{array}{cccc|c} 2 & 1 & 1 & -2 & 4 \\ 1 & 2 & 2 & -4 & 5 \\ 3 & 3 & 3 & -6 & 9 \end{array}\right]$$

$$\sim \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 2 & -4 & 5 \\ 2 & 1 & 1 & -2 & 4 \\ 3 & 3 & 3 & -6 & 9 \end{array}\right] \begin{matrix} R_1 \\ R_2 \\ \\ \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & -4 & 5 \\ 0 & \textcircled{-3} & -3 & 6 & -6 \\ 0 & -3 & -3 & 6 & -6 \end{array}\right] \begin{matrix} \\ -2R_1 + R_2 \\ -3R_1 + R_3 \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & 2 & -4 & 5 \\ 0 & \textcircled{1} & 1 & -2 & 2 \\ 0 & -3 & -3 & 6 & -6 \end{array}\right] \begin{matrix} \\ -\frac{1}{3}R_2 \\ \\ \end{matrix}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{matrix} -2R_2 + R_1 \\ \\ 3R_2 + R_3 \end{matrix}$$

→ Infinitely many solutions

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$x_1 = 1$

$x_2 + x_3 - 2x_4 = 2$

Let $x_4 = r$, $r \in R$

$x_3 = s$ $s \in R$

$x_2 + s - 2r = 2$

$\rightarrow x_2 = 2 + 2r - s$

$$\therefore X = \begin{bmatrix} 1 \\ 2+2r-s \\ s \\ r \end{bmatrix} \qquad r, s \in R$$

$$\left.\begin{matrix} r = 0 \\ s = 1 \end{matrix}\right\} \longrightarrow X = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\left.\begin{matrix} r = 1 \\ s = 1 \end{matrix}\right] \longrightarrow X = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix} \begin{matrix} \rightarrow x_1 \\ \rightarrow x_2 \\ \rightarrow x_3 \\ \rightarrow x_4 \end{matrix}$$

$\vdots$

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Gauss- Elimination Method

$$AX = B$$

$$[A : B] \longrightarrow REF$$

e.g: Using Gauss elimination method, solve

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$$x - y + 5z = 10$$

sol:

$$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & 1 & 4 \\ 3 & 1 & 2 & 8 \\ 1 & -1 & 5 & 10 \end{array}\right] \quad \text{Pivot}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & \textcircled{-2} & -1 & -4 \\ 0 & -2 & 4 & 6 \end{array}\right] \begin{matrix} \\ -3R_1 + R_2 \\ -R_1 + R_3 \end{matrix}$$

$$\frac{-(-2)}{-2} = -1$$

(Pivot)

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & -1 & -4 \\ 0 & 0 & 5 & 10 \end{array}\right] \begin{matrix} \\ \\ -1R_2 + R_3 \end{matrix}$$

$$5z = 10 \rightarrow \boxed{z = 2}$$

$$-2y - z = -4 \rightarrow -2y - 2 = -4 \rightarrow -2y = -2 \rightarrow \boxed{y = 1}$$

$$x + y + z = 4$$
$$x + 1 + 2 = 4 \rightarrow x = 1$$

$$\therefore X = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

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e.g: Find all the value(s) of (if any) for which, the system $AX = B$

where $[A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & a^2-4 & a-2 \end{array}\right]$

has, a) infinite solutions b) Non solution

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Sol:

$a^2 - 4 = 0 \longrightarrow a = \pm 2$

$a - 2 = 0 \longrightarrow a = 2$

a) $a = 2$

b) $a = -2$

c) $a \in \mathbb{R} \setminus \{-2, 2\}$

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Finding the inverse of a matrix by elementary row operations

Recall

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ the multiplicative identity (unit) matrix for 2x2-matrices

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ — — — — — — — for 3x3-matrices

$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ — — — — — — — for 4x4-matrices

⋮

To find A^{-1} by elementary row operations

$[A \mid I]$

↓ RREF

$[I \mid B]$

A has inverse (namely $A^{-1} = B$)

(invertable)

(non-singular)

$\left[\begin{array}{c|c} 0\,0\,\cdots\,0 & \cdots \end{array}\right]$

A has no inverse

(not invertable)

(singular)

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Notice that:

If A has inverse, then

$$A^{-1}A = I = AA^{-1}$$

e.g: Find the inverse of the given matrix (if it exists) by using elementary row operations

1) $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right] -3R_1 + R_2$$

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right] -R_2$$

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array}\right] -2R_2 + R_1$$

$$\therefore A^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$$

Check: $AA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓

$2\times 2 \quad 2\times 2$

$A^{-1}A = I$ Ex.

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2) $B = \begin{bmatrix} 1 & 3 \\ 4 & 12 \end{bmatrix}$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 4 & 12 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 0 & -4 & 1 \end{array}\right] -4R_1 + R_2$$

$\therefore$ B has no inverse (not invertable)
(singular)

e.g: Given that $A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix}$,

1) find the inverse of A by elementary row operations

2) and hence, solve the system

$$x + y - z = 4$$
$$y - 2z = -2$$
$$2x + z = 10$$

by using inverse method

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4

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & -2 & 3 & -2 & 0 & 1 \end{array}\right] -2R_1 + R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & -1 & -2 & 2 & 1 \end{array}\right] \begin{array}{l} -R_2 + R_1 \\ \\ 2R_2 + R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & -2 & -1 \end{array}\right] -R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 1 \\ 0 & 1 & 0 & 4 & -3 & -2 \\ 0 & 0 & 1 & 2 & -2 & -1 \end{array}\right] \begin{array}{l} -R_3 + R_1 \\ 2R_3 + R_2 \end{array}$$

$$\therefore A^{-1} = \left[\begin{array}{ccc} -1 & 1 & 1 \\ 4 & -3 & -2 \\ 2 & -2 & -1 \end{array}\right]$$

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2) $\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 10 \end{bmatrix}$

$A \quad X \quad B$

$AX = B \rightarrow X = A^{-1}B$

$\therefore X = \begin{bmatrix} -1 & 1 & 1 \\ 4 & -3 & -2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 10 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$

$3\times3 \quad 3\times1$

$\rightarrow x = 4, \; y = 2, \; z = 2$

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* Solving Linear systems by LU-decomposition method

$$AX = B \longrightarrow A = LU$$

L: lower triangular matrix.

U: upper triangular matrix.

$$L(UX) = B$$

$$LY = B \quad \text{—} (1)$$

$$UX = Y \quad \text{—} (2)$$

* Doolittle's method

A 2×2 $\longrightarrow$

$$\begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} = A$$

L $\quad$ U

A; 3×3 - matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = A$$

L $\quad$ U

$\vdots$

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e.g: Solve the following linear system using LU-decomposition method.

$$4x_1 + 5x_2 = 7$$
$$12x_1 + 14x_2 = 18$$

Sol:

$$\begin{bmatrix} 4 & 5 \\ 12 & 14 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

A X B

$AX = B$

$$\begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 12 & 14 \end{bmatrix}$$

2x2 2x2

$u_{11} = 4$ $u_{12} = 5$

$l_{21}u_{11} = 12 \rightarrow 4l_{21} = 12 \rightarrow l_{21} = 3$

$l_{21}u_{12} + u_{22} = 14$

$(3)(5) + u_{22} = 14 \rightarrow u_{22} = -1$

check:

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 4 & 5 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 12 & 14 \end{bmatrix}$$

L U A

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$LY = B$ — ①

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 18 \end{bmatrix}$$

$L \quad Y \quad B$

$y_1 = 7$

$3y_1 + y_2 = 18 \longrightarrow 3(7) + y_2 = 18$

$\rightarrow y_2 = -3$

$\therefore Y = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$

$UX = Y$ — ②

$$\begin{bmatrix} 4 & 5 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$$

$-x_2 = -3 \rightarrow x_2 = 3$

$4x_1 + 5x_2 = 7$

$4x_1 + 5(3) = 7$

$4x_1 + 15 = 7 \longrightarrow 4x_1 = -8 \rightarrow x_1 = -2$

Sol. $X = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

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e.g: Using LU-decomposition method

to solve, the linear system

$$3x_1 + 5x_2 + 2x_3 = 8$$
$$8x_2 + 2x_3 = -7$$
$$6x_1 + 2x_2 + 8x_3 = 26$$

sol:

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ 26 \end{bmatrix}$$

A X B

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{bmatrix}$$

L U A

$u_{11} = 3$ $u_{12} = 5$ $u_{13} = 2$

$l_{21} u_{11} = 0 \rightarrow 3 l_{21} = 0 \rightarrow l_{21} = 0$

$l_{12} u_{12} + u_{22} = 8$

$(0)(5) + u_{22} = 8 \rightarrow u_{22} = 8$

$l_{21} u_{13} + u_{23} = 2 \rightarrow (0)(2) + u_{23} = 2$

$\rightarrow u_{23} = 2$

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$l_{31} u_{11} = 6 \rightarrow 3 l_{31} = 6 \rightarrow l_{31} = 2$ (10)

$l_{31} u_{12} + l_{32} u_{22} = 2$

$(2)(5) + 8 l_{32} = 2 \rightarrow 8 l_{32} = -8$

$\rightarrow l_{32} = -1$

$l_{31} u_{13} + l_{32} u_{23} + u_{33} = 8$

$(2)(2) + (-1)(2) + u_{33} = 8$

$4 - 2 + u_{33} = 8 \rightarrow u_{33} = 6$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 0 & 0 & 6 \end{bmatrix} = A$$

$L \qquad U$

check!!

$LY = B \rightarrow$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ 26 \end{bmatrix}$$

$L \qquad Y \qquad B$

$y_1 = 8$

$y_2 = -7$

$2y_1 - y_2 + y_3 = 26$

$2(8) - (-7) + y_3 = 26$

$16 + 7 + y_3 = 26 \rightarrow y_3 = 3$

$$\therefore Y = \begin{bmatrix} 8 \\ -7 \\ 3 \end{bmatrix}$$

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$UX = Y$ (2)

$$\begin{bmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -7 \\ 3 \end{bmatrix}$$

$U \qquad X$

$6x_3 = 3 \rightarrow \boxed{x_3 = \frac{1}{2}}$

$8x_2 + 2x_3 = -7$

$8x_2 + 2(\frac{1}{2}) = -7 \rightarrow 8x_2 = -8$

$\rightarrow \boxed{x_2 = -1}$

$3x_1 + 5x_2 + 2x_3 = 8$

$3x_1 + 5(-1) + 2(\frac{1}{2}) = 8$

$3x_1 - 5 + 1 = 8$

$3x_1 = 12 \rightarrow \boxed{x_1 = 4}$

$\therefore \quad X = \begin{bmatrix} 4 \\ -1 \\ \frac{1}{2} \end{bmatrix}$

END .

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Finding the inverse of a matrix by using LU-decomposition method:

Recall

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity for 2x2 - matrices

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the identity for 3x3 - matrix.

$AX = I$ $\qquad X = A^{-1}$

$L(UX) = I$ ($UX = Y$)

$LY = I$

$UX = Y$

For 2x2 - matrix

First Column of A^{-1}

$LY = e_1$ — ①

$UX = Y$ — ②

$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} x & \\ & \end{bmatrix}$

2nd column of A^{-1}

$LY = e_2$ — ①

$UX = Y$ — ②

$A^{-1} = \begin{bmatrix} x & x \\ & \end{bmatrix}$

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A is

* If 3x3 - matrix

e_1 e_2 e_3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1st column

$LY = e_1$ — (1)

$UX = Y$ — (2)

$$A^{-1} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$ 3x3 - matrix

2nd column

$LY = e_2$

$UX = Y$

3rd column

$LY = e_3$

$UX = Y$

---

e.g: Using LU-decomposition, find A^{-1} ?

1) $A = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 \\ l_{21} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$$

$\rightarrow u_{11} = 1$ $u_{12} = 3$

$l_{21} u_{11} = 1 \rightarrow l_{21} = 1$

$l_{21} u_{12} + u_{22} = 2 \rightarrow (1)(3) + u_{22} = 2$

$\rightarrow u_{22} = -1$

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$$\therefore \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$$

$L \quad U \quad A$

$e_1 \quad e_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

First column of the inverse

$$LY = e_1 \quad -①$$

$$UX = Y \quad -②$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$y_1 = 1$

$y_1 + y_2 = 0 \rightarrow y_2 = -1$

$$\therefore Y = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$UX = Y \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$-x_2 = -1 \rightarrow x_2 = 1$

$x_1 + 3x_2 = 1$

$x_1 + 3(1) = 1 \rightarrow x_1 = -2$

$$\therefore X = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

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2nd ~~row~~ Column

$$A^{-1} = \begin{bmatrix} -2 & \\ 1 & 0 \end{bmatrix}$$

$$LY = e_2$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$y_1 = 0$

$y_1 + y_2 = 1$

$\rightarrow$ $0 + y_2 = 1$

$\therefore y_2 = 1$

$\therefore Y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$UX = Y \rightarrow \begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$-x_2 = 1 \rightarrow x_2 = -1$

$x_1 + 3x_2 = 0$

$x_1 + 3(-1) = 0 \rightarrow x_1 = 3$

$\therefore X = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

$$\therefore A^{-1} = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$$

check that $AA^{-1} = I = A^{-1}A$

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2) $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 0 & 2 & 0 \end{bmatrix}$ $R_3 - R_1$

$LU = A$

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \quad R_3 - R_1$$

$L \qquad U \qquad A$

$\therefore$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{1}{3} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

1st column of A^{-1}

$e_1 \quad e_2 \quad e_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$LY = e_1 \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$A^{-1} = \begin{bmatrix} \quad \end{bmatrix}$

$y_1 = 1$

$2y_1 + y_2 = 0 \rightarrow y_2 = -2$

$-\frac{1}{3} y_2 + y_3 = 0 \rightarrow \frac{2}{3} + y_3 = 0 \rightarrow y_3 = \frac{-2}{3}$

$$\therefore Y = \begin{bmatrix} 1 \\ -2 \\ \frac{-2}{3} \end{bmatrix}$$

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$UX = Y$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ \frac{-2}{3} \end{bmatrix}$$

$\frac{1}{3}x_3 = \frac{-2}{3} \rightarrow (x_3 = -2)$

$-3x_2 + x_3 = -2$

$-3x_2 - 2 = -2 \rightarrow -3x_2 = 0$

$\rightarrow (x_2 = 0)$

$x_1 + 2x_2 = 1$

$x_1 + 2(0) = 1 \rightarrow (x_1 = 1)$

$\therefore X = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$

2nd Column ~~row~~ of A^{-1}

$$A^{-1} = \begin{bmatrix} 1 & (0) & \\ 0 & & \\ -2 & & \end{bmatrix}$$

$LY = e_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow Y = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{3} \end{bmatrix}$$

$$UX = Y \longrightarrow X = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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3rd Column ~~row~~ of the inverse

$$LY = e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & \\ 0 & 0 & \\ -2 & 1 & \end{bmatrix}$$

$$\longrightarrow \quad Y = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$UX = Y \longrightarrow X = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 1 \\ -2 & 1 & 3 \end{bmatrix}$$

Check! $AA^{-1} = I = A^{-1}A$

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LU-decomposition by Gaussian elimination method.

e.g: Find the LU-decomposition of the given matrix, by using Gaussian elimination method

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 4 & 1 & 9 \\ -2 & 10 & -3 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 0 & 9 & 1 \end{bmatrix} \begin{matrix} -2R_1 + R_2 \\ 1R_1 + R_3 \end{matrix} \qquad \begin{matrix} \frac{-4}{2} R_1 + R_2 \\ \frac{2}{2} R_1 + R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{bmatrix} -3R_2 + R_3 \qquad \frac{-9}{3} R_2 + R_3$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

check

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 4 \\ 4 & 1 & 9 \\ -2 & 10 & -3 \end{bmatrix} \checkmark$$

END.

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Special Matrices

1) Tridiagonal Matrix (Thomas method)

$$\begin{bmatrix} f_1 & g_1 & 0 \\ e_2 & f_2 & g_2 \\ 0 & e_3 & f_3 \end{bmatrix}$$

3x3 – tridiagonal matrix

$$\begin{bmatrix} f_1 & g_1 & 0 & 0 \\ e_2 & f_2 & g_2 & 0 \\ 0 & e_3 & f_3 & g_3 \\ 0 & 0 & e_4 & f_4 \end{bmatrix}$$

4x4 – tridiagonal matrix

$$\begin{bmatrix} f_1 & g_1 & 0 & 0 & 0 \\ e_2 & f_2 & g_2 & 0 & 0 \\ 0 & e_3 & f_3 & g_3 & 0 \\ 0 & 0 & e_4 & f_4 & g_4 \\ 0 & 0 & 0 & e_5 & f_5 \end{bmatrix}$$

5x5 matrix.

$$AX = B$$

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Thomas method

the aim is to find the LU-decomposition of the tridiagonal matrix.

$$e_k^{New} = \frac{e_k^{old}}{f_{k-1}}$$

$$f_k^{New} = -e_k g_{k-1} + f_k^{New}$$

$k = 2, 3, \dots, n$

e.g.: Find the LU-decomposition of

1) $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -2 & -4 \\ 0 & -2 & 5 \end{bmatrix}$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 2 & -4 & -4 \\ 0 & \frac{1}{2} & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 2 & -4 & -4 \\ 0 & \frac{1}{2} & 7 \end{bmatrix}$$

e: $\frac{-2}{-4}$, f

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & -4 & -4 \\ 0 & 0 & 7 \end{bmatrix}$$

L $\quad$ U

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2) $A = \begin{bmatrix} 2 & 2 & 0 & 0 \\ -2 & 3 & 2 & 0 \\ 0 & 6 & 5 & 4 \\ 0 & 0 & -2 & 2 \end{bmatrix}$ (3)

$\sim \begin{bmatrix} 2 & 2 & 0 & 0 \\ -1 & 5 & 2 & 0 \\ 0 & \frac{6}{5} & \frac{13}{5} & 4 \\ 0 & 0 & \frac{-10}{13} & \frac{66}{13} \end{bmatrix}$

$\frac{-6}{5} \times 2 + 5 = \frac{-12}{5} + \frac{25}{5} = \frac{13}{5}$

$-2 + \frac{13}{5} = -2 * \frac{5}{13}$

$\frac{10}{13} \times 4 + 2 = \frac{40}{13} + \frac{26}{13} = \frac{66}{13}$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & \frac{6}{5} & 1 & 0 \\ 0 & 0 & \frac{-10}{13} & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 0 & 0 \\ 0 & 5 & 2 & 0 \\ 0 & 0 & \frac{13}{5} & 4 \\ 0 & 0 & 0 & \frac{66}{13} \end{bmatrix}$

L U

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[2] Symmetric Matrix (Cholesky Method)

Recall

A is symmetric if $A^T = A$

$$L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}, \quad U = L^T \quad \text{for } 2\times 2 \text{ matrix}$$

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}, \quad U = L^T \quad \text{for } 3\times 3 \text{ matrix}$$

$$L = \begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{bmatrix}, \quad U = L^T, \quad \text{for } 4\times 4 \text{ matrix}$$

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e.g: Use Cholesky method to find the LU-decomposition of

1) $A = \begin{bmatrix} 4 & 6 \\ 6 & 25 \end{bmatrix}$

$$\begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 25 \end{bmatrix}$$

L $\quad$ U

$l_{11}^2 = 4 \rightarrow l_{11} = \sqrt{4} = 2 \rightarrow \boxed{l_{11} = 2}$

$l_{11} l_{21} = 6 \rightarrow 2 l_{21} = 6 \rightarrow \boxed{l_{21} = 3}$

$l_{21}^2 + l_{22}^2 = 25 \rightarrow 9 + l_{22}^2 = 25$

$\rightarrow l_{22}^2 = 16 \rightarrow \boxed{l_{22} = 4}$

$$\begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix} = A$$

L $\quad$ U

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2) $A = \begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix}$

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} = \begin{bmatrix} 4 & 2 & 14 \\ 2 & 17 & -5 \\ 14 & -5 & 83 \end{bmatrix}$$

L $\quad$ U $\quad$ A

$l_{11}^2 = 4 \longrightarrow \boxed{l_{11} = 2}$

$l_{11} l_{21} = 2 \longrightarrow 2 l_{21} = 2 \rightarrow \boxed{l_{21} = 1}$

$l_{11} l_{31} = 14 \rightarrow 2 l_{31} = 14 \rightarrow \boxed{l_{31} = 7}$

$l_{21}^2 + l_{22}^2 = 17 \Rightarrow 1 + l_{22}^2 = 17$

$\Rightarrow l_{22}^2 = 16 \rightarrow \boxed{l_{22} = 4}$

$l_{21} l_{31} + l_{22} l_{32} = -5$

$(1)(7) + 4 l_{32} = -5 \Rightarrow 4 l_{32} = -12$

$\Rightarrow \boxed{l_{32} = -3}$

$l_{31}^2 + l_{32}^2 + l_{33}^2 = 83$

$49 + 9 + l_{33}^2 = 83 \rightarrow l_{33}^2 = 25 \rightarrow \boxed{l_{33} = 5}$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 7 & -3 & 5 \end{bmatrix} \begin{bmatrix} 2 & 1 & 7 \\ 0 & 4 & -3 \\ 0 & 0 & 5 \end{bmatrix} = A$$

L $\quad$ U

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Gauss - Seidel method (iterative method)

Linear system $AX = B$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

This method will be convergent, if

$$|a_{ii}| > \sum_{j=1}^{n} |a_{ij}|$$

e.g.: By Gauss - Seidel method, solve the system

$$6x - y - 2z = 12$$
$$3x + 5y + z = 3$$
$$2x - y + 7z = 36$$

initial guess $x_0 = 0$, $y_0 = 0$, $z_0 = 0$

Do two iterations ?

Sol:

$$x = \frac{12 + y + 2z}{6}$$

$$y = \frac{3 - 3x - z}{5}$$

$$z = \frac{36 - 2x + y}{7}$$

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1st iteration $x_0 = 0, y_0 = 0, z_0 = 0$

$$x_1 = \frac{12 + 0 + 2(0)}{6} \rightarrow \boxed{x_1 = 2}$$

$$y_1 = \frac{3 - 3(2) - 0}{5} \Rightarrow \boxed{y_1 = -0.6}$$

$$z_1 = \frac{36 - 2(2) + (-0.6)}{7} = \boxed{4.485714 = z_1}$$

2nd iteration $x_1 = 2, \; y_1 = -0.6, \; z_1 = 4.485714$

$$x_2 = \frac{12 + (-0.6) + 2(4.485714)}{6} \rightarrow \boxed{x_2 = 3.39524}$$

$$y_2 = \frac{3 - 3(3.39524) - (4.485714)}{5}$$

$$\boxed{y_2 = -2.334286}$$

$$z_2 = \frac{36 - 2(3.39524) + (-2.334286)}{7}$$

$$\boxed{z_2 = 3.89319}$$

Convergent

True solution $(\overset{x}{3}, \overset{y}{-2}, \overset{z}{4})$

· END ·

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Least squares Regression (Best fitting curve)

Before starting

* Sigma - Notation $\sum$ (summation)

1) $\sum_{i=1}^{n} f(i) = f(1) + f(2) + \cdots + f(n)$

* $\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$

* $\sum_{i=1}^{5} x_i = x_1 + x_2 + x_3 + x_4 + x_5$

* $\sum_{i=1}^{3} (a + bx_i) = (a + bx_1) + (a + bx_2) + (a + bx_3)$

2) $\sum_{i=1}^{n} a = na$

* $\sum_{i=1}^{4} 5 = 5 + 5 + 5 + 5 = 20$

* $\sum_{i=1}^{100} 2 = \underbrace{2 + 2 + 2 + \cdots + 2}_{100 \text{ times}} = (100)(2) = 200$

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3) $\sum_{i=1}^{n}(f(i) \mp g(i)) = \sum_{i=1}^{n} f(i) \mp \sum_{i=1}^{n} g(i)$

* $\sum_{i=1}^{3}(i^2+2i-3) = (1^2+2(1)-3)+(2^2+(2)(2)-3)+(3^2+2(3)-3)$

$= 0+5+12$

$= 17$ ✓

or $\sum_{i=1}^{3}(i^2+2i-3) = \sum_{i=1}^{3} i^2 + \sum_{i=1}^{3} 2i - \sum_{i=1}^{3} 3$

$= (1^2+2^2+3^2)+(2(1)+2(2)+2(3))-(3+3+3)$

$= 14+12-9$

$= 17$ ✓

4) $\sum_{i=1}^{n} c\,f(i) = c\sum_{i=1}^{n} f(i)$ c: Constant

* $\sum_{i=1}^{4}(5i^2+1) = (5(1^2)+1)+(5(2^2)+1)+(5(3^2)+1)$
$+(5(4^2)+1)$

$= (5+1)+(20+1)+(45+1)+(80+1)$

$= 6+21+46+81$

$= 154$

or $\sum_{i=1}^{4}(5i^2+1) = 5\sum_{i=1}^{4} i^2 + \sum_{i=1}^{4} 1$

$= 5(1^2+2^2+3^2+4^2)+4$

$= 5(30)+4 = 154$ ✓

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* Partial Derivatives

→ function in one variable.

$y = f(x) \longrightarrow \frac{dy}{dx} = f'(x)$ → ordinary derivative of f w.r.t x

→ Functions in two or more variables

* $Z = f(x, y) \longrightarrow \frac{\partial z}{\partial x} = f_x$ (Partial derivative of f w.r.t x)

$\frac{\partial z}{\partial y} = f_y$ (Partial derivative of f w.r.t y)

* $w = f(x, y, z) \longrightarrow \frac{\partial w}{\partial x} = f_x$ (Partial derivative of f w.r.t x)

$\frac{\partial w}{\partial y} = f_y$ (Partial derivative of f w.r.t y)

$\frac{\partial w}{\partial z} = f_z$ (Partial derivative of f w.r.t z)

⋮

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e.9 Find all the partial derivatives of

1) $f(x,y) = x^3 + y^2 + xy + 1$

$f_x = 3x^2 + y$

$f_y = 2y + x$

2) $f(x,y,z) = xy + xz + yz$

$f_x = y + z$

$f_y = x + z$

$f_z = x + y$

3) $f(a,b,c) = (ax_i + by_i + cz_i)^2$ $i = 1, 2, 3, \ldots, n$

x_i, y_i, z_i are fixed

$f_a = 2(ax_i + by_i + cz_i)(x_i)$

$= 2x_i(ax_i + by_i + cz_i)$

$f_b = 2(ax_i + by_i + cz_i)(y_i)$

$= 2y_i(ax_i + by_i + cz_i)$

$f_c = 2(ax_i + by_i + cz_i)(z_i)$

$= 2z_i(ax_i + by_i + cz_i)$

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Min. or Max. of any functions (in one or more variables) occur when all the first derivative(s) are (is) zeros (zero).

e.g: Find the minimum of $f(x,y) = (x-2)^2 + (y-3)^2 + 1$

$f_x = 2(x-2)$

$f_y = 2(y-3)$

$f_x = 0 \rightarrow x = 2$

$f_y = 0 \rightarrow y = 3$

$\therefore$ the min. of f occurs at $(2,3)$

then min. $= f(2,3) = 1$.

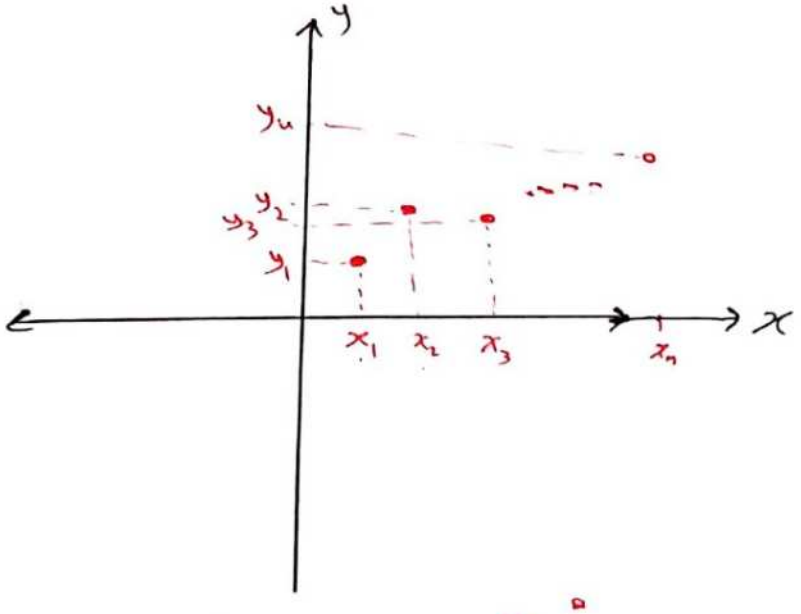
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Now: Least-squares regression.

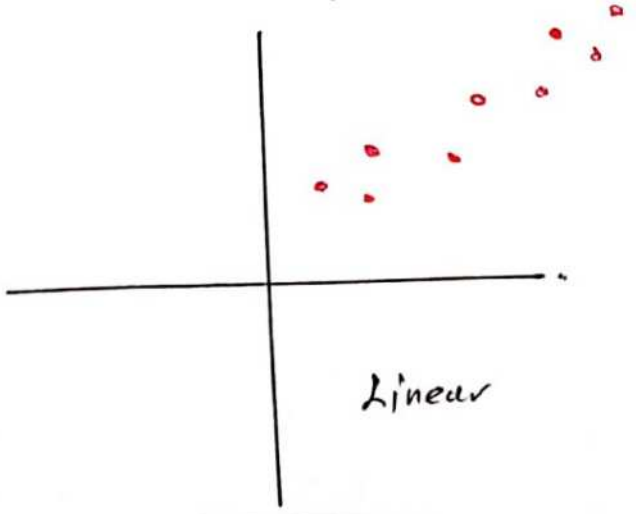
Assume we want to find a relationship (model) between two variables x, y after recording measurements of corresponding values of x and y. And assume we get n ordered pairs (data), namely

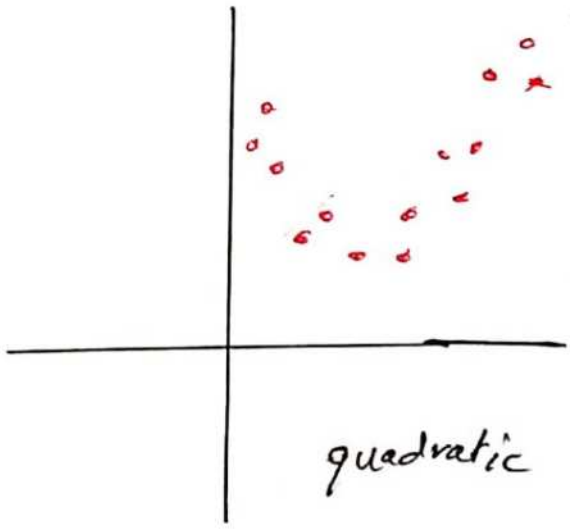
$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$.

to see the shape of the better model, plot these points in the xy-plane.



scatter diagram



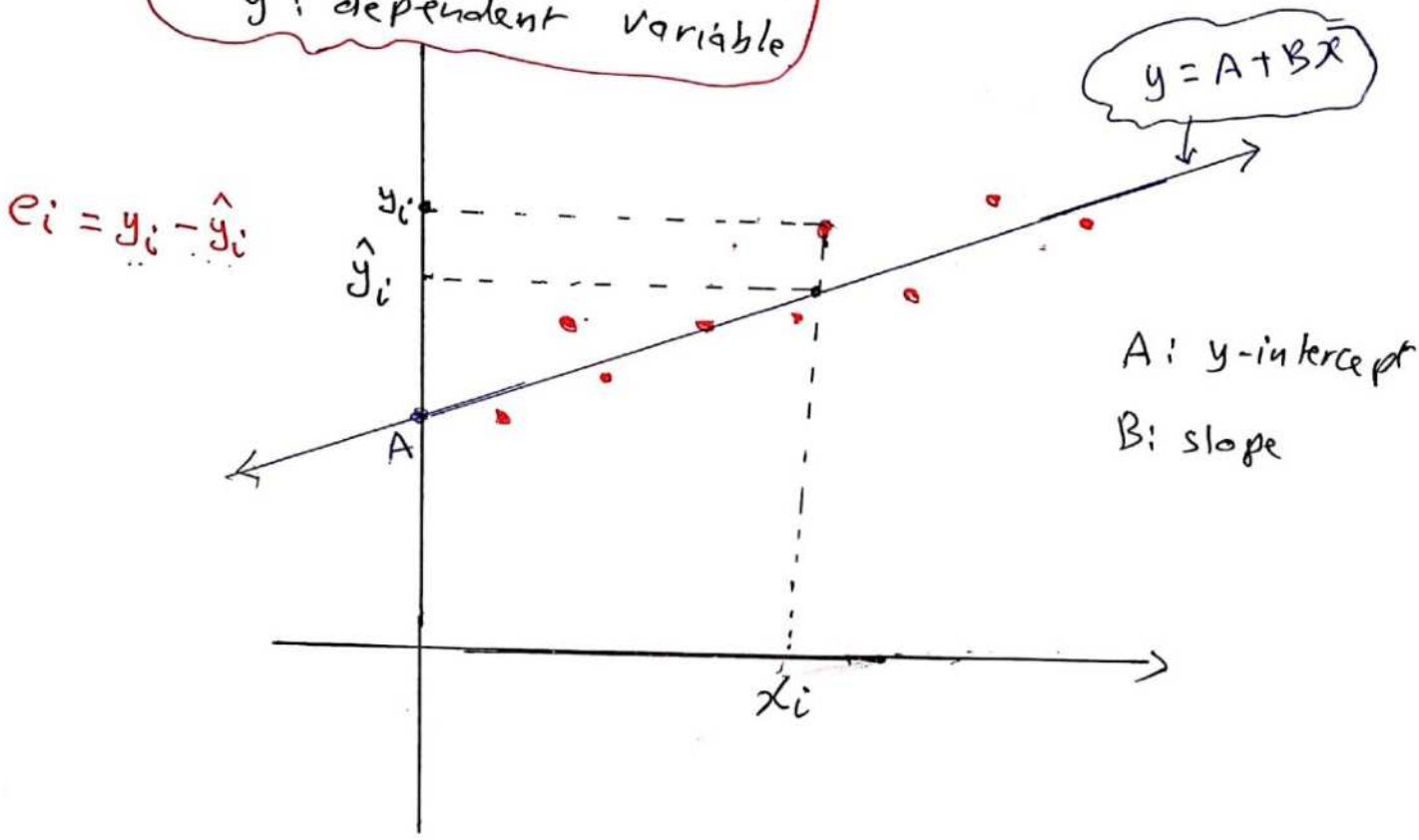


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* Linear Regression

Data : $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) = \{(x_i, y_i)\}_{i=1}^{n}$

x : independent variable
y : dependent variable



y_i = the measured value of y when $x = x_i$

$\hat{y}_i = A + Bx_i$ the calculated value (from the model) when $x = x_i$

$\hat{y}_i$ is also called the predicted value of y when $x = x_i$

$$e_i = y_i - \hat{y}_i$$ ← e_i : error (residual)

Note
One of the important aims of the regression curve is to predict the value of dependent variable y at a given value (in the range) of the independent variable x

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To obtain the best-fitting line (or curve) by least squares method, is, to minimize the sum of the squares of residuals

$$S_r = \sum_{i=1}^{n} e_i^2$$

Now, to find the best fitting line

$$y = A + Bx \quad \text{(Also called linear regression)}$$

$$S_r = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$S_r = \sum_{i=1}^{n} (y_i - A - Bx_i)^2$$

$$\frac{\partial S_r}{\partial A} = \sum_{i=1}^{n} 2(y_i - A - Bx_i)(-1)$$

$$= -2 \sum_{i=1}^{n} (y_i - A - Bx_i)$$

$$= -2 \left[\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} A - \sum_{i=1}^{n} Bx_i \right]$$

$$= -2 \left[\sum_{i=1}^{n} y_i - nA - B \sum_{i=1}^{n} x_i \right]$$

$$\frac{\partial S_r}{\partial A} = 0 \rightarrow \boxed{nA + B \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i} \quad (1)$$

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$$\frac{\partial S_r}{\partial B} = \sum_{i=1}^{n} 2\,(y_i - A - Bx_i)(-x_i)$$

$$= -2 \sum_{i=1}^{n} (x_i y_i - A x_i - B x_i^2)$$

$$= -2 \left[\sum_{i=1}^{n} x_i y_i - A \sum_{i=1}^{n} x_i - B \sum_{i=1}^{n} x_i^2 \right]$$

$$\frac{\partial S_r}{\partial B} = 0 \rightarrow \boxed{A \sum_{i=1}^{n} x_i + B \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i} \quad (2)$$

So we get the following linear system

$$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

we use Cramer's rule to find B.

$$B = \frac{\begin{vmatrix} n & \Sigma y_i \\ \Sigma x_i & \Sigma x_i y_i \end{vmatrix}}{\begin{vmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{vmatrix}} = \frac{n \Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)}{n \Sigma x_i^2 - (\Sigma x_i)^2}$$

$$\boxed{B = \frac{n \Sigma x_i y_i - (\Sigma x_i)(\Sigma y_i)}{n \Sigma x_i^2 - (\Sigma x_i)^2}}$$

- from statistics, $\bar{y} = A + B\bar{x}$

$$\rightarrow \boxed{A = \bar{y} - B\bar{x}}$$

where $\bar{y} = \frac{\Sigma y_i}{n}$, $\bar{x} = \frac{\Sigma x_i}{n}$.

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e.g: Fit a least-squares line to the following data. And hence, find the predicted value of y and the residual (error) when $x = 7$

x	1	3	5	7	9
y	4	2	6	8	10

Sol:

$y = A + Bx$ or $y_i = A + Bx_i + e_i$

or $y_i = \hat{y}_i + e_i$

$$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

	x_i	y_i	x_i^2	$x_i y_i$
1	1	4	1	4
2	3	2	9	6
3	5	6	25	30
4	7	8	49	56
5	9	10	81	90
Σ	25	30	165	186

$$\Rightarrow \begin{bmatrix} 5 & 25 \\ 25 & 165 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 30 \\ 186 \end{bmatrix}$$

$\Rightarrow A = 1.5$, $B = 0.9$

A ↓ y-intercept; B ↓ slope

$y = 1.5 + 0.9x$

$n = 5$

$x = 7 = x_4$

$\hat{y}_4 = 1.5 + 0.9(7) = 7.8$

$e_4 = y_4 - \hat{y}_4 = 8 - 7.8 = 0.2$

END

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Polynomial Regression

* Least squares Parabola (Quadratic)

Fitting the data $\{(x_i, y_i)\}_{i=1}^{n}$ by Least squares parabola

$\rightarrow$ $y = a + bx + cx^2 + e$ — residual (error)

$$y_i = a + bx_i + cx_i^2 + e_i$$

$$S_r = \sum_{i=1}^{n} e_i^2$$ sum of the squares of residuals

$$S_r = \sum_{i=1}^{n} (y_i - a - bx_i - cx_i^2)^2$$

solving the equations

$$\frac{\partial S_r}{\partial a} = 0 \quad , \quad \frac{\partial S_r}{\partial b} = 0 \quad \text{and} \quad \frac{\partial S_r}{\partial c} = 0$$

we get the following linear system

$$n a + b\Sigma x_i + c\Sigma x_i^2 = \Sigma y_i \quad -(1)$$

$$a\Sigma x_i + b\Sigma x_i^2 + c\Sigma x_i^3 = \Sigma x_i y_i \quad -(2)$$

$$a\Sigma x_i^2 + b\Sigma x_i^3 + c\Sigma x_i^4 = \Sigma x_i^2 y_i \quad -(3)$$

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Using matrices we can write the above system as

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \Sigma x_i^2 y_i \end{bmatrix}$$

In the same manner, we can find the linear system, if we want a cubic regression

$$y_i = a + bx_i + cx_i^2 + dx_i^3 + e_i$$

$$S_r = \sum_{i=1}^{n} (y_i - a - bx_i - cx_i^2 - dx_i^3)^2$$

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 & \Sigma x_i^5 \\ \Sigma x_i^3 & \Sigma x_i^4 & \Sigma x_i^5 & \Sigma x_i^6 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \Sigma x_i^2 y_i \\ \Sigma x_i^3 y_i \end{bmatrix}$$

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e.g: Using least squares method, fit the following data by a quadratic regression

x_i	1.1	2	3.2	4	4.5	5.2
y_i	0.5	2.5	2	4	3.5	6.5

Sol: $y_i = a + bx_i + cx_i^2 + e_i$

$n = 6$,

$$\begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \\ \Sigma x_i^2 y_i \end{bmatrix}$$

	x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
1	1.1	0.5	1.21	1.331	1.4641	0.55	0.605
2	2	2.5	4	8	16	5	10
3	3.2	2	10.24	32.768	104.8576	6.4	20.48
4	4	4	16	64	256	16	64
5	4.5	3.5	20.25	91.125	410.0625	15.75	70.875
$n \to 6$	5.2	6.5	27.04	140.608	731.1616	33.8	175.76
	Σx_i	Σy_i	Σx_i^2	Σx_i^3	Σx_i^4	$\Sigma x_i y_i$	$\Sigma x_i^2 y_i$
Σ	20	19	78.74	337.832	1519.5458	77.5	341.72

$$\begin{bmatrix} 6 & 20 & 78.74 \\ 20 & 78.74 & 337.832 \\ 78.74 & 337.832 & 1519.5458 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 19 \\ 77.5 \\ 341.72 \end{bmatrix}$$

$a = 1.178264$, $b = -0.388651$, $c = 0.250234$

$\therefore y_i = 1.178264 - 0.388651 x_i + 0.250234 x_i^2 + e_i$

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Multiple Linear regression (Plane)

To fit the data

$$\{(x_i, y_i, z_i)\}_{i=1}^{n} = \{(x_1, y_1, z_1), (x_2, y_2, z_2), \dots (x_n, y_n, z_n)\}$$

by multiple linear regression, namely

$$z_i = a + bx_i + cy_i + e_i \qquad i = 1, 2, \dots, n$$

write $S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (z_i - a - bx_i - cy_i)^2$

and then, put $\frac{\partial S_r}{\partial a} = 0$, $\frac{\partial S_r}{\partial b} = 0$ and $\frac{\partial S_r}{\partial c} = 0$

to get the following linear system

$$\begin{aligned} n a + b\Sigma x_i + c\Sigma y_i &= \Sigma z_i \quad (1) \\ a\Sigma x_i + b\Sigma x_i^2 + c\Sigma x_i y_i &= \Sigma x_i z_i \quad (2) \\ a\Sigma y_i + b\Sigma x_i y_i + c\Sigma y_i^2 &= \Sigma y_i z_i \quad (3) \end{aligned}$$

or by matrices

$$\begin{bmatrix} n & \Sigma x_i & \Sigma y_i \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i y_i \\ \Sigma y_i & \Sigma x_i y_i & \Sigma y_i^2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \Sigma z_i \\ \Sigma x_i z_i \\ \Sigma y_i z_i \end{bmatrix}$$

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e.g : Use multiple Linear regression (plane), to fit the following data : (x, y, z)

x	1	3	5	7	9
y	2	6	4	8	10
z	7	10	16	20	30

and hence, find the predicted value of z when $x = 7, y = 8$ and the residual ?

Sol:

$$z_i = a + bx_i + cy_i + e_i$$

$$\begin{bmatrix} n & \Sigma x_i & \Sigma y_i \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i y_i \\ \Sigma y_i & \Sigma x_i y_i & \Sigma y_i^2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \Sigma z_i \\ \Sigma x_i z_i \\ \Sigma y_i z_i \end{bmatrix}$$

$n = 5$

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	x_i	y_i	z_i	x_i^2	$x_i y_i$	$x_i z_i$	y_i^2	$y_i z_i$
1	1	2	7	1	2	7	4	14
2	3	6	10	9	18	30	36	60
3	5	4	16	25	20	80	16	64
4	7	8	20	49	56	140	64	160
$n=5$	9	10	30	81	90	270	100	300
	25	30	83	165	186	527	220	598

$$\begin{bmatrix} 5 & 25 & 30 \\ 25 & 165 & 186 \\ 30 & 186 & 220 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 83 \\ 527 \\ 598 \end{bmatrix}$$

$$a = \frac{262}{95} \quad , \quad b = \frac{55}{19} \quad , \quad c = -\frac{2}{19}$$

$$\therefore \boxed{Z_i = \frac{262}{95} + \frac{55}{19} x_i - \frac{2}{19} y_i + e_i}$$

Now

when $x = 7 = x_4$, $y = 8 = y_4$

the predicted value of $Z_4 = \hat{Z}_4 = \frac{262}{95} + \frac{55}{19}(7) - \frac{2}{19}(8)$

$$\therefore \hat{Z}_4 \simeq 22.17895 \quad (\text{to 5 d.p's})$$

$$\therefore \text{the residual } (e_4) = Z_4 - \hat{Z}_4$$

$$= 20 - 22.17895$$

$$= -0.17895$$

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e.g: By using least squares method, to fit the data $\{(x_i, y_i)\}_{i=1}^{n}$ by the model $y = Ax^4$, derive a formula to compute A?

$$y_i = Ax_i^4 + e_i$$

$$S_r = \sum_{i=1}^{n} e_i^2$$

$$= \sum_{i=1}^{n} (y_i - Ax_i^4)^2$$

$$\frac{dS_r}{dA} = \sum_{i=1}^{n} 2(y_i - Ax_i^4)^1(-x_i^4)$$

$$= -2 \sum_{i=1}^{n} (x_i^4 y_i - Ax_i^8)$$

$$\frac{dS_r}{dA} = -2\left[\sum_{i=1}^{n} x_i^4 y_i - A\sum x_i^8\right]$$

$$\frac{dS_r}{dA} = 0 \rightarrow -2\left[\sum_{i=1}^{n} x_i^4 y_i - A\sum_{i=1}^{n} x_i^8\right] = 0$$

$$\rightarrow \sum_{i=1}^{n} x_i^4 y_i - A\sum_{i=1}^{n} x_i^8 = 0$$

$$A = \frac{\sum_{i=1}^{n} x_i^4 y_i}{\sum_{i=1}^{n} x_i^8}$$

$$A = \frac{\sum_{i=1}^{n} x_i^4 y_i}{\sum_{i=1}^{n} x_i^8}$$

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Linearization of Nonlinear relationships:

① $y = A e^{Bx}$ exponential model

$\ln y = \ln(A e^{Bx})$

$\ln y = \ln A + \ln e^{Bx}$

$\ln y = \ln A + Bx \ln e$

$\ln e = 1$

$\ln y = \ln A + Bx$

$y = \ln A + Bx$

$Y = \ln A + BX$

B: slope

$\ln A$: y-intercept

② $y = A x^{B}$ simple power model

$\ln y = \ln(A x^{B})$

$\ln y = \ln A + \ln x^{B}$

$\ln y = \ln A + B \ln x$

$Y = \ln A + BX$

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3. $y = \frac{Ax}{B+x}$ saturation-growth-rate

$$\rightarrow \frac{1}{y} = \frac{B+x}{Ax}$$

$$\frac{1}{y} = \frac{B}{Ax} + \frac{x}{Ax}$$

$$\frac{1}{y} = \frac{B}{A}\left(\frac{1}{x}\right) + \left(\frac{1}{A}\right)$$

$$Y = \frac{B}{A}X + \frac{1}{A}$$

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e.g: Fit the following data by the exponential model $y = Ae^{Bx}$ (by linearization)

x	2	4	6	8
y	2	3	3.5	4.5

Sol:

$$y = Ae^{Bx} \longrightarrow \ln y = \ln(Ae^{Bx})$$

$$\Rightarrow \boxed{\ln y = \ln A + Bx}$$

$$\begin{bmatrix} n & \Sigma x \\ \Sigma x & \Sigma x^2 \end{bmatrix} \begin{bmatrix} \ln A \\ B \end{bmatrix} = \begin{bmatrix} \Sigma \ln y \\ \Sigma x \ln y \end{bmatrix}$$

$n = 4$

	x	y	$\ln y$	x^2	$x \ln y$
1	2	2	0.693147	4	1.386294
2	4	3	1.098612	16	4.394448
3	6	3.5	1.252763	36	7.516578
$n=4$	8	4.5	1.504077	64	12.032616
Σ	20		4.548599	120	25.329936

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$$\begin{bmatrix} 4 & 20 \\ 20 & 120 \end{bmatrix} \begin{bmatrix} \ln A \\ B \end{bmatrix} = \begin{bmatrix} 4.548599 \\ 25.329936 \end{bmatrix}$$

$\Rightarrow$ $\ln A = 0.4904145$ and $B = 0.129347$

$$A = e^{0.4904145}$$

$$A = 1.632993$$

$$\therefore y = 1.632993\, e^{0.129347x}$$

END

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Interpolation:

→ Polynomial interpolation:

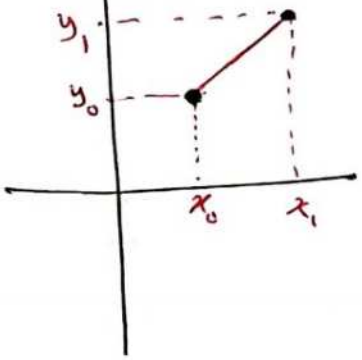
If $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ $-(n+1)$ points $-$ (precise data), then there exists a unique polynomial of the nth degree (n^{th} order) that passing through these $n+1$ points.

The polynomial interpolation is to estimate the value of y corresponding to the intermediate value x between $x_0, x_1, \dots, x_n$, by substituting x in this function. i.e $y = f_n(x)$

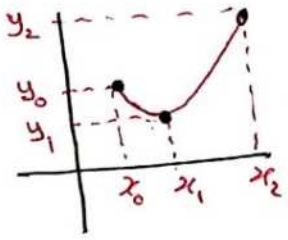
Also, we can approximate a complicated function that passes through the points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ $\left(f(x_i) = y_i\right)_{i=0,1,2,\dots,n}$ by polynomial of degree n, namely $f_n(x)$ $x \in [x_0, x_n]$

Linear interpolation

$f_1(x) = a_0 + a_1 x$

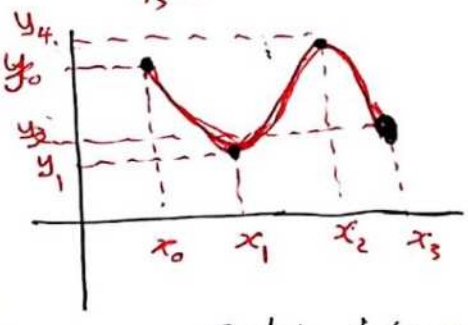


$f_2(x) = a_0 + a_1 x + a_2 x^2$



Quadratic interpolation
(Prabola)

$f_3(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$



Cubic interpolation

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1) Newton's Divided Difference Interpolation Polynomials:

Def:

If we have $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\cdots$, $(x_n, f(x_n))$

1) First finite divided difference

$$f[x_k, x_{k-1}] = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}} \quad , k = 1, 2, \cdots, n.$$

2) Second finite divided difference

$$f[x_k, x_{k-1}, x_{k-2}] = \frac{f[x_k, x_{k-1}] - f[x_{k-1}, x_{k-2}]}{x_k - x_{k-2}}$$

$$k = 2, 3, \cdots, n.$$

3) third finite divided difference

$$f[x_k, x_{k-1}, x_{k-2}, x_{k-3}] = \frac{f[x_k, x_{k-1}, \cdots x_{k-2}] - f[x_{k-1}, \cdots x_{k-3}]}{x_k - x_{k-3}}$$

$$k = 3, 4, 5, \cdots, n.$$

$\vdots$

m) the nth finite divided difference

$$f[x_n, x_{n-1}, x_{n-2}, \cdots x_0] = \frac{f[x_n, x_{n-1}, \cdots, x_1] - f[x_{n-1}, x_{n-2}, \cdots x_0]}{x_n - x_0}$$

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Examples

* $f[x_1, x_0] = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}$

* $f[x_2, x_1] = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$

* $f[x_5, x_4] = \dfrac{f(x_5) - f(x_4)}{x_5 - x_4}$

* $f[x_2, x_1, x_0] = \dfrac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$

* $f[x_3, x_2, x_1] = \dfrac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$

* $f[x_4, x_3, x_2] = \dfrac{f[x_4, x_3] - f[x_3, x_2]}{x_3 - x_2}$

* $f[x_3, x_2, x_1, x_0] = \dfrac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$

* $f[x_4, x_3, x_2, x_1] = \dfrac{f[x_4, x_3, x_2] - f[x_3, x_2, x_1]}{x_4 - x_1}$

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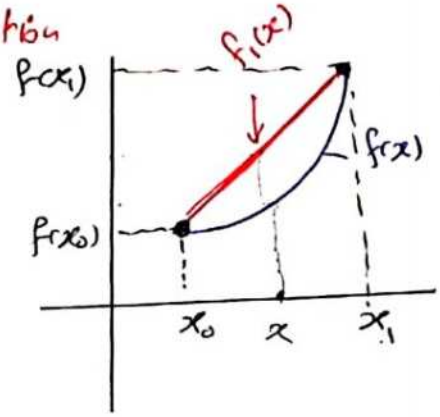
* Linear Interpolation

Data : $(x_0, f(x_0))$, $(x_1, f(x_1))$

$$f_1(x) = b_0 + b_1(x - x_0)$$ Linear interpolation formula

where $b_0 = f(x_0)$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$



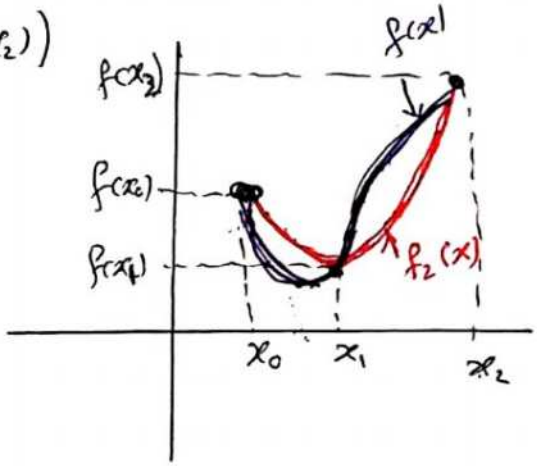
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* Quadratic Interpolation (Parabola)

Data : $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

↳ Quadratic formula



where $b_0 = f(x_0)$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

x_i	$f(x_i)$	First D.D	Second D.D
x_0	$f(x_0)$ b_0		
		$f[x_1, x_0]$ b_1	
x_1	$f(x_1)$		$f[x_2, x_1, x_0]$ b_2
		$f[x_2, x_1]$	
x_2	$f(x_2)$		

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* Cubic Interpolation

Data : $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$, $(x_3, f(x_3))$

$$f_3(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2)$$

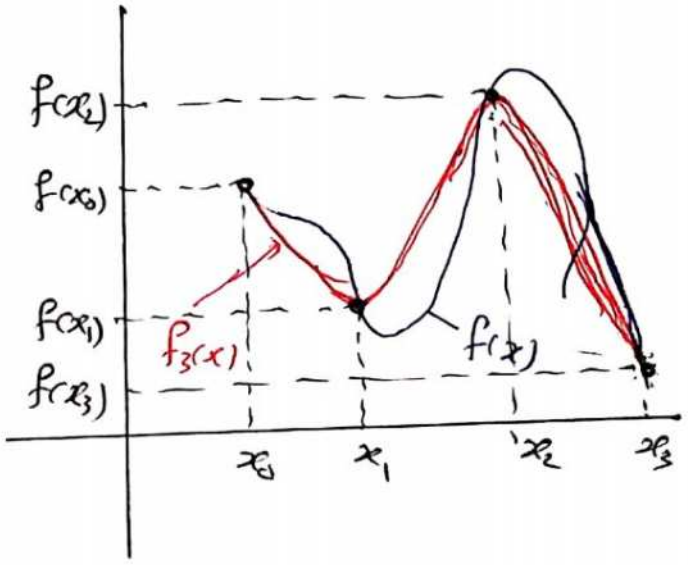
Cubic interpolation formula

when $b_0 = f(x_0)$

$b_1 = f[x_1, x_0]$

$b_2 = f[x_2, x_1, x_0]$

$b_3 = f[x_3, x_2, x_1, x_0]$



x_i	$f(x_i)$	First D.D	Second D.D	Third D.D
x_0	$f(x_0)$ ← b_0			
		$f[x_1, x_0]$ ← b_1		
x_1	$f(x_1)$		$f[x_2, x_1, x_0]$ ← b_2	
		$f[x_2, x_1]$		$f[x_3, x_2, x_1, x_0]$ ← b_3
x_2	$f(x_2)$		$f[x_3, x_2, x_1]$	
		$f[x_3, x_2]$		
x_3	$f(x_3)$			

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* n^{th} (order) degree polynomial Interpolation.

Data: $(x_0, f(x_0))$, $(x_1, f(x_1))$, ... , $(x_n, f(x_n))$

$$f_n(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + \cdots + b_n(x-x_0)(x-x_1)\cdots(x-x_{n-1})$$

nth degree polynomial interpolation formula

where $b_0 = f(x_0)$, $b_1 = f[x_1, x_0]$, $b_2 = f[x_2, x_1, x_0]$, ...

$$b_n = f[x_n, x_{n-1}, \ldots, x_0]$$

---

e.g: Estimate $(\ln 3)$ using linear interpolation by taking a) $x_0 = 1$, $x_1 = 5$

b) $x_0 = 2$, $x_2 = 4$

and compute the percent absolute relative true error assume the true value of ln3 is 1.098612

Sol: $f(x) = \ln x$

a) $x_0 = 1 \rightarrow f(x_0) = 0$

$x_1 = 5 \longrightarrow f(x_1) = 1.609438$

$f_1(x) = b_0 + b_1(x-x_0)$

$b_0 = 0$, $b_1 = f[x_1, x_0] = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} = \dfrac{1.609438 - 0}{5 - 1}$

$\therefore b_1 = 0.402360$

$\therefore f_1(x) = 0 + 0.402360(x - 1)$

$f_1(x) = 0.402360(x-1)$ ✓

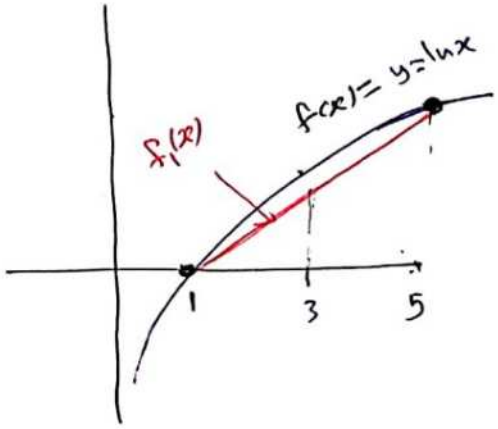
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$\therefore \ln 3 \cong f_1(3) = 0.402360\,(3-1) = 0.804719$ (7)

$\therefore \ln 3 = 0.804719$

$\therefore |\epsilon_t| * 100\% = \left| \frac{1.098612 - 0.804719}{1.098612} \right| * 100\%$

$= 26.75\%$



b) $x_0 = 2 \Rightarrow f(x_0) = 0.693147$

$x_1 = 4 \Rightarrow f(x_1) = 1.386294$

$f_1(x) = b_0 + b_1(x-2)$

$b_0 = f(x_0) = 0.693147$

$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1.386294 - 0.693147}{4-2}$

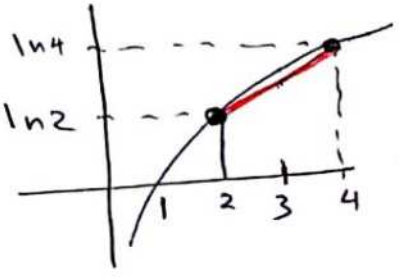
$= 0.3465735$

$\therefore f_1(x) = 0.693147 + 0.3465735(x-2)$

$f_1(x) = 0.693147 + 0.3465735(x-2)$

$f(3) = \ln 3 \cong f_1(3) = 1.0397205$

$|\epsilon_t| = 5.36\%$



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e.g: Use the given data, to find the Newton's 3rd-order interpolating polynomial, and hence estimate $f(3)$?

x	-2	2	4	8
$f(x)$	3	7	13	17

Sol:

$$f_3(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2)$$

x_i	$f(x_i)$	1st FDD	2nd FDD	3rd FDD
x_0 -2	3 (b_0)			
		1 (b_1)		
x_1 2	7		$\frac{1}{3}$ (b_2)	
		3		$-\frac{1}{15}$ (b_3)
x_2 4	13		$-\frac{1}{3}$	
		1		
x_3 8	17			

$$f_3(x) = 3 + 1(x+2) + \frac{1}{3}(x+2)(x-2) - \frac{1}{15}(x+2)(x-2)(x-4)$$

$$f(3) \cong f_3(3) = 3 + (5) + \frac{1}{3}(5)(1) - \frac{1}{15}(5)(1)(-1)$$

$$= 10$$

– END –

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2) Lagrange Interpolating Polynomials:

Introduction: Product Notation (Π-notation)

* $\prod_{i=1}^{3} i^2 = (1^2)(2^2)(3^2) = (1)(4)(9) = 36$

* $\prod_{i=0}^{4} (x_i - 5) = (x_0 - 5)(x_1 - 5)(x_2 - 5)(x_3 - 5)(x_4 - 5)$

* $\prod_{\substack{i=0 \\ i \neq 3}}^{5} (x_i - x_3) = (x_0 - x_3)(x_1 - x_3)(x_2 - x_3)(x_4 - x_3)(x_5 - x_3)$

If $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\cdots$ $(x_n, f(x_n))$ $n+1$ - points (precise data)

then the Lagrange Interpolating polynomial of n-order is

$$f_n(x) = \sum_{i=0}^{n} L_i(x) f(x_i)$$

$$= L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2) + \cdots + L_n(x) f(x_n)$$

where $L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$

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* Lagrange Interpolating Polynomial of first order

Data : $(x_0, f(x_0))$, $(x_1, f(x_1))$

$$f_1(x) = L_0(x) f(x_0) + L_1(x) f(x_1)$$

where $L_0(x) = \frac{x - x_1}{x_0 - x_1}$, $L_1(x) = \frac{x - x_0}{x_1 - x_0}$

* Lagrang Interpolating Polynomial of second order

Data : $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$

$$f_2(x) = L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2)$$

where

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

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* Lagrange Interpolation Polynomial of third-order.

Data: $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$, $(x_3, f(x_3))$

$$f_3(x) = L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2) + L_3(x) f(x_3)$$

where

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}$$

$$L_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

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e.g: If $f(x) = \frac{x^2+4}{x^2+1}$, $x_0 = -1$, $x_1 = 2$, $x_3 = 3$

use the data $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$

to find the Lagrange interpolating function (Polynomial) of 2nd order. Then use it to estimate $f(1)$ and compute the absolute percent relative true error $(|\epsilon_t| * 100\%)$ ..

Sol:

x_0 $f(x_0)$, x_1 $f(x_1)$, x_2 $f(x_2)$

$(-1, 2.5)$, $(2, 1.6)$, $(3, 1.3)$

$$f_2(x) = L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2)$$

i.e $$f_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

$$\therefore f_2(x) = \frac{(x-2)(x-3)}{(-1-2)(-1-3)}(2.5) + \frac{(x+1)(x-3)}{(2+1)(2-3)}(1.6) + \frac{(x+1)(x-2)}{(3+1)(3-2)}(1.3$$

$$f_2(x) = \frac{5}{24}(x-2)(x-3) - \frac{8}{15}(x+1)(x-3) + \frac{13}{40}(x+1)(x-2) \checkmark$$

$$f(1) \cong f_2(1) = 1.9 \checkmark$$

True value $= f(1) = \frac{1^2+4}{1^2+1} = \frac{5}{2} = 2.5 \checkmark$

$$\epsilon_t = \left|\frac{2.5 - 1.9}{2.5}\right| * 100\% = 24\%$$

- END -

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Numerical Integration:

Trapezoidal Rule

Introduction

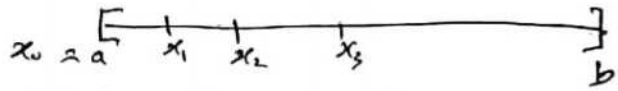
If $[a,b]$ is a closed interval

$x_0 = a < x_1 < x_2 < \cdots < x_n = b$ then

$P = \{x_0, x_1, \ldots, x_n\}$ is called a uniform partition of the closed interval $[a,b]$ if

(step size) $\Delta x = h = x_1 - x_0 = x_2 - x_1 = \cdots = x_n - x_{n-1} = \frac{b-a}{n}$

and $[x_0, x_1], [x_1, x_2], \ldots, [x_{n-1}, x_n]$ are called subintervals or segments



Notes: ① The number of segments $= n$ and (subintervals)

the number of the points $x_0, x_1, \ldots, x_n$ equals $n+1$

i.e

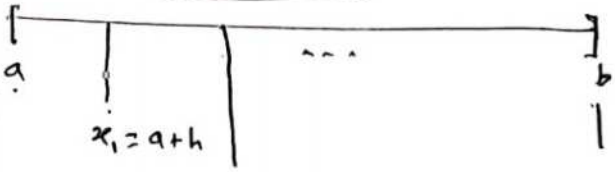
the number of segments + 1 = number of points

② $h = \frac{b-a}{n}$

$x_i = a + ih \quad i = 0, 1, 2, \ldots, n$

$x_0 = a + 0h = a$

$x_1 = a + h$ ... $x_n = a + nh = a + n\frac{b-a}{n} = b$



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e.g: Find the step size and the uniform partition for the interval $[-1,3]$, if number of subintervals (segments) is equal to 4?

Sol:

$a = -1$, $b = 3$, $n = 4$

$\Rightarrow \Delta x_i = h = \frac{3-(-1)}{4} = 1$ (step size)

$x_0 = a = -1$

$x_1 = a + h = -1 + 1 = 0$

$x_2 = x_1 + h = a + 2h = 0 + 1 = 1$

$x_3 = x_2 + h = 1 + 1 = 2$

$x_4 = x_3 + h = 2 + 1 = 3$

$P = \{-1, 0, 1, 2, 3\}$.

e.g: If P is a uniform partition of the interval $[1,2]$ and the step size $= 0.1$, find the number of elements of P.

Sol: $a = 1$, $b = 2$, $h = 0.1$, $n =$ number of segments.

$h = \frac{b-a}{n} = \frac{2-1}{n} \Rightarrow \frac{1}{n} = 0.1$

$\Rightarrow 0.1n = 1 \Rightarrow n = 10$ segments

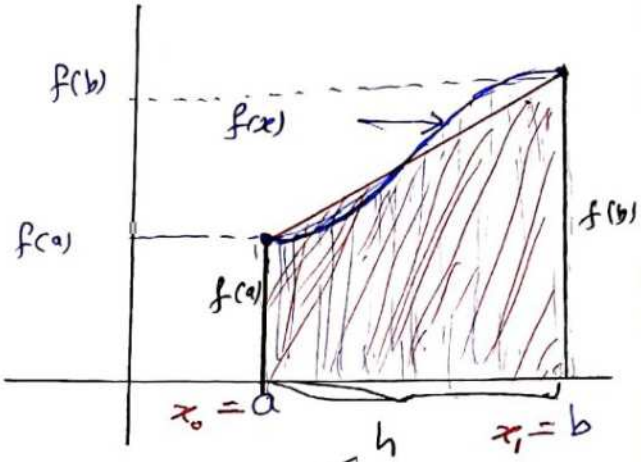
$\therefore$ the number of the elements of P is 11.

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Trapezoidal Rule: (simple T.R)

$f(x) \geq 0 \quad \forall x \in [a,b]$

$\int_a^b f(x)\,dx$ = Area under curve and above x-axis.



$$\Rightarrow \int_a^b f(x)dx \simeq \text{Area of the trapezoid}$$

$$\simeq \frac{h}{2}[f(a)+f(b)]$$

$$\simeq \frac{b-a}{2}(f(a)+f(b)) \quad \checkmark$$

$$h = \frac{x_1 - x_0}{1} = \frac{b-a}{1}$$

$$h = b-a$$

$$\int_a^b f(x)dx = \frac{h}{2}(f(a)+f(b)) \qquad h = \frac{b-a}{1}, \; n=1$$

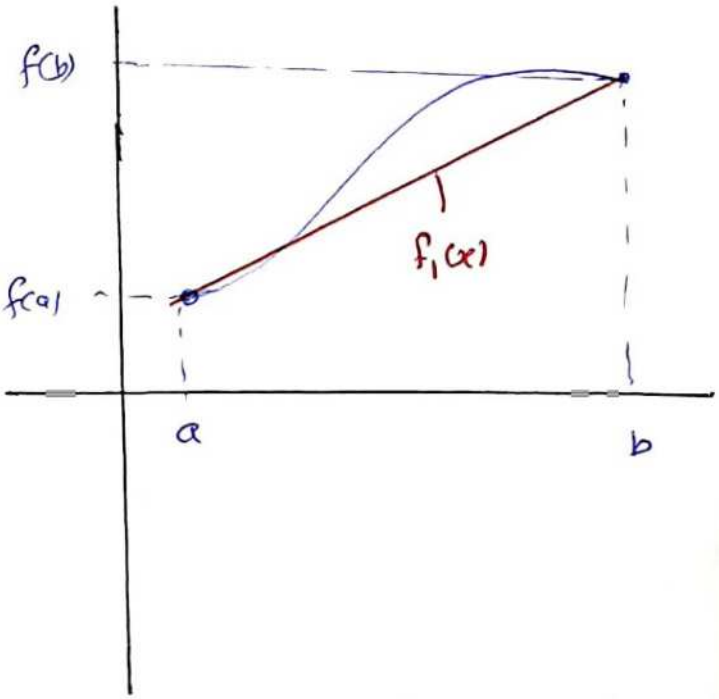
$$= \frac{h}{2}(f(x_0)+f(x_1))$$

or

$$\int_a^b f(x)dx = \frac{b-a}{2}(f(a)+f(b))$$

Also

$$\int_a^b f(x)dx \simeq \int_a^b f_1(x)\,dx$$



$$\simeq \int_a^b \left[f(a) + \frac{f(b)-f(a)}{b-a}(x-a)\right]dx$$

$$\simeq \frac{b-a}{2}(f(a)+f(b)) \quad \checkmark$$

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e.g: Use simple trapezoidal rule, to approximate $\int_0^1 x^3 dx$ and find ~~find~~ the absolute, relative error ?

Sol:

$$n = 1 \Rightarrow h = \frac{b-a}{1} = \frac{1-0}{1} = 1$$

$$a = 0 \,, \quad b = 1 \,, \quad f(x) = x^3$$

$$\Rightarrow \int_0^1 x^3 dx = \frac{1}{2}[f(0) + f(1)]$$

$$= \frac{1}{2}[0 + 1]$$

$$= 0.5$$

True value $\int_0^1 x^3 dx = \frac{x^4}{4}\Big|_0^1 = 0.25$.

$$\epsilon_t = \frac{|0.25 - 0.5|}{0.25} * 100\% = 100\% .$$

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Multiple - Application Trapezoidal Rule

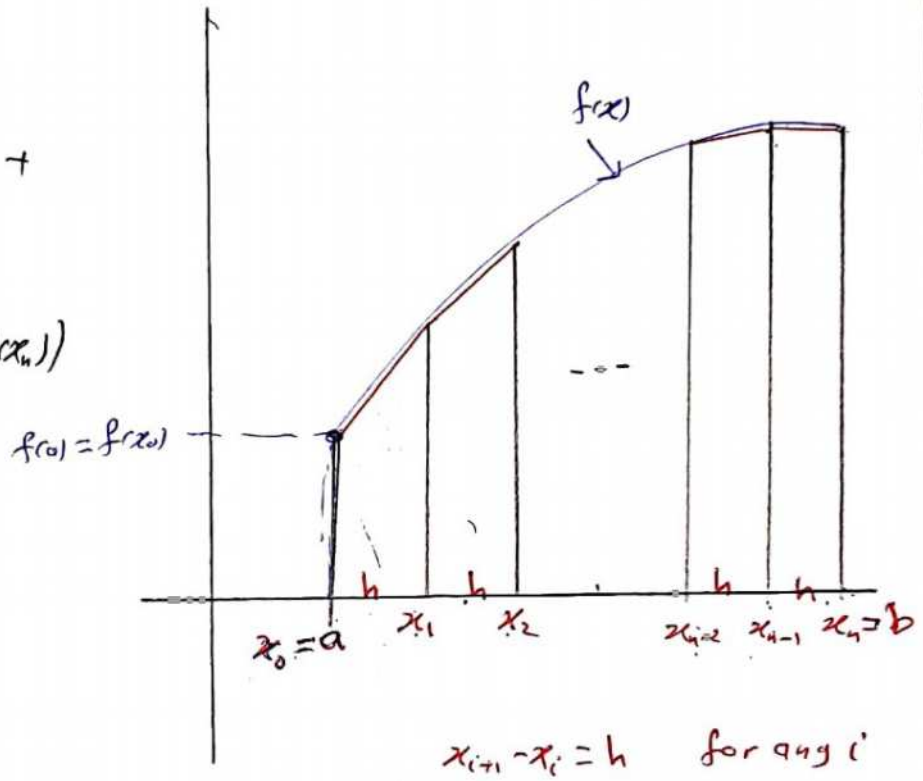
Composite integration formula :T, R

$$\int_a^b f(x)dx \cong \frac{h}{2}(f(x_0)+f(x_1))$$

$$+\frac{h}{2}(f(x_1)+f(x_2))+$$

$$\frac{h}{2}(f(x_2)+f(x_3))+\cdots+$$

$$\frac{h}{2}(f(x_{n-2})+f(x_{n-1}))+\frac{h}{2}(f(x_{n-1})+f(x_n))$$



$x_{i+1} - x_i = h$ for any i

$$\Rightarrow \int_a^b f(x)dx \cong \frac{h}{2}[f(x_0)+f(x_1) + f(x_1)+f(x_2) + f(x_2)+f(x_3) +$$

$$\cdots + f(x_{n-2})+f(x_{n-1}) + f(x_{n-1})+f(x_n)]$$

$$\Rightarrow \int_a^b f(x)dx \cong \frac{h}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)]$$

$$\Rightarrow \int_a^b f(x)dx \cong \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]$$

$$h = \frac{b-a}{n}$$

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e.g.: Use Multiple-Application Trapezoidal Rule, take the number of segments to be 4, to approximate $\int_0^1 x^3 dx$ and compute $\epsilon_t\%$.

Sol:

$n = 4 \Rightarrow h = \frac{b-a}{4} = \frac{1-0}{4} = 0.25$, $f(x) = x^3$

$x_0 = a = 0$, $x_1 = 0.25$, $x_2 = 0.5$, $x_3 = 0.75$, $x_4 = 1 = b$.

$$\therefore \int_0^1 x^3 dx \simeq \frac{0.25}{2}\left[0^3 + 2\left(0.25^3 + 0.5^3 + 0.75^3\right) + 1^3\right]$$

$$\simeq 0.265625$$

True value: $\int_0^1 x^3 dx = 0.25$

$$\therefore \quad \epsilon_t\% = \left|\frac{0.25 - 0.265625}{0.25}\right| * 100\%$$

$$= 6.25\%$$

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e.g : Use Multiple-Application Trapezoidal formula (take 4 points) to approximate

$$\int_0^{\pi/2} \sin x^2 \, dx$$ and compute $\epsilon_t\%$?

Sol:

$$f(x) = \sin x^2$$

$$a = 0, \quad b = \frac{\pi}{2}, \quad n = 3$$

$$h = \frac{b-a}{3} = \frac{\frac{\pi}{2} - 0}{3} = \frac{\pi}{6}$$

$$\Rightarrow x_0 = 0, \quad x_1 = \frac{\pi}{6}, \quad x_2 = \frac{\pi}{3}, \quad x_3 = \frac{\pi}{2}$$

$$\int_0^{\pi/2} \sin x^2 dx \simeq \frac{h}{2}\left[f(x_0) + 2(f(x_1) + f(x_2)) + f(x_3)\right]$$

$$\simeq \frac{\pi}{12}\left[\sin 0^2 + 2\left(\sin\left(\frac{\pi}{6}\right)^2 + \sin\left(\left(\frac{\pi}{3}\right)^2\right)\right) + \sin\left(\left(\frac{\pi}{2}\right)^2\right)\right]$$

$\sin\left(\left(\frac{\pi}{2}\right)^2\right)$

$$\therefore \int_0^{\pi/2} \sin x^2 \, dx \simeq 1.5420377743$$

True value = 0.8281163288

$$\epsilon_t = \left|\frac{\text{True} - \text{app.}}{\text{True}}\right| * \%$$

END

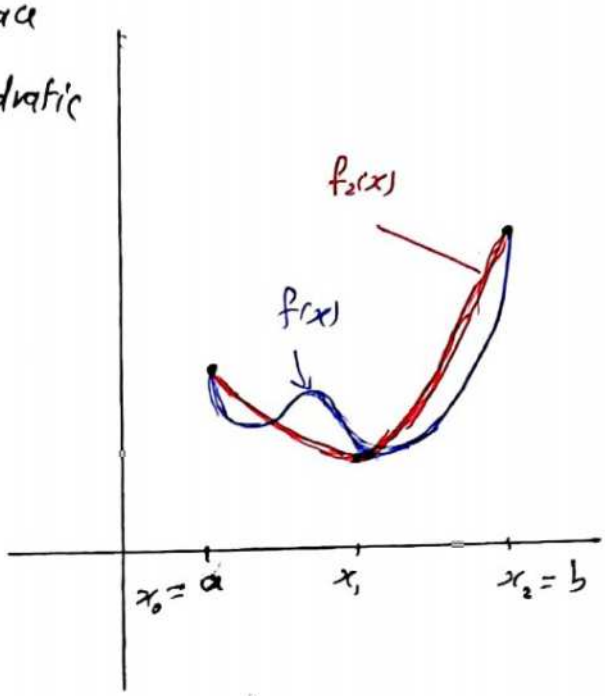
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Simpson's Rules

Simple Simpson's $\frac{1}{3}$-rule

$n=2$ in order to replace the integrand by a quadratic function.

$$\int_a^b f(x)dx \cong \int_a^b f_2(x)\,dx$$



$n=2 \Rightarrow h = \frac{b-a}{2}$

$\Rightarrow x_0 = a \quad , x_1 = a + \frac{b-a}{2} = \frac{a+b}{2} \quad , x_2 = b$

$$f_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) +$$

$$\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

After integrating $f_2(x)$ over the interval $[a,b]$

we get the following rule (Simpson $\frac{1}{3}$-rule)

$$\int_a^b f(x)dx = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2) \right].$$

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e.g: Use simple Simpson's $\frac{1}{3}$-rule to approximate $\int_1^2 x^2 dx$.

Sol: $f(x) = x^2$, $n=2$, $a=1$, $b=2 \Rightarrow h = \frac{2-1}{2} = 0.5$

$x_0 = 1$, $x_1 = 1.5$, $x_2 = 2$.

$$\therefore \int_1^2 x^2 dx \simeq \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

$$\simeq \frac{0.5}{3}\left[1^2 + 4(1.5^2) + 2^2\right]$$

$$\simeq \frac{7}{3}.$$

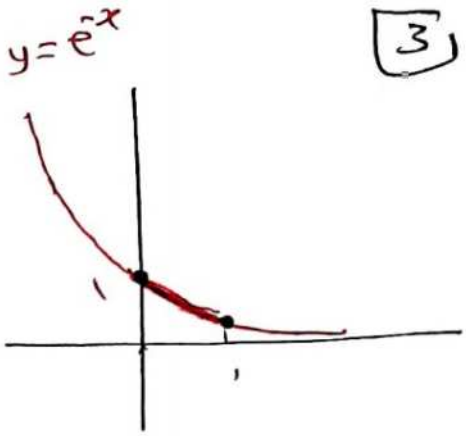
e.g: Find the length of the curve of $y = e^{-x}$ from $x=0$ to $x=1$, by using simple simpson's $\frac{1}{3}$-rule to approximate the integral.

Hint: the length of the curve of y from $x=a$ to $x=b$ is given by

$$L = \int_a^b \sqrt{\left(\frac{dy}{dx}\right)^2 + 1}\, dx$$

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Sol:



$$L = \int_0^1 \sqrt{\left(\frac{dy}{dx}\right)^2 + 1}\, dx$$

$$\frac{dy}{dx} = -e^{-x}$$

$$\Rightarrow L = \int_0^1 \sqrt{(-e^{-x})^2 + 1}\, dx$$

$$L = \int_0^1 \sqrt{e^{-2x} + 1}\, dx$$

Now $a = 0$, $b = 1$, $n = 2$, $f(x) = \sqrt{e^{-2x}+1}$

$$\Rightarrow h = \frac{1-0}{2} = 0.5 \Rightarrow x_0 = 0,\ x_1 = 0.5,\ x_2 = 1$$

$$\therefore L \simeq \frac{0.5}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

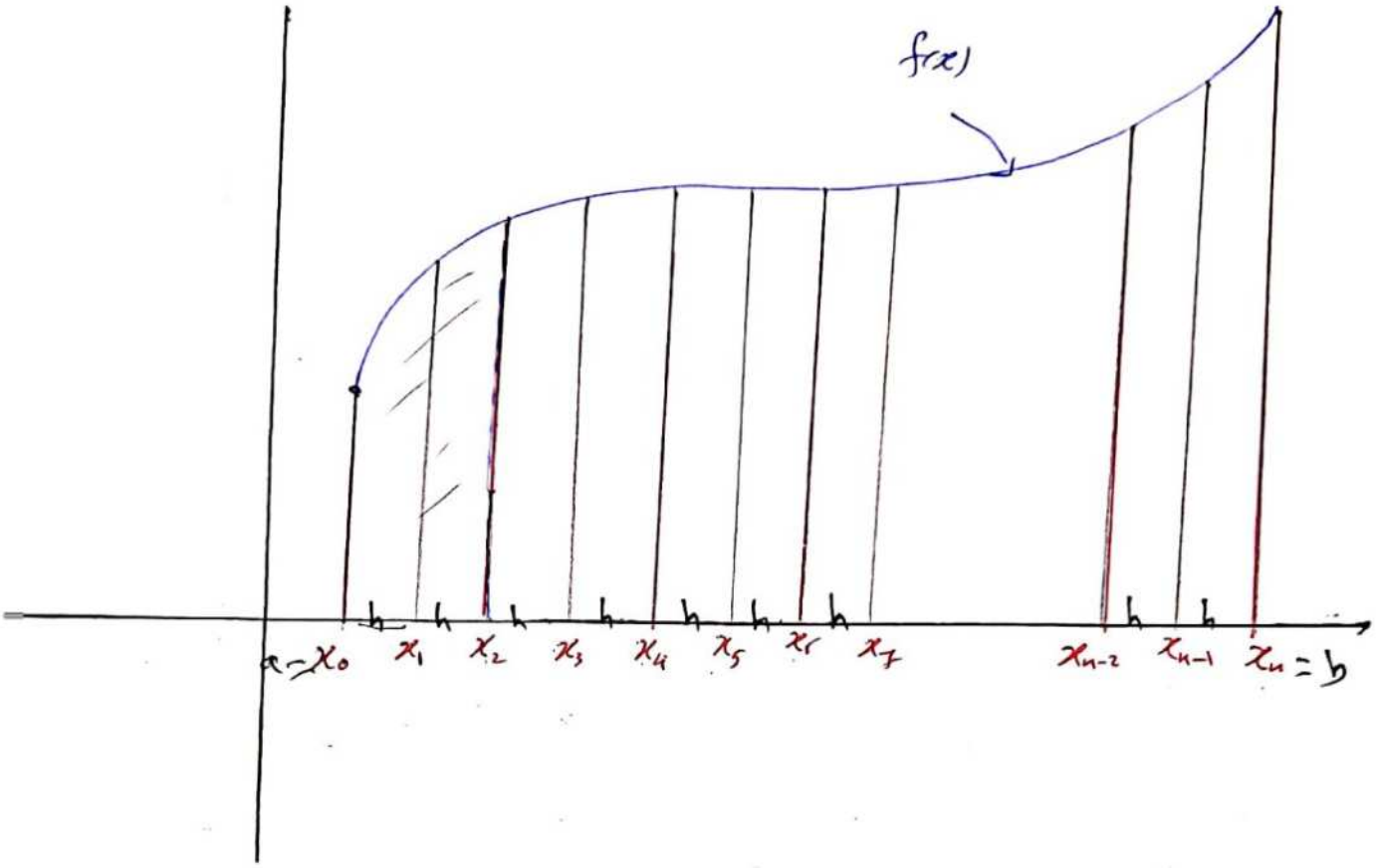
$$\simeq \frac{0.5}{3}\left[\sqrt{1+1} + 4\sqrt{e^{-1}+1} + \sqrt{e^{-2}+1}\right]$$

$$L \simeq 1.192998304$$

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The Multiple Applications of Simpson's $\frac{1}{3}$-Rule:

n should be even. i.e $n = 2m$, $m \in \mathbb{N}$



$$h = \frac{b-a}{n}$$

$$\int_a^b f(x)dx \simeq \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right] + \frac{h}{3}\left(f(x_2) + 4f(x_3) + f(x_4)\right)$$

$$+ \frac{h}{3}\left(f(x_4) + 4f(x_5) + f(x_6)\right) + \cdots + \frac{h}{3}\left(f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\right)$$

$$\Rightarrow \int_a^b f(x)dx = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2) + f(x_2) + 4f(x_3) + f(x_4) + f(x_4) + 4f(x_5)\right.$$

$$\left. + f(x_6) + \cdots + f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\right]$$

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(5)

$$\Rightarrow \int_a^b f(x)\,dx = \frac{h}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + 2f(x_6) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big]$$

$$\Rightarrow \int_a^b f(x)\,dx = \frac{h}{3}\Big[f(x_0) + 4\big(f(x_1) + f(x_3) + \cdots + f(x_{n-1})\big) + 2\big(f(x_2) + f(x_4) + \cdots + f(x_{n-2})\big) + f(x_n)\Big]$$

$$\int_a^b f(x)\,dx \doteq \frac{h}{3}\Big[f(x_0) + 4\sum_{i=1}^{m} f(x_{2i-1}) + 2\sum_{i=1}^{m-1} f(x_{2i}) + f(x_n)\Big]$$

$n = 2m$

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e.g : Using Multiple - Application Simpson's $\frac{1}{3}$-rule

taking $n=6$, approximate $\int_0^3 x^2 dx$.

Sol:

$n=6$, $f(x)=x^2$, $a=0$, $b=3$

$$\Longrightarrow h = \frac{3-0}{6} = 0.5$$

$x_0=0$, $x_1=0.5$, $x_2=1$, $x_3=1.5$, $x_4=2$, $x_5=2.5$

$x_6=3$

$$\therefore \int_0^3 x^2 dx \cong \frac{h}{3}\left[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+2f(x_4)+4f(x_5)+f(x_6)\right]$$

$$\cong \frac{0.5}{3}\left[0^2+4(0.5^2)+2(1^2)+4(1.5^2)+2(2^2)+4(2.5^2)+3^2\right]$$

$$\cong 9$$

True value $\int_0^3 x^2 dx = \frac{x^3}{3}\Big|_0^3 = 9$

END

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Simple Simpson's $\frac{3}{8}$ - Rule : $(n=3)$

To approximate the integral $\int_a^b f(x)\,dx$ using this method, we replace the integrand $f(x)$ by cubic function $f_3(x)$ (using Lagrange polynomial of order 3)

$$\Rightarrow \int_a^b f(x)\,dx \cong \int_a^b f_3(x)\,dx$$

Notice that

$n=3 \rightarrow h=\frac{b-a}{3}$

$x_0=a,\ x_1=a+h,\ x_2=a+2h$

$x_3=a+3h=b$

After Calculations

$$\Rightarrow \int_a^b f(x)\,dx = \frac{3h}{8}\left[f(x_0)+3f(x_1)+3f(x_2)+f(x_3)\right]$$

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e.g: Apply Simple Simpson's $\frac{3}{8}$-rule to approximate $\int_0^1 \sec\sqrt{x}\,dx$

Sol: $n=3$, $a=0$, $b=1 \longrightarrow h=\frac{b-a}{n}=\frac{1-0}{3}=\frac{1}{3}$

$\Rightarrow x_0=0$, $x_1=\frac{1}{3}$, $x_2=\frac{2}{3}$ and $x_3=1$.

$$f(x)=\sec\sqrt{x}=\frac{1}{\cos\sqrt{x}}$$

$$\Rightarrow \int_0^1 \sec\sqrt{x}\,dx \cong \frac{3h}{8}\left[f(x_0)+3f(x_1)+3f(x_2)+f(x_3)\right]$$

$$\cong \frac{3(\frac{1}{3})}{8}\left[\frac{1}{\cos\sqrt{0}}+\frac{3}{\cos\sqrt{\frac{1}{3}}}+\frac{3}{\cos\sqrt{\frac{2}{3}}}+\frac{1}{\cos\sqrt{1}}\right]$$

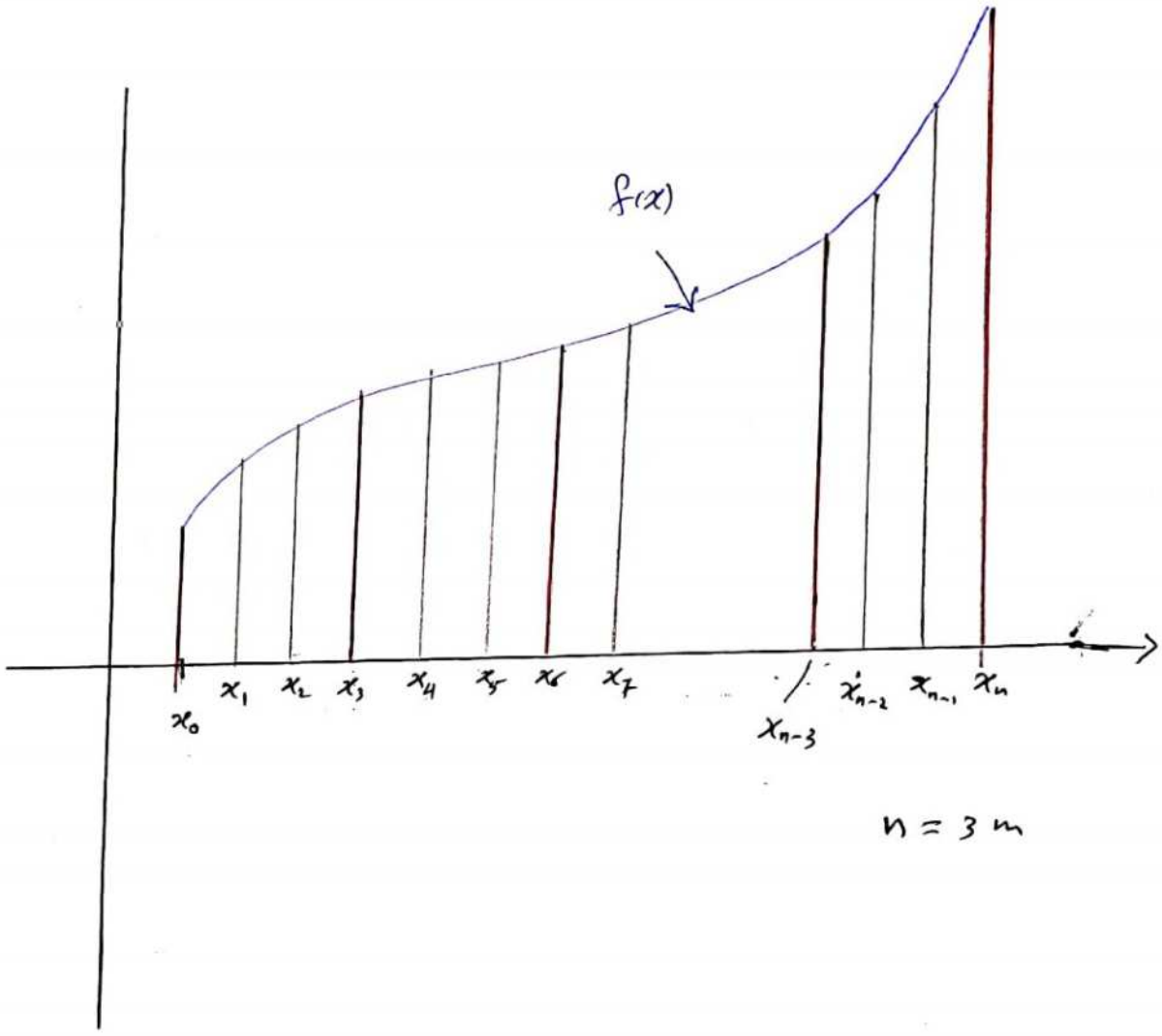
$$\cong 1.35154 \quad \checkmark$$

True value (by calculator) 1.351069947.

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Multiple - Application Simpson's $\frac{3}{8}$ - Rule

$n = 3m \quad , m \in \mathbb{N} \quad (n \equiv \text{multiple of } 3)$



$$\int_a^b f(x)dx = \frac{3h}{8}\left(f(x_0)+3f(x_1)+3f(x_2)+f(x_3)\right) + \frac{3h}{8}\left(f(x_3)+3f(x_4)+3f(x_5)+f(x_6)\right)$$

$$+ \quad \cdots \quad + \quad \frac{3h}{8}\left(f(x_{n-3})+3f(x_{n-2})+3f(x_{n-1})+f(x_n)\right)$$

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$$\Rightarrow \int_a^b f(x)\,dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + 2f(x_3) + 3f(x_4) + 3f(x_5)\right.$$
$$+ 2f(x_6) + \cdots + 2f(x_{n-3}) + 3f(x_{n-2}) +$$
$$\left. 3f(x_{n-1}) + f(x_n)\right]$$

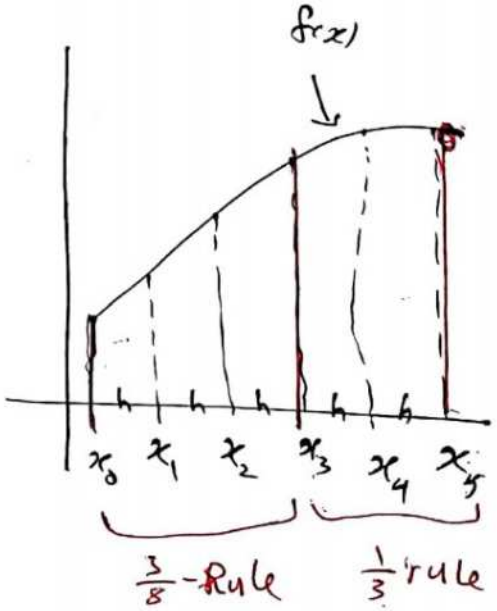
$$n = 6 \rightarrow \int_a^b f(x)\,dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + 2f(x_3) + 3f(x_4)\right.$$
$$\left. + 3f(x_5) + f(x_6)\right]$$

---

use Simpson's rules if

$n = 5$

$$\Rightarrow \int_a^b f(x)\,dx = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$
$$+ \frac{h}{3}\left[f(x_3) + 4f(x_4) + f(x_5)\right]$$

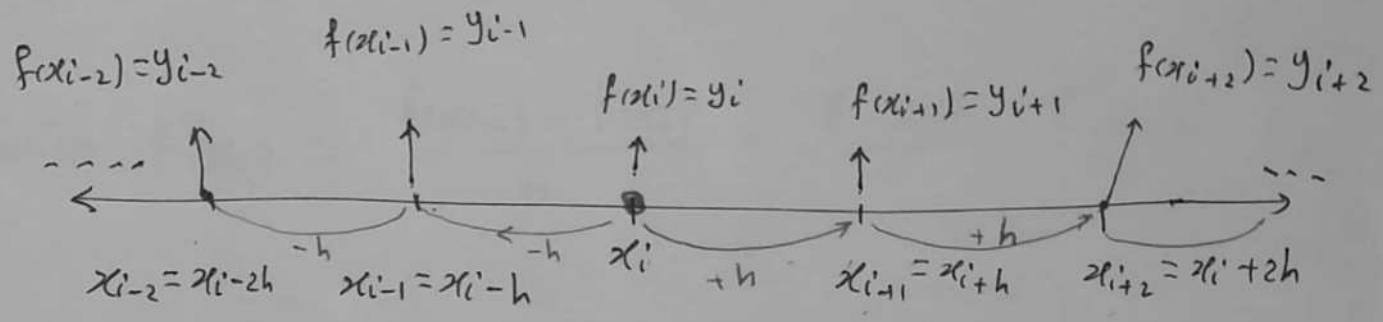


- END -

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Numerical Differentiation.

Notations: If $y = f(x)$



Taylor series: about $x = x_i$

$$f(x) = f(x_i) + f'(x_i)(x - x_i) + \frac{f''(x_i)(x-x_i)^2}{2!} + \frac{f'''(x_i)(x-x_i)^3}{3!} + \cdots \quad \cdots (1)$$

* If $x = x_{i+1} \rightarrow x - x_i = x_{i+1} - x_i = h$

$$\Rightarrow f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \cdots \quad \cdots (2)$$

* If $x = x_{i-1} \rightarrow x - x_i = x_{i-1} - x_i = -h$

sub. in (1)

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \frac{f'''(x_i)h^3}{3!} + \cdots \quad (3)$$

* If $x = x_{i+2} \rightarrow x_{i+2} - x_i = x - x_i = 2h$

$$\Rightarrow f(x_{i+2}) = f(x_i) + f'(x_i)(2h) + \frac{f''(x_i)(4h^2)}{2!} + \cdots \quad (4)$$

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from ② ⟹

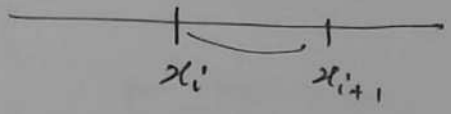
$$f'(x_i)\,h = f(x_{i+1}) - f(x_i) - \frac{f''(x_i)h^2}{2!} - \frac{f'''(x_i)h^3}{3!} - \cdots$$

$$\xrightarrow{\div h} f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - \underbrace{\frac{f''(x_i)h}{2} - \frac{f'''(x_i)h^2}{6} - \cdots}_{-O(h)}$$

$$\Rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - O(h)$$

or

$$\boxed{f'(x_i) \cong \frac{f(x_{i+1}) - f(x_i)}{h}}$$



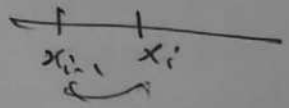
from ③ ⟹

$$f'(x_i)\,h = f(x_i) - f(x_{i-1}) + \frac{f''(x_i)h^2}{2!} - \frac{f'''(x_i)h^3}{3!} + \cdots$$

$$\xrightarrow{\div h} f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + \frac{f''(x_i)h}{2!} - \frac{f'''(x_i)h^2}{2!} + \cdots$$

$$\Rightarrow f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$

$$\boxed{f'(x_i) \cong \frac{f(x_i) - f(x_{i-1})}{h}}$$



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(2) − (3) ⟹

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + 2\frac{f'''(x_i)h^3}{3!} + 2\frac{f^{(5)}(x_i)h^5}{5!} + \cdots$$

$\div 2h$

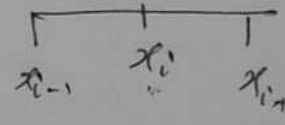
$$\Rightarrow \frac{f(x_{i+1}) - f(x_{i-1})}{2h} = f'(x_i) + \frac{f'''(x_i)h^2}{3} + \frac{f^{(5)}(x_i)h^4}{5!} + \cdots$$

$$\Rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} - \frac{f'''(x_i)h^2}{3} - \frac{f^{(5)}(x_i)h^4}{5!} - \cdots$$

$$\rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} - O(h^2)$$

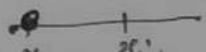
or

$$f'(x_i) \cong \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$$



To approximate the first derivative we can use

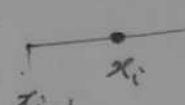
1) $f'(x_i) \cong \frac{f(x_{i+1}) - f(x_i)}{h}$ — Forward divided Difference Formula



2) $f'(x_i) \cong \frac{f(x_i) - f(x_{i-1})}{h}$ — Backward divided Difference Formula



3) $f'(x_i) \cong \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$ — Centered divided difference formula



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(2) + (3) $\Rightarrow$

$$f(x_{i+1}) + f(x_{i-1}) = 2f(x_i) + 2\frac{f''(x_i)h^2}{2!} + 2\frac{f^{(4)}(x_i)h^4}{4!} + \cdots$$

$$\Rightarrow f''(x_i)h^2 = f(x_{i+1}) - 2f(x_i) + f(x_{i-1}) - \frac{f^{(4)}(x_i)h^4}{12} - \cdots$$

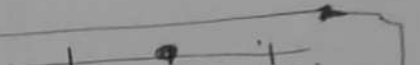
$$\div h^2 \Rightarrow f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} - \frac{f^{(4)}(x_i)h^2}{12} - \cdots$$

$$\Rightarrow f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} - O(h^2)$$

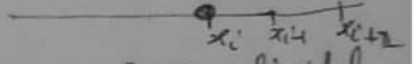
$$\Rightarrow \boxed{f''(x_i) \cong \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2}}$$

To approximate $f''(x_i)$ we can use

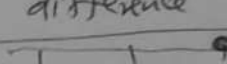
1) $f''(x_i) \cong \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2}$ centered finite divided difference formula



2) $f''(x_i) \cong \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$ Forward finite divided difference formula



3) $f''(x_i) \cong \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2}$ Backward finite divided difference formula

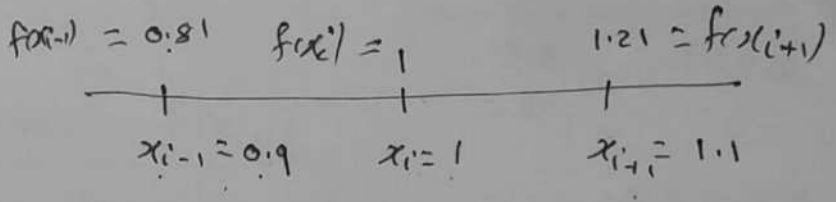


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e.g: Given that $f(x) = x^2$, approximate $f'(1)$ using Forward and centered divided difference formulas. take the step size $h = 0.1$

Sol:

$f(x) = x^2$



Forward: $f'(1) \cong \dfrac{f(x_{i+1}) - f(x_i)}{h}$

$\cong \dfrac{1.21 - 1}{0.1}$

$\cong \dfrac{0.21}{0.1} = \dfrac{21}{10} = 2.1$

True value
$f'(x) = 2x$.
$f'(1) = 2$

Centered: $f'(1) \cong \dfrac{f(x_{i+1}) - f(x_{i-1})}{2h}$

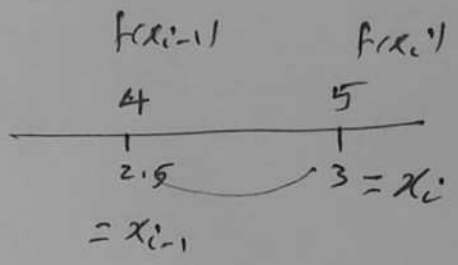
$\cong \dfrac{1.21 - 0.81}{2(0.1)}$

$\cong 2$ ..

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e.g: Given that $f(2.5) = 4$ and $f(3) = 5$, use a suitable divided difference formula to approximate $f'(3)$?

Sol:



$$f'(3) = \frac{f(x_i) - f(x_{i-1})}{h}$$

$h = 3 - 2.5 = 0.5$

$$= \frac{5 - 4}{0.5} = \frac{1}{0.5}$$

$$= 2$$

e.g: Given that $f(x) = x^3$, approximate $f''(2)$ use centered divided difference formula take the step size to be 0.5 ?

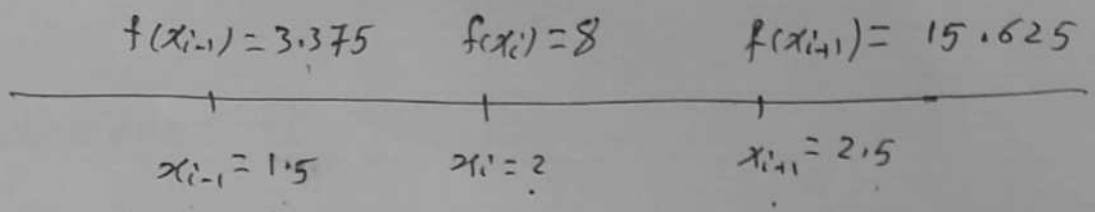
$h = 0.5$

$x_i = 2$

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Sol: $f(x) = x^3$

(7)



$$f''(2) \cong \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2}$$

$$\cong \frac{15.625 - 2(8) + 3.375}{0.5^2}$$

$$f''(2) \cong 12$$

True value $= f'(x) = 3x^2 \Rightarrow f'(2) = 12$

Also $f''(2) = 12$

Error $= 0$

END.

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Euler's Method

Introduction to Differential Equations:

Def:

A differential Equation (D.E) is an equation that containts derivative(s) of unknown function, and its order is the order of the highest derivatives that it contains.

To solve a D.E is to find the unknown function that satisfies it.

Note: If the differential equation contains partial derivative(s), it is called partial differential equation (PDE)

but if it contains ordinary derivative(s) it is called ordinary differential equation (ODE)

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Examples:

* $y'' + xy' + y = e^x$ ODE of 2nd order

* $\frac{\partial^2 u}{\partial y \partial x} + \frac{\partial^2 u}{\partial x^2} = 0$ PDE of 2nd order

* $x\,dy + y\,dx = 0$ 1st D.E

* $y^{(4)} + y^5 = 0$ 4th order D.E

* $(x^2 + y^2)\,dy + 2xy\,dx = 0$ 1st order D.E

* $y''' = \cos x$ D.E of 3rd order

* $f'(x) = 6x + 1$ D.E of 1st order

* $u_t = 9u_{xx}$ PDE of 2nd order. (Heat equation).

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e.g.: Show that $y = \sin 2x + \cos 2x$ is a solution for the differential equation

$$y'' + 4y = 0$$

Soln:

$$y = \sin 2x + \cos 2x$$

$$y' = 2\cos 2x - 2\sin 2x$$

$$y'' = -4\sin 2x - 4\cos 2x$$

$$y'' + 4y = -4\sin 2x - 4\cos 2x + 4(\sin 2x + \cos 2x)$$

$$= -4\sin 2x - 4\cos 2x + 4\sin 2x + 4\cos 2x$$

$$= 0$$

$\therefore y = \sin 2x + \cos 2x$ satisfies the D.E.

Hence, it is a solution for the D.E.

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e.g: Solve the differential equation $\frac{dy}{dx} = 2x$, $y(0) = 4$

Sol:

$\Rightarrow dy = 2x\,dx$

$\Rightarrow \int dy = \int 2x\,dx$

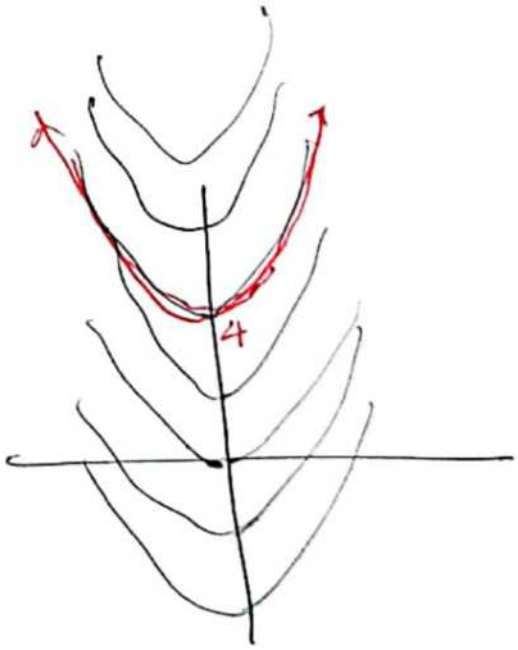
$y = x^2 + C$ general solution

, $y(0) = 4$ ← condition

$y(0) = 4 \rightarrow 0 + C = 4$

$\rightarrow C = 4$

$\therefore y = x^2 + 4$



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D.E + condition(s) at the same value of the independent variable ⟶ Initial Value Problem (IVP)

$$y'' + y = 0 \quad , \quad y(0) = 5 \; , \quad y'(0) = 4$$

$$y' = 2x + 1 \quad , \quad y(1) = 7$$

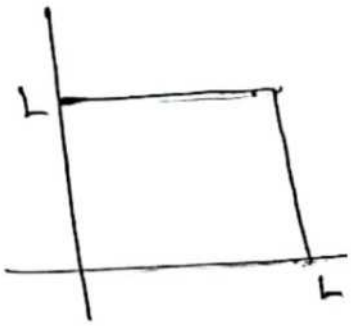
D.E + conditions at different values of the independent variable(s) ⟶ Boundary Value Problem (BVP)

$$y'' + y = 0 \quad , \quad y(0) = 2 \; , \quad y(\pi) = 7$$

$$u_{xx} + u_{yy} = 0$$

$$u(x,0) = 0 \; , \quad u(x,L) = 0$$

$$u(0,y) = 0 \quad , \quad u(L,y) = f(y)$$



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Euler's Method:

To solve (approximation) the IVP $\frac{dy}{dx} = f(x,y)$, $y(x_0) = y_0$ by Euler's method

$$\frac{y_{i+1} - y_i}{h} = f(x_i, y_i)$$

$$\Longrightarrow \boxed{y_{i+1} = y_i + h\, f(x_i, y_i)}$$

$x_0 \longrightarrow y_0$

$+h \downarrow$ x_1 $\quad y_1$

$+h \downarrow$ x_2 $\quad y_2$

$+h \downarrow$ x_3 $\quad y_3$

$\vdots$

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E.g: Using Euler's method, Solve the D.E (IVP)

$y' = 2x$, $y(0) = 1$ over the interval $[0, 4]$, take $h = 1$ (step size) then find the analytic solution of this IVP.

Sol:

$y' = 2x$ $\quad [0, 4]$

$y_{i+1} = y_i + h(2x_i)$ $\quad , y(0) = 1$

$x_0 = 0$	$y_0 = 1$
$x_1 = 1$	$y_1 = 1$
$x_2 = 2$	$y_2 = 3$
$x_3 = 3$	$y_3 = 7$
$x_4 = 4$	$y_4 = 13$

$h = 1 \rightarrow$

$y_{i+1} = y_i + 2x_i$

$y_1 = y_0 + 2x_0$

$= 1 + 2(0) = 1$

$y_2 = y_1 + 2x_1$

$y_2 = 1 + 2(1) = 3$

$y_3 = y_2 + 2x_2$

$y_3 = 3 + 2(2) = 7$

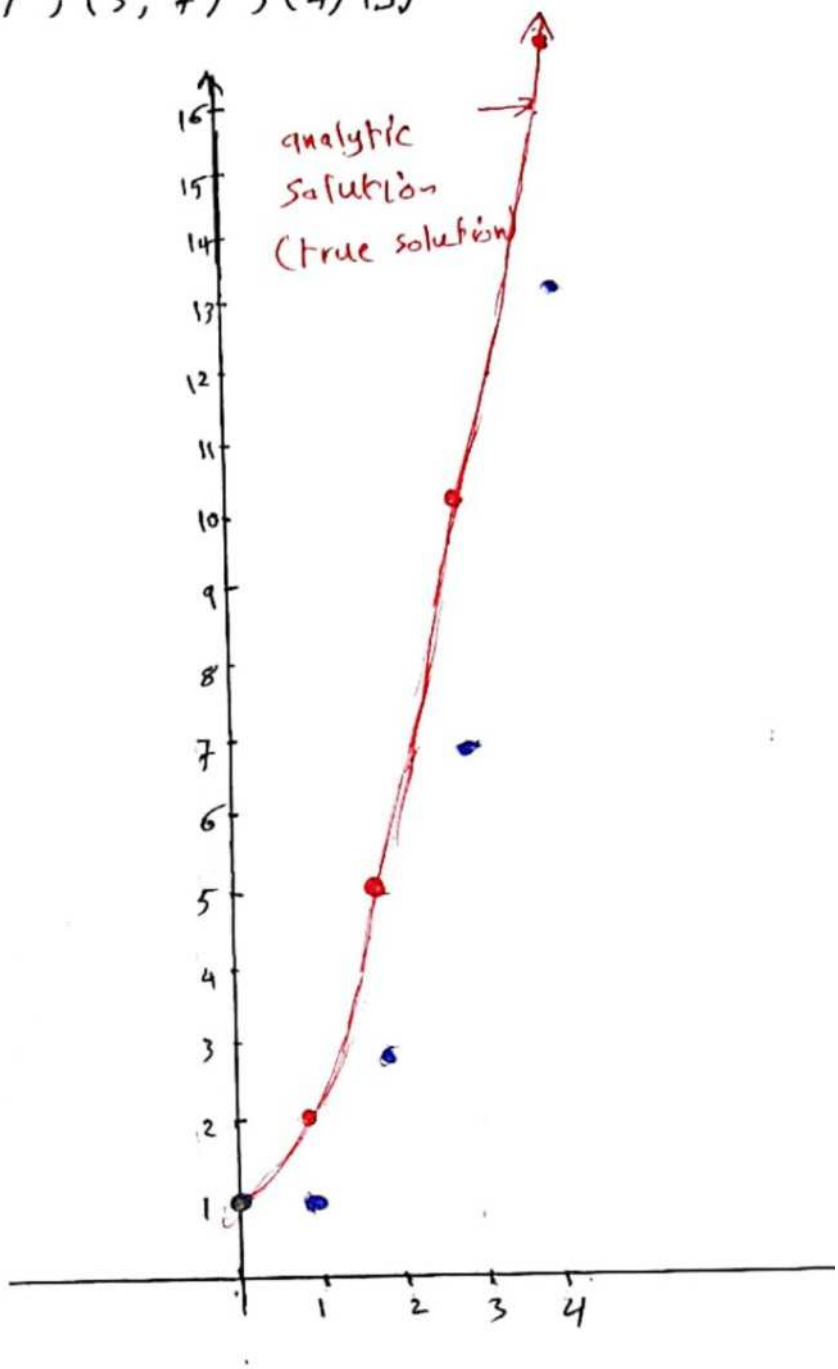
$y_4 = y_3 + 2x_3$

$= 7 + 2(3)$

$= 13$

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$\Rightarrow$ (0,1) , (1,1) , (2,3) , (3, 7) , (4, 13)



Analytic solution:

$y' = 2x$, $y(0) = 1$

$\Rightarrow dy = 2x\,dx$

$\Rightarrow \int dy = \int 2x\,dx$

$y = x^2 + C$

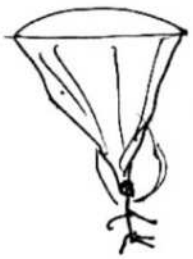
$y(0) = 1 \rightarrow C = 1$

$\Rightarrow y = x^2 + 1$

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Recall ; Parachutist Problem:

$$\frac{dv}{dt} = g - \frac{c}{m}v \qquad v(0)=0$$



$$mg - cv = m\frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = g - \frac{c}{m}v$$

Analytic solution

$$v = \frac{gm}{c}\left(1 - e^{-\frac{c}{m}t}\right)$$

Ex: A parachutist of mass 68.1 kg, jumps out of a stationary hot air baloon. if the drag coefficient $c = 12.5$ kg/s $g = 9.81\ m/s^2$

then $\frac{dv}{dt} = 9.81 - \frac{12.5}{68.1}v$

$$v = \frac{(9.81)(68.1)}{12.5}\left(1 - e^{-\frac{12.5}{68.1}t}\right)$$

Complete the following table

Sol:

Notice that

$$v_{i+1} = v_i + h\left(9.81 - \frac{12.5}{68.1}v_i\right)$$

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$$V = \frac{(9.81)(68.1)}{12.5}\left(1 - e^{-\frac{12.5}{68.1}(t)}\right)$$

$$V_{i+1} = V_i + 2\left(9.81 - \frac{12.5}{68.1} V_i\right)$$

t (sec)	V_i (Euler)	V (analytic)
0	0	0
2		
4		
6		
8		
10		
12		
14		
16		
18		
20		

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e.g: Use Euler's method to approximate $y(1.6)$ by solving the IVP

$$\frac{dy}{dx} = x^2 + y^2 \quad , y(1) = 2$$

take the step size to be 0.2

Solution:

$$\frac{dy}{dx} = x^2 + y^2 \quad , y(1) = 2$$

$$\Rightarrow f(x,y) = x^2 + y^2 \quad \text{(slope)} \quad , h = 0.2$$

$$\Rightarrow \boxed{y_{i+1} = y_i + 0.2(x_i^2 + y_i^2)}$$

$y(1.6) \cong ??$

i

x_i	y_i
$x_0 = 1$	$y_0 = 2$
$x_1 = 1.2$	$y_1 = 3$
$x_2 = 1.4$	$y_2 = 5.088$
$x_3 = 1.6$	$y(1.6) = 10.657\,5488 = y_3$

END

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Runge-Kutta 2nd order Methods:

To solve (numerically) the IVP

$$\frac{dy}{dx} = f(x,y) \qquad , y(x_0) = y_0$$

Find

1) $K_1 = f(x_i, y_i)$

2) $K_2 = f(x_i + ph \ , \ y_i + ph\,K_1)$ h: step size

where $a_1 + a_2 = 1$

$a_2 p = \frac{1}{2}$

then apply recurrence formula

$$y_{i+1} = y_i + h(a_1 K_1 + a_2 K_2) \qquad i = 0, 1, 2, \ldots$$

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① Heun's Method:

$a_2 = \frac{1}{2}$

(Predictor - corector)

$a_2 = \frac{1}{2} \longrightarrow a_1 + \frac{1}{2} = 1 \longrightarrow a_1 = \frac{1}{2}$

$a_2 p = \frac{1}{2} \longrightarrow \frac{1}{2} p = \frac{1}{2} \longrightarrow p = 1$

$\Rightarrow k_1 = f(x_i, y_i)$

$k_2 = f(x_i + h, y_i + k_1 h)$ ← Predictor

$$y_{i+1} = y_i + h\left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)$$

or $$y_{i+1} = y_i + \frac{h}{2}(k_1 + k_2)$$

Corrector

$i = 0, 1, 2, 3, \ldots$

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(2) Midpoint Method :

$a_2 = 1$

$a_2 = 1 \longrightarrow a_1 + 1 = 1 \longrightarrow a_1 = 0$

$a_2 p = \frac{1}{2} \longrightarrow 1p = \frac{1}{2} \longrightarrow p = \frac{1}{2}$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \tfrac{1}{2}h, y_i + \tfrac{1}{2}hk_1)$$

$$y_{i+1} = y_i + hk_2$$

$i = 0, 1, 2, 3, \ldots$

(3) Ralston Method :

$a_2 = \frac{2}{3}$

$a_1 + a_2 = 1 \longrightarrow a_1 + \frac{2}{3} = 1 \longrightarrow a_1 = \frac{1}{3}$

$a_2 p = \frac{1}{2} \longrightarrow \frac{2}{3} p = \frac{1}{2} \longrightarrow p = \frac{3}{4}$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \tfrac{3}{4}h, y_i + \tfrac{3}{4}hk_1)$$

$$y_{i+1} = y_i + h\left(\tfrac{1}{3}k_1 + \tfrac{2}{3}k_2\right)$$

$i = 0, 1, 2, \ldots$

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e.g: Use Heun's method to solve the (IVP)

$$\frac{dy}{dx} = y\sin x \quad , \quad y(0) = 1$$

over the interval $[0, 1]$, assume the step size $h = 0.5$.

Sol: $f(x, y) = y\sin x$

$x_0 = 0$, $y_0 = 1$, $h = 0.5$

$x_0 = 0 \quad y_0 = 1$

$x_1 = 0.5 \quad y_1$

$x_2 = 1 \quad y_2 = ?$

1st iteration:

$x_0 = 0$, $y_0 = 1$

$K_1 = f(x_i, y_i) \longrightarrow K_1 = y_0 \sin x_0$

$K_1 = 1 \sin 0 \rightarrow \boxed{K_1 = 0}$

$$K_2 = f(x_0 + \underset{h}{0.5}, y_0 + K_1(\underset{h}{0.5}))$$

$K_2 = f(0.5, 1 + 0(0.5))$

$K_2 = f(0.5, 1.0) = 1.0 \sin(0.5) \rightarrow \boxed{K_2 = 0.4794255}$

$$y_1 = y_0 + h\left(\frac{1}{2}K_1 + \frac{1}{2}K_2\right)$$

$$y_1 = 1 + \frac{0.5}{2}(0 + 0.4794255)$$

$$\rightarrow \boxed{y_1 = 1.1198564}$$

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$x_1 = 0.5$, $y_1 = 1.1198564$

The analytic solution: $y = e^{1-\cos x}$

$y(0.5) = e^{1-\cos(0.5)} = 1.1302258$

2nd iteration

$x_1 = 0.5$, $y_1 = 1.1198564$

$K_1 = f(0.5, 1.1198564)$

$= 1.1198564 \sin 0.5$

$\Rightarrow \boxed{K_1 = 0.5368878}$

$K_2 = f(0.5 + 0.5, 1.1198564 + 0.5(0.5368878))$

$K_2 = f(1, 1.3883003)$

$= 1.3883003 \sin 1$

$\boxed{K_2 = 1.1682442}$

$\therefore y_2 = 1.1198564 + \frac{0.5}{2}[0.5368878 + 1.1682442]$

$y_2 = 1.5461394$

True value

$y(1) = e^{1-\cos 1}$

$= 1.583595183$

$(0, 1)$, $(0.5, 1.1198564)$ and $(1, 1.5461394)$

Ex: Solve the above example by Midpoint and Ralston methods

- END -

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Numerical Solution for BVP

by

Finite Difference Method:

Idea: replace

y'' by $\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$	centered F.D formula for 2nd derivative
y' by $\frac{y_{i+1} - y_{i-1}}{2h}$	center F.D formula for 1st derivative
y by y_i	

$h = \Delta x$ = step size

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e.g: By using finite difference method, solve the BVP:

$$y'' - y = 0 \quad , \quad y(0) = 1 \quad , \quad y(1) = 2$$

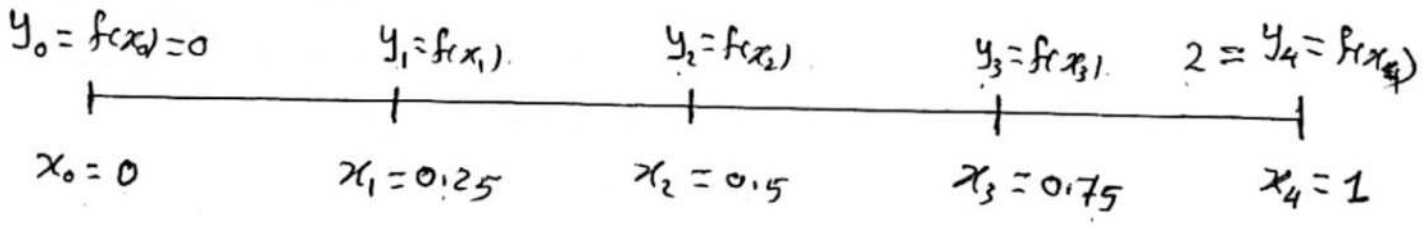
Use 4-segments (5 point (values)(nodes))

Sol:

$$n = 4 \quad , \quad a = 0 \quad , \quad b = 1$$

$$h = \frac{b-a}{n} \rightarrow h = \frac{1-0}{4} \longrightarrow h = 0.25$$

$$\Rightarrow a = x_0 = 0 \quad , x_1 = 0.25 \quad , x_2 = 0.5 \quad , x_3 = 0.75 \quad , x_4 = 1 = b$$



Now replace y'' by $\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$ and y by y_i in the D.E $y'' - y = 0$, to get

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} - y_i = 0$$

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$* h^2 \Rightarrow \quad y_{i+1} - 2y_i + y_{i-1} - h^2 y_i = 0$

$$y_{i-1} - (2+h^2)y_i + y_{i+1} = 0$$

but $h^2 = 0.125^2 = 0.0625$

$$\Rightarrow \boxed{y_{i-1} - 2.0625 y_i + y_{i+1} = 0 \qquad i = 1, 2, 3}$$

$i=1 \rightarrow y_0 - 2.0625 y_1 + y_2 = 0$

$\Rightarrow 1 - 2.0625 y_1 + y_2 = 0$

$$\Rightarrow \boxed{-2.0625 y_1 + y_2 = -1} \qquad (1)$$

$$i=2 \rightarrow \boxed{y_1 - 2.0625 y_2 + y_3 = 0} \qquad (2)$$

$i=3 \longrightarrow y_2 - 2.0625 y_3 + y_4 = 0$

$\rightarrow y_2 - 2.0625 y_3 + 2 = 0$

$$\Rightarrow \boxed{y_2 - 2.0625 y_3 = -2} \qquad (3)$$

$$\begin{bmatrix} -2.0625 & 1 & 0 \\ 1 & -2.0625 & 1 \\ 0 & 1 & -2.0625 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix}$$

by solving this system we get

$$y_1 = 1.130193 \;, \quad y_2 = 1.331023 \;, \quad y_3 = 1.615041$$

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The analytic solution of the BVP

$$y = 0.69440\, e^{x} + 0.30560\, e^{-x}$$

$$y(0) = y_0 = 1$$

$1.12962876 = y(0.25) \cong y_1 = 1.130193$

$1.33022782 = y(0.5) \cong y_2 = 1.331023$

$1.61440003 = y(0.75) \cong y_3 = 1.615041$

$1.99999859 = y(1) \cong y_4 = 2$

2

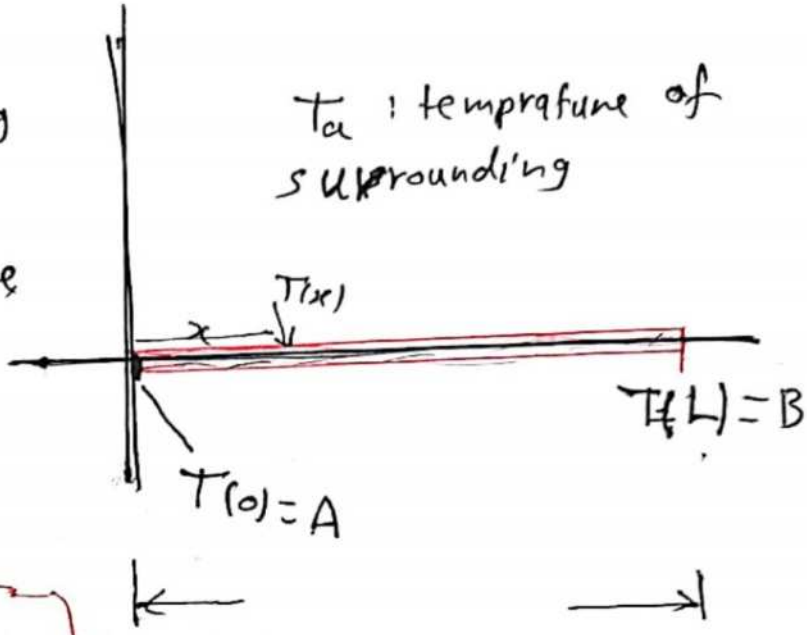
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Application:

A noninsulated uniform rod positioned between two bodies of constant but different temperature.

If L is non-insulated along its length and the system in the steady state the equation



$$\frac{d^2T}{dx^2} + h'(T_a - T) = 0$$

$$T(0) = A \ , \ T(L) = B$$

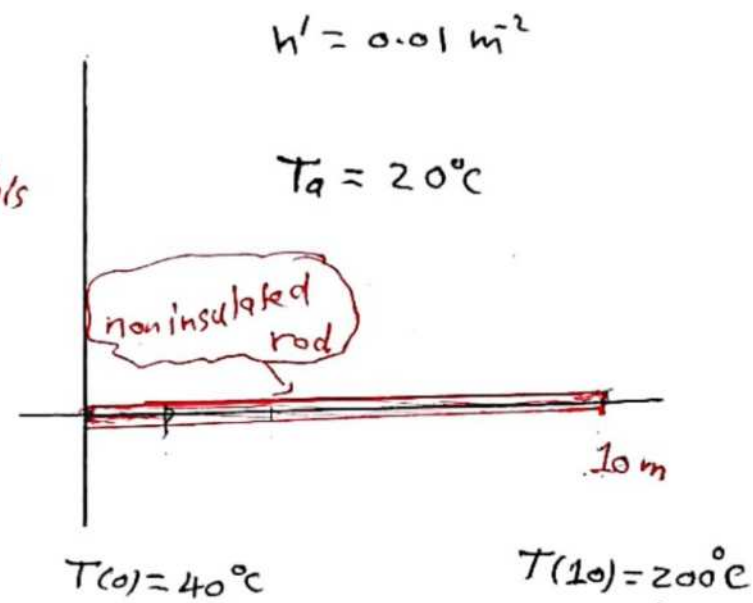
h': heat transfer coefficient

T_a: the temperature of the surroundling.

$T = T(x)$ = the temperature at x.

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e.g: If the system in the steady state. and the differential equation of this model:



$$\frac{d^2T}{dx^2} + h'(T_a - T) = 0$$

$T(0) = 40^\circ C$, $T(10) = 200^\circ C$

and the ~~trans~~ heat transfer coefficient $= 0.01\ m^{-2}$

and the surrounding temperature $= 20^\circ C$

Find the temperature when $x = 2, 4, 6, 8$ m

using finite difference method.

Sol $h = 2$ (we have 5-segments)

The diff. equation

$$\frac{d^2T}{dx^2} + 0.01(20 - T) = 0 \quad , T(0) = 40,$$

$$T(10) = 200$$

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$$\frac{d^2T}{dx^2} + 0.2 - 0.01T = 0$$

$$\Rightarrow \frac{d^2T}{dx^2} - 0.01T = -0.2$$

$$\Rightarrow \frac{T_{i+1} - 2T_i + T_{i-1}}{h^2} - 0.01\,T_i = -0.2$$

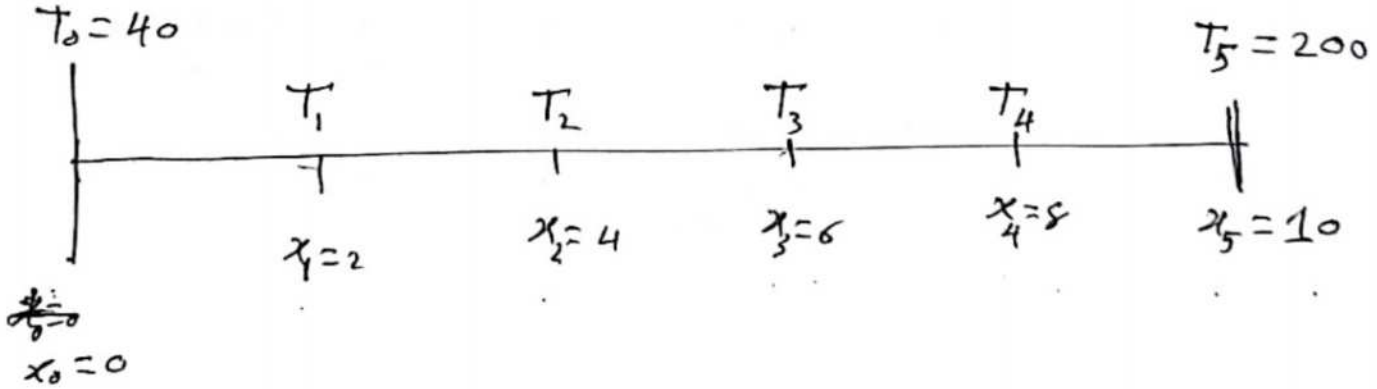
but $h = 2 \rightarrow h^2 = 4$

$$\Rightarrow \frac{T_{i+1} - 2T_i + T_{i-1}}{4} - 0.01\,T_i = -0.2$$

$$*4 \Rightarrow T_{i+1} - 2T_i + T_{i-1} - 0.04\,T_i = -0.8$$

$$\boxed{T_{i-1} - 2.04\,T_i + T_{i+1} = -0.8}$$

$T(0) = 40, \quad T(10) = 200$



4 - unknowns

$\Rightarrow i = 1, 2, 3, 4$

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$$T_{i-1} - 2.04\,T_i + T_{i+1} = -0.8$$

$i = 1 \rightarrow T_0 - 2.04T_1 + T_2 = -0.8$

$40 - 2.04T_1 + T_2 = -0.8$

$$-2.04T_1 + T_2 = -40.8 \quad -(1)$$

$i = 2 \rightarrow$ $T_1 - 2.04T_2 + T_3 = -0.8$ $-(2)$

$i = 3 \rightarrow$ $T_2 - 2.04T_3 + T_4 = -0.8$ $-(3)$

$i = 4 \rightarrow T_3 - 2.04T_4 + T_5 = -0.8$

$\rightarrow T_3 - 2.04T_4 + 200 = -0.8$

$\Rightarrow$ $T_3 - 2.04T_4 = -200.8$ (4)

$$\begin{bmatrix} -2.04 & 1 & 0 & 0 \\ 1 & -2.04 & 1 & 0 \\ 0 & 1 & -2.04 & 1 \\ 0 & 0 & 1 & -2.04 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} -40.8 \\ -0.8 \\ -0.8 \\ -200.8 \end{bmatrix}$$

Solving this system we get

$T_1 = 65.9698°C$; $T_2 = 93.7785°C$, $T_3 = 124.5382°C$

$T_4 = 159.4795°C$

END

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